

COVARIANT STRETCHING–DIFFUSION RIGIDITY AND GLOBAL REGULARITY FOR THE THREE-DIMENSIONAL NAVIER–STOKES EQUATIONS

GUILLEM DURAN BALLESTER

ABSTRACT. We give a covariant proof of global regularity for the three-dimensional incompressible Navier–Stokes equations organized around one normalized observer approaching a putative singularity. Translation, parabolic scale, constant Galilean velocity, and pressure gauge are fixed coherently, while a backward adjoint localization retains a scale-critical quotient oscillation. Every retained unit window has a uniform positive viscous price in normalized critical units.

The moving-root local-energy identity records exactly how that price is financed. Its signed current contains pressure, nonlinear transport, dilation, translation, and cutoff motion. Summation over successive windows shows that sublinear cumulative current cannot support infinitely many active events. Persistent current is retained together with the observer as marked PDE data. Loss of compactness in center, pressure, scale, or defect variables produces a classified local output; repeated regeneration of active centers produces a sequence-realized logarithmic critical tail. The diffuse, homogeneous, defect, log-periodic, and aperiodic tail alternatives are rigid, so every persistent-current output is trivial. This contradicts the retained oscillation and excludes finite-time blow-up.

The proof is developed through five structural theorems: canonical observation, a positive critical dissipation price, the exact financing identity, persistent-current compactification, and critical-tail rigidity. The appendices derive these statements directly from the suitable local-energy inequality, pressure reconstruction, parabolic compactness, and ε -regularity. Material coordinates provide the geometric interpretation: vortex stretching and viscous diffusion are governed by the dual tensors ∇X and $(\nabla X^T \nabla X)^{-1}$, and zero production is spatially homogeneous modulo the Galilean quotient.

CONTENTS

1. Introduction	3
The five structural theorems	5
Proof architecture	5
2. Suitable solutions and the singular target	7
2.1. Suitable weak solutions	7
2.2. Scale-critical concentration and local kinetic energy	8
3. Euler transport and the Stokes gradient	9
3.1. The solenoidal Gelfand triple	9
3.2. The finite energy inequality	10
4. Material deformation and stretching–diffusion duality	11
5. Covariance under scale–translation charts	12
5.1. The repaired equation	12
5.2. Covariant energy and pressure	13
6. Monotone charts and recurrent pressure cancellation	13

2020 *Mathematics Subject Classification.* 35Q30, 35B44, 35B40, 35K55, 76D05.

Key words and phrases. Navier–Stokes equations, global regularity, suitable weak solutions, material deformation, stretching–diffusion duality, covariant dissipation, concentration compactness, rigidity, blow-up coordinates.

6.1.	A monotone interpolation through ordered active cylinders	14
6.2.	The invariant mean identity	15
7.	Scale-critical quotient geometry	20
7.1.	Physical action and critical action	20
7.2.	Galilean and pressure quotients	20
7.3.	The quotient topology	21
8.	Backward adjoint localizations and entropy	22
8.1.	The normalized finite-horizon adjoint problem	22
8.2.	Transport identity and the Gaussian equality model	23
9.	Stokes-metric projection onto localized kinetic variations	25
9.1.	Scale-critical kinetic probes	25
9.2.	The Stokes-metric projection inequality	27
9.3.	Alignment, coercivity, and coherent probe selection	29
10.	Canonical critical orbits	30
10.1.	Symmetry reduction and target-retaining windows	30
10.2.	Canonical observation	31
10.3.	The two classical scale regimes	31
11.	Exact critical financing	32
11.1.	Normalized shell decomposition	33
11.2.	Finite-depth concentration decomposition	34
12.	Covariant adjoint geometry and the moving-cutoff energy	38
12.1.	The explicit nonnegative components	38
12.2.	The uncoupled entropy balance	39
12.3.	The moving-cutoff kinetic energy	42
12.4.	Finite-block use of the moving cutoff	43
12.5.	Equality states	43
13.	The positive critical price	43
14.	Covariant exclusion of singularities	44
	Appendix A. Weak covariant calculus and compact stability	46
A.1.	Regularization of the equation	46
A.2.	Compactness of the variational data	46
	Appendix B. Singular-germ extraction	47
B.1.	Velocity concentration	47
B.2.	Compactness selection	48
B.3.	Marked descendant structure for regenerated current	54
B.4.	Closure of the amplitude-renormalizing residual	59
B.5.	Terminal concentration and exclusion of the record profile	69
B.6.	A single forward orbit	72
B.7.	Compactness and persistence	72
	Appendix C. Gauge, pressure, and adjoint estimates	73
C.1.	Scale-center normalization	73
C.2.	Local pressure decomposition	73
C.3.	Adjoint localization on marked blocks	74
	Appendix D. Critical flux-dissipation balance	74
D.1.	Compatible nested parabolic cells	74
D.2.	Summed local energy identity	74
	Appendix E. Target-visible coercivity and zero-dissipation limits	74
E.1.	Zero-dissipation limits are target-null	75
E.2.	Coercivity on the target quotient	75

E.3. Divergence on a persistent target sequence	76
Appendix F. Localized equality and the role of the Gaussian	76
References	76

1. INTRODUCTION

Consider the incompressible Navier–Stokes equations on $\mathbb{R}^3$,

$$(1.1) \quad \partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \operatorname{div} u = 0, \quad \nu > 0.$$

Three geometric observations organize the argument. Arnold’s description of Euler flow treats incompressible transport as conservative motion on the group of volume-preserving diffeomorphisms [2, 4]. Onsager’s variational description of irreversible evolution identifies a dissipative current with a gradient in a mobility metric [8]. For Navier–Stokes these statements fit together exactly:

$$(1.2) \quad \underbrace{\partial_t u + \mathbb{P}((u \cdot \nabla)u)}_{\text{Euler-covariant derivative}} = - \underbrace{\nu(-\mathbb{P}\Delta)}_{\text{Stokes mobility}} D\mathcal{E}(u), \quad \mathcal{E}(u) = \frac{1}{2}\|u\|_2^2.$$

The conservative term is orthogonal to $D\mathcal{E}(u) = u$, while the squared metric norm of the right-hand side is $\nu\|\nabla u\|_2^2$. Thus the energy inequality is the energy–dissipation identity associated with (1.2).

The third observation is specific to incompressible vorticity. In material coordinates the deformation tensor $\mathbf{F} = \nabla_a X$ transports vorticity, while viscosity is pulled back by the inverse Cauchy–Green metric $\mathbf{C}^{-1} = (\mathbf{F}^T \mathbf{F})^{-1}$. Since $\det \mathbf{F} = 1$, stretching and enhanced diffusion are dual aspects of the same volume-preserving deformation. This is the structural mechanism developed in Section 4.

Ricci-flow viewpoint	Navier–Stokes realization
Canonical observer near a singularity	Scale, center, Galilean mode, and pressure gauge
Conjugate heat kernel	Backward adjoint localization
Entropy or reduced-volume monotonicity	Covariant flux–dissipation identity
Noncollapsing	Retained scale-critical quotient oscillation
Blow-up limit	Ancient or recurrent normalized flow
Soliton at equality	Zero-production covariant equality state
Soliton rigidity	Spatial homogeneity modulo the Galilean quotient
Singular-scale analysis	Marked logarithmic current and critical-tail analysis

The physical Stokes action alone does not exclude concentration because its dissipation on a normalized event at scale r is weighted by r . A normalized gap and a physical lower bound are therefore distinct assertions. In particular, cancellation of fluxes between nested cells cannot by itself replace this factor by a scale-independent constant. The required quantity is instead the net dissipation in a scale-critical localized-energy balance. The proof has four steps.

1. *Canonical observation.* Terminal loss generates one coherent scale–center path. Translation, dilation, Galilean velocity, and pressure gauge are fixed, while a backward adjoint localization retains a uniform quotient-oscillation floor.
2. *Critical dissipation price.* Compactness and zero-production rigidity turn retained nonconstant oscillation into

$$\nu \int_{I_j} \int \zeta^2 |\nabla V|^2 \geq d_* > 0$$

on every active window. This is scale invariant; the corresponding physical energy is only $r_j d_*$.

3. *Exact financing.* The moving-root local-energy equality is

$$K_j^+ - K_j^- + D_j + M_j = H_j,$$

where H_j contains the complete signed pressure, transport, dilation, translation, cutoff, and transition current. After summation, bounded endpoint storage and the fixed price rule out sublinear cumulative current.

4. *Recurrence–tail rigidity.* Persistent current either loses a first center, pressure, scale, exterior, trace, or defect coordinate, or returns through another active center. The first failed coordinate is retained at its first parabolic scale. In the recurrent case the observer and current are retained as marks. Both constructions produce a realized logarithmic critical tail, and critical-tail rigidity makes that tail target-null.

Thus the entire mechanism is

$$(1.3) \quad \boxed{\begin{array}{l} \text{retained critical activity} \implies \text{positive critical price} \\ \implies \text{persistent signed current} \implies \text{recurrent critical tail} \implies \text{rigidity.} \end{array}}$$

This separation starts from the localized Navier–Stokes energy identity. Its convection, pressure, cutoff-motion, chart, dissipation, and defect terms all depend linearly on the localization test. A compatible parent–child partition therefore prevents double counting but does not, by itself, cancel only the signed terms. Selective cancellation is a statement about oriented normal traces on a disjoint spacetime partition. Viscous dissipation and the suitable-energy defect then retain their sign on the disjoint interiors. Section 11 develops this local energy decomposition entirely within the Navier–Stokes equation and isolates the target-visible zero-dissipation problem.

The rigidity in the fourth step is the covariant equality calculation. Ordered active cylinders admit one exact chart with nonincreasing scale. On an autonomous compact recurrent limit, covariant spatial averaging removes convection, translation, and every nonzero spatial pressure mode, leaving

$$\nu \langle |\nabla V|^2 \rangle - a_* \langle |V - \Pi_0 V|^2 \rangle = 0.$$

For $a_* \leq 0$ this forces spatial constancy and hence loss of the mean-subtracted singular target. This pressure-cancellation argument is proved in Section 6.

The proof uses one coherent observer and one accounting identity. The observer retains scale-critical quotient oscillation; the identity says that its fixed viscous price must be financed by signed current. Persistent current either appears as a classified loss of compactness or returns through smaller scales and generates a logarithmic critical tail. Tail rigidity supplies the final contradiction.

Theorem 1.1 (Global regularity). *Let $m > 5/2$, $\nu > 0$, and let $u_0 \in H^m(\mathbb{R}^3; \mathbb{R}^3)$ be divergence free. Then (1.1) has a unique global classical solution*

$$u \in C([0, \infty); H^m) \cap C^\infty(\mathbb{R}^3 \times (0, \infty)).$$

Every Leray–Hopf solution with initial datum u_0 agrees with this classical solution.

The proof of Theorem 1.1 is given in Section 14. It invokes the following five structural theorems, each proved in this paper:

- (i) canonical observation, Theorem 10.2;
- (ii) the positive critical price, Theorem 13.2;
- (iii) the exact financing identity, Theorem 11.2;
- (iv) persistent-current compactification, Theorem 11.13;
- (v) critical-tail rigidity, Theorem 11.14.

Remark 1.2 (Logical scope). No regularity theorem from another manuscript is an input. The appendices derive the five structural theorems from the suitable local-energy inequality, pressure reconstruction, parabolic compactness, ε -regularity, and the ancient-solution rigidity statements cited there. The finite-depth source decomposition proves that persistent-current compactification is exhaustive; each of its outputs is resolved inside critical-tail rigidity.

The five structural theorems.

Theorem	Geometric content	Derivation in this paper
Canonical observation	One scale–center–gauge path retains quotient activity	Appendix B
Critical price	Every retained normalized window pays $d_* > 0$	Sections E and 13
Exact financing	Stored energy plus dissipation and defect equals complete signed current	Sections D and 11
Persistent-current compactification	Sublinear current is impossible; persistent current yields a classified loss or recurrent tail	Sections B.3, B.4 and 11.2
Critical-tail rigidity	Every realized diffuse, defect, homogeneous, log-periodic, or aperiodic tail is target-null	Sections B.5 and E

Proof architecture. Rectangles are analytic states, diamonds are exhaustive alternatives, and solid ellipses are contradictions. Every edge belongs to the single observer–price–current–tail argument.

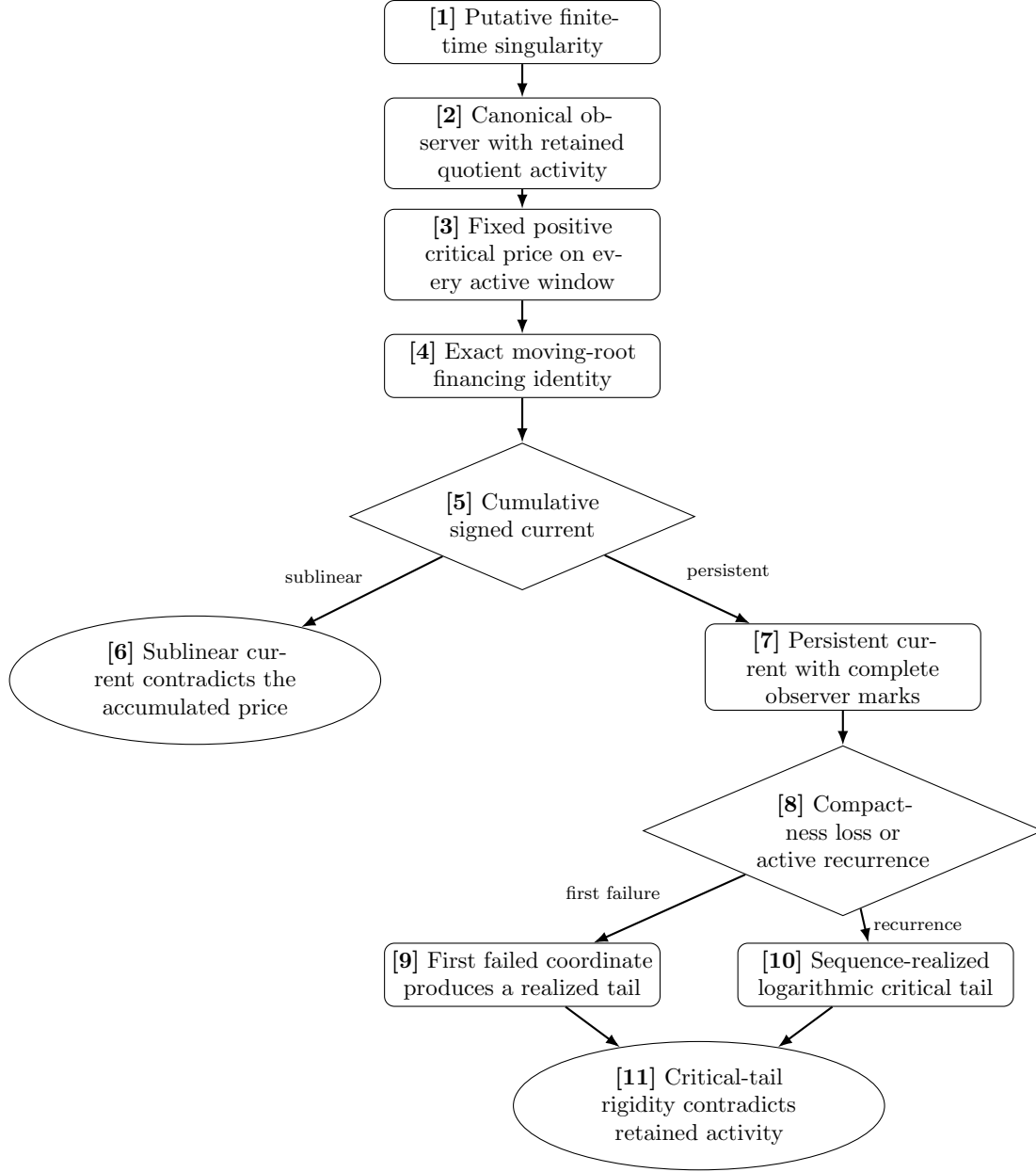
Proof-dependency diagram.

FIGURE 1. The single covariant proof spine. Persistent current cannot be lost: its first failed coordinate is retained as a classified PDE output, while continued active recurrence produces a realized critical tail.

Proof-state source table.

State	Analytic content	Exact source and successor
[1]	Terminal loss	Theorem 2.1 ; continue to [2].
[2]	Coherent normalization	Theorem 10.2 ; continue to [3].
[3]	Critical price	Theorem 13.2 ; continue to [4].
[4]	Exact accounting	Theorem 11.2 ; continue to [5].
[5]	Current growth test	Lemma B.35 ; outputs [6] or [7].
[6]	Insufficient financing	The left side grows at least like Nd_* , while the right side is $o(N)$; terminal.
[7]	Persistent current	Lemma B.40 ; continue to [8].
[8]	Exhaustive compactification	Theorem 11.13 ; outputs [9] or [10].
[9]	First failed coordinate	Theorem 11.7 and Proposition B.4 ; continue to [11].
[10]	Recurrent marked path	Theorem B.42 ; produces a realized tail and continues to [11].
[11]	Tail rigidity	Theorem 11.14 ; terminal.

The facts carried from [2] onward are the fixed quotient-oscillation floor, realization by the same physical terminal sequence, compatible pressure gauges, the exact scale–center composition law, and the complete signed local-energy current. Thus neither a subsequence nor a recurrent limit is permitted to forget the moving root or its pressure contribution.

Exhaustion table.

Case	Retained output	Closure used by Theorem 1.1
Sublinear current	The complete signed current is $o(N)$ over N retained events	Lemma B.35 .
Compactness loss	Pressure, center, scale, exterior, trace, or defect coordinate first fails	Theorems 11.7 and 11.14 and Proposition B.4 .
Active recurrence	Persistent current repeatedly reaches another retained center	Lemma B.40 and Theorems B.42 and 11.14 .

First-failure source table.

The following table expands state [9]. A failed coordinate is retained as PDE data and passed to the critical-tail theorem; it is never treated as an unproved terminal alternative.

Output	PDE state	Status
F1	Homogeneous velocity in the Galilean quotient	Target-null by definition of $\mathcal{C}_\Gamma^\circ$.
F2	Compact nonexpanding recurrence	Zero-production rigidity by Theorem 6.6 .
F3	Pressure, exterior, scale, trace, or defect loss	First visible parabolic scale gives a sequence-realized tail by Theorem 11.7 and Proposition B.4 .
F4	Same-origin scale collapse with persistent current	Eventwise critical price and marked recurrence give a realized tail by Theorem B.43 .

2. SUITABLE SOLUTIONS AND THE SINGULAR TARGET

2.1. Suitable weak solutions. We use the standard finite-energy suitable class [6, 3, 7]. Thus

$$(2.1) \quad u \in L^\infty(0, T; L_\sigma^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}_\sigma^1(\mathbb{R}^3)), \quad p \in L_{\text{loc}}^{3/2}(\mathbb{R}^3 \times (0, T)),$$

the equation holds in distributions, the global Leray–Hopf energy inequality holds, and the local energy inequality holds for every nonnegative compactly supported smooth test function. Here the subscript σ denotes divergence-free fields. A terminal point (x_0, T) is regular when u is locally bounded in a backward parabolic neighborhood of that point. We write

$$(2.2) \quad \Sigma_u(T) := \{x_0 \in \mathbb{R}^3 : (x_0, T) \text{ is not regular}\}.$$

For $z_0 = (x_0, t_0)$ and $r > 0$, write $Q_r(z_0) = B_r(x_0) \times (t_0 - r^2, t_0)$ and define

$$(2.3) \quad A(z_0, r) := r^{-1} \operatorname{ess\,sup}_{t_0 - r^2 < t < t_0} \int_{B_r(x_0)} |u(x, t)|^2 dx,$$

$$(2.4) \quad E(z_0, r) := r^{-1} \int_{Q_r(z_0)} |\nabla u|^2 dx dt,$$

$$(2.5) \quad C(z_0, r) := r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt,$$

$$(2.6) \quad D(z_0, r) := r^{-2} \int_{t_0 - r^2}^{t_0} \int_{B_r(x_0)} |p - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

These quantities are invariant under the Navier–Stokes scaling. The pressure mean in D fixes the local pressure gauge.

Theorem 2.1 (Velocity concentration at a singular point). *There is a threshold $\varepsilon_v = \varepsilon_v(\nu) > 0$ such that every $x_0 \in \Sigma_u(T)$ admits radii $r_j \downarrow 0$ for which*

$$(2.7) \quad C((x_0, T), r_j) \geq \varepsilon_v \quad \text{for every } j.$$

Proof. Use the one-scale velocity regularity criterion of Gustafson–Kang–Tsai [5] with spatial and temporal exponents equal to three. In the present normalization it gives a constant $\varepsilon_{\text{reg}} > 0$ such that $C(z_0, r) < \varepsilon_{\text{reg}}$ implies boundedness in $Q_{r/2}(z_0)$. The statement is invariant under (7.1). Therefore a singular point satisfies

$$\limsup_{r \downarrow 0} C((x_0, T), r) \geq \varepsilon_{\text{reg}}.$$

Take $\varepsilon_v = \varepsilon_{\text{reg}}/2$ and choose a sequence converging to zero on which $C \geq \varepsilon_v$. $\square$

Appendix B.1 records the additional pressure normalization and oscillation-retention statements needed by the later compactness construction; these do not enter the preceding concentration lemma.

2.2. Scale-critical concentration and local kinetic energy. The target of the variational interpretation is the set of germs retaining the velocity concentration forced by singularity.

Definition 2.2 (Solution-specific singular concentration target). Let $\varepsilon_v(\nu) > 0$ be the threshold in the velocity ε -regularity theorem and put $\eta_* := \varepsilon_v/2$. For a fixed solution u and terminal time T , let $\mathfrak{S}_{\text{sing}, T}(u)$ be the set of pairs $(x_0, \{r_j\}_{j \geq 1})$, with $r_j \downarrow 0$, such that

$$(2.8) \quad C((x_0, T), r_j) \geq \eta_* \quad \text{for every } j.$$

Every $x_0 \in \Sigma_u(T)$ occurs in at least one element of $\mathfrak{S}_{\text{sing}, T}(u)$ by Theorem 2.1. Conversely, if u is bounded in a backward neighborhood of (x_0, T) , then $C((x_0, T), r) = O(r^3)$, so no sequence based at x_0 belongs to $\mathfrak{S}_{\text{sing}, T}(u)$.

The following estimate explains why localized kinetic probes are sufficient to measure the energy side of this target.

Lemma 2.3 (Local interpolation). *For every suitable weak solution and every interior cylinder $Q_r(z_0)$,*

$$(2.9) \quad C(z_0, r) \leq c(A(z_0, r)^{3/4}E(z_0, r)^{3/4} + A(z_0, r)^{3/2}),$$

where c is universal.

Proof. After translating the cylinder, the Sobolev and interpolation inequalities on B_r give, for almost every time,

$$\int_{B_r} |u|^3 \leq c \left(\int_{B_r} |u|^2 \right)^{3/4} \left(\int_{B_r} |\nabla u|^2 \right)^{3/4} + cr^{-3/2} \left(\int_{B_r} |u|^2 \right)^{3/2}.$$

Integrate over a time interval of length r^2 , apply Hölder's inequality to the first term, and divide by r^2 . Substitution of (2.3) and (2.4) yields (2.9). $\square$

Corollary 2.4 (Concentration requires store or dissipation). *Suppose $C(z_j, r_j) \geq \eta > 0$. Then*

$$\max\{A(z_j, r_j), E(z_j, r_j)\} \geq c\eta^{2/3} > 0.$$

If $E(z_j, r_j) \rightarrow 0$, then $\liminf_j A(z_j, r_j) \geq c\eta^{2/3}$.

Proof. Both assertions follow directly from (2.9). $\square$

Thus a singular target retains local kinetic energy, viscous dissipation, or both. The canonical orbit constructed in Section 10 sharpens this alternative by subtracting the weighted velocity mean. Pressure remains part of the local regularity criterion and of the conservative flux; viscosity supplies the dissipative current.

3. EULER TRANSPORT AND THE STOKES GRADIENT

3.1. The solenoidal Gelfand triple. Let

$$\mathcal{V}_\sigma := \dot{H}_\sigma^1(\mathbb{R}^3), \quad \mathcal{H}_\sigma := L_\sigma^2(\mathbb{R}^3), \quad \mathcal{V}_\sigma^* := \dot{H}_\sigma^{-1}(\mathbb{R}^3).$$

The Stokes operator $A_S : \mathcal{V}_\sigma \rightarrow \mathcal{V}_\sigma^*$ is defined weakly by

$$(3.1) \quad \langle A_S v, \varphi \rangle = \int_{\mathbb{R}^3} \nabla v : \nabla \varphi \, dx.$$

On smooth fields, $A_S = -\mathbb{P}\Delta$. Put

$$(3.2) \quad K := \nu A_S.$$

This is the Riesz isomorphism associated with the Dirichlet form multiplied by ν . For $g, h \in \mathcal{V}_\sigma^*$, define

$$(3.3) \quad (g, h)_{K^{-1}} := \langle g, K^{-1}h \rangle, \quad \|g\|_{K^{-1}}^2 = (g, g)_{K^{-1}}.$$

The projected convection operator is

$$(3.4) \quad B(v, w) := \mathbb{P}((v \cdot \nabla)w), \quad \langle B(v, w), \varphi \rangle = - \int_{\mathbb{R}^3} v \otimes w : \nabla \varphi \, dx.$$

For sufficiently regular divergence-free v ,

$$(3.5) \quad \langle B(v, v), v \rangle = 0.$$

Definition 3.1 (Euler-covariant derivative). Along a divergence-free curve $u(t)$, define

$$(3.6) \quad \mathfrak{D}_t^E u := \partial_t u + B(u, u).$$

This is the material derivative after projecting away the pressure constraint.

Proposition 3.2 (Exact covariant gradient identity). *Let u be a smooth solution of (1.1). For $\mathcal{E}(u) = \frac{1}{2}\|u\|_2^2$,*

$$(3.7) \quad \mathfrak{D}_t^E u = -K D\mathcal{E}(u) = -Ku.$$

If $J_\nu := -Ku$, then

$$(3.8) \quad \|J_\nu\|_{K^{-1}}^2 = \langle Ku, u \rangle = \nu \|\nabla u\|_2^2,$$

and

$$(3.9) \quad \frac{d}{dt} \mathcal{E}(u(t)) = -\|J_\nu(t)\|_{K^{-1}}^2.$$

Proof. Apply $\mathbb{P}$ to (1.1). Since $\mathbb{P}\nabla p = 0$, this gives $\partial_t u + B(u, u) + Ku = 0$, which is (3.7). The definition of the inverse metric gives

$$\|Ku\|_{K^{-1}}^2 = \langle Ku, u \rangle = \nu \int_{\mathbb{R}^3} |\nabla u|^2 dx.$$

Pairing (3.7) with u and using (3.5) proves (3.9). $\square$

3.2. The finite energy inequality. For a Leray–Hopf solution, choose its standard weakly continuous representative and set

$$(3.10) \quad M(t) := \mathcal{E}(u(t)) + \nu \int_0^t \|\nabla u(s)\|_2^2 ds.$$

The global energy inequality says that M is nonincreasing after changing it on a null set. Its distributional derivative therefore defines a nonnegative Radon measure.

Theorem 3.3 (Energy dissipation for suitable weak solutions). *There is a nonnegative measure μ_{en} on $(0, T)$ such that, at every pair of times $0 \leq t_1 < t_2 < T$ for which the strong energy inequality is represented,*

$$(3.11) \quad \mathcal{E}(u(t_2)) + \int_{t_1}^{t_2} \|J_\nu(t)\|_{K^{-1}}^2 dt + \mu_{\text{en}}((t_1, t_2]) = \mathcal{E}(u(t_1)).$$

In particular,

$$(3.12) \quad \int_0^T \|J_\nu(t)\|_{K^{-1}}^2 dt \leq \mathcal{E}(u(0)).$$

For every classical solution, $\mu_{\text{en}} = 0$ on each compact interval of classical existence.

Proof. Define $\mu_{\text{en}} := -D_t M$ in distributions. Monotonicity of M gives $\mu_{\text{en}} \geq 0$, and the increment formula for M is exactly (3.11). The bound (3.12) follows by discarding the two nonnegative terminal terms. Smooth solutions satisfy the energy equality (3.9), so their defect measure vanishes. $\square$

Equation (3.11) is the finite dissipation bound used in the rest of the paper. The available quantity is the initial kinetic energy, and irreversible motion is measured by the Stokes action $\|J_\nu\|_{K^{-1}}^2$.

4. MATERIAL DEFORMATION AND STRETCHING–DIFFUSION DUALITY

Let u be smooth on a time interval and let $X(a, t)$ be its material flow,

$$(4.1) \quad \partial_t X(a, t) = u(X(a, t), t), \quad X(a, t_0) = a.$$

Write

$$(4.2) \quad \mathbf{F} := \nabla_a X, \quad \mathbf{C} := \mathbf{F}^T \mathbf{F}, \quad \mathbf{\Omega}(a, t) := \omega(X(a, t), t), \quad \omega := \nabla \times u.$$

The tensor $\mathbf{C}$ is the pullback of the Euclidean metric by the volume-preserving map X . Denote the Eulerian strain by $S = \frac{1}{2}(\nabla u + \nabla u^T)$ and put $S_L = S \circ X$.

Proposition 4.1 (Exact material pullback). *For every smooth solution and every time on which (4.1) is defined,*

$$(4.3) \quad \det \mathbf{F} = 1, \quad \partial_t \mathbf{F} = (\nabla u \circ X) \mathbf{F},$$

$$(4.4) \quad \partial_t \mathbf{C} = 2\mathbf{F}^T S_L \mathbf{F}, \quad \det \mathbf{C} = 1.$$

For every smooth scalar field q ,

$$(4.5) \quad (\Delta_x q) \circ X = \partial_{a_i} (\mathbf{C}^{ij} \partial_{a_j} (q \circ X)),$$

where $(\mathbf{C}^{ij}) = \mathbf{C}^{-1}$. The pulled-back vorticity satisfies

$$(4.6) \quad \partial_t \mathbf{\Omega} = S_L \mathbf{\Omega} + \nu \partial_{a_i} (\mathbf{C}^{ij} \partial_{a_j} \mathbf{\Omega}).$$

Equivalently, the viscous Cauchy invariant $\xi := \mathbf{F}^{-1} \mathbf{\Omega}$ satisfies

$$(4.7) \quad \partial_t \xi = \nu \mathbf{F}^{-1} \partial_{a_i} (\mathbf{C}^{ij} \partial_{a_j} (\mathbf{F} \xi)).$$

Proof. Differentiating (4.1) in a gives the evolution of $\mathbf{F}$. Jacobi's formula and incompressibility give

$$\partial_t \det \mathbf{F} = (\operatorname{div} u) \circ X \det \mathbf{F} = 0,$$

and the initial determinant is one. Differentiating $\mathbf{C} = \mathbf{F}^T \mathbf{F}$ proves (4.4). Formula (4.5) is the Laplace–Beltrami formula with volume density $\sqrt{\det \mathbf{C}} = 1$.

The vorticity equation is

$$(\partial_t + u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The antisymmetric part of ∇u annihilates ω , so $(\omega \cdot \nabla) u = S \omega$. Composition with X and (4.5) give (4.6). Finally, $\partial_t \mathbf{F}^{-1} = -\mathbf{F}^{-1} (\nabla u \circ X)$; the inviscid terms cancel after substituting $\mathbf{\Omega} = \mathbf{F} \xi$, which proves (4.7). $\square$

Corollary 4.2 (Material enstrophy balance). *If the fields decay sufficiently at infinity, then*

$$(4.8) \quad \frac{1}{2} \frac{d}{dt} \int_{\mathbb{R}^3} |\mathbf{\Omega}|^2 da + \nu \int_{\mathbb{R}^3} \mathbf{C}^{ij} \partial_{a_i} \mathbf{\Omega} \cdot \partial_{a_j} \mathbf{\Omega} da = \int_{\mathbb{R}^3} \mathbf{\Omega} \cdot S_L \mathbf{\Omega} da.$$

Proof. Pair (4.6) with $\mathbf{\Omega}$, integrate in a , and integrate the divergence term by parts. The material volume element is $da = dx$ because $\det \mathbf{F} = 1$. $\square$

The duality in (4.6) is algebraic. If the singular values of $\mathbf{F}$ are $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$, then

$$(4.9) \quad \sigma_1 \sigma_2 \sigma_3 = 1, \quad \lambda(\mathbf{C}^{-1}) = \{\sigma_1^{-2}, \sigma_2^{-2}, \sigma_3^{-2}\}, \quad \sigma_3^{-2} \geq \sigma_1.$$

Thus strong material stretching produces a strongly diffusive compressed direction. This observation alone is not a coercive estimate: the pulled-back vorticity could avoid variation in that direction. The equality mechanism is therefore geometric. Avoiding the enhanced direction requires simultaneous alignment of vorticity, strain, and the material eigenframe. The central

rigidity problem is to show that a target-retaining orbit cannot sustain this alignment indefinitely unless it approaches a stretching-depleted, lower-dimensional state.

For the regularity argument, this geometric alignment is tested by the Eulerian critical production. Equality means $\nabla V = 0$ on the retained connected core, and therefore spatial homogeneity modulo the Galilean quotient; see [Theorem 12.6](#). Thus the material duality explains the sign of the production, while the compact Eulerian equality theorem supplies the rigorous rigidity statement used in the proof.

5. COVARIANCE UNDER SCALE-TRANSLATION CHARTS

5.1. The repaired equation. Let $X(\tau) \in \mathbb{R}^3$ and $\Lambda(\tau) > 0$ be absolutely continuous, and define

$$(5.1) \quad x = X(\tau) + \Lambda(\tau)y, \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

$$(5.2) \quad u(x, t) = \Lambda(\tau)^{-1}V(y, \tau), \quad p(x, t) = \Lambda(\tau)^{-2}P(y, \tau).$$

Put

$$(5.3) \quad a(\tau) = \frac{\Lambda'(\tau)}{\Lambda(\tau)}, \quad b(\tau) = \frac{X'(\tau)}{\Lambda(\tau)}.$$

Proposition 5.1 (Covariant equation in a repaired chart). *The physical equation (1.1) is equivalent, in distributions on the image of the chart, to*

$$(5.4) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(V + y \cdot \nabla V) + b \cdot \nabla V, \quad \operatorname{div} V = 0.$$

Consequently the chart-covariant derivative

$$(5.5) \quad \mathfrak{D}_\tau^{X, \Lambda} V := \partial_\tau V + B(V, V) - a(V + y \cdot \nabla V) - b \cdot \nabla V$$

satisfies

$$(5.6) \quad \mathfrak{D}_\tau^{X, \Lambda} V = -\nu A_S V.$$

Proof. The change of variables in (5.1)–(5.2) gives the same factor Λ^{-3} in every term of the physical equation. After removing that factor, the time derivative contributes $a(V + y \cdot \nabla V) + b \cdot \nabla V$, giving (5.4). Applying the Helmholtz projection gives (5.6). $\square$

For the centered Type I variables

$$t = -e^{-\tau}, \quad \Lambda(\tau) = e^{-\tau/2}, \quad X = 0,$$

one has

$$(5.7) \quad a = -\frac{1}{2}, \quad b = 0,$$

and hence

$$(5.8) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V - \frac{1}{2}(V + y \cdot \nabla V).$$

This is the standard centered Type I convention. In a repaired Type II chart, the required coefficient regularity is

$$(5.9) \quad a, b \in L^1_{\text{loc}}(d\tau).$$

Whenever a compact-window argument requires bounded modulation, that bound is supplied by the observer construction; failure of the bound is retained as an observer-current output.

5.2. Covariant energy and pressure. On a whole-space chart, physical kinetic energy and action take the form

$$(5.10) \quad \mathcal{E}(u(t(\tau))) = \frac{\Lambda(\tau)}{2} \int_{\mathbb{R}^3} |V(y, \tau)|^2 dy,$$

$$(5.11) \quad \|J_\nu(t)\|_{K^{-1}}^2 dt = \Lambda(\tau) \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 dy d\tau.$$

Proposition 5.2 (Energy identity in moving coordinates). *For a smooth whole-space represented solution,*

$$(5.12) \quad \frac{d}{d\tau} \left(\frac{\Lambda}{2} \|V\|_2^2 \right) = -\Lambda \nu \|\nabla V\|_2^2.$$

Thus (3.9) and (5.12) are the same scalar identity in different coordinates.

Proof. Pair (5.4) with V . Convection, translation, and pressure give zero. Moreover,

$$\int_{\mathbb{R}^3} V \cdot (V + y \cdot \nabla V) dy = -\frac{1}{2} \|V\|_2^2.$$

Since $\Lambda' = a\Lambda$, the derivative of the prefactor in (5.10) cancels this dilation term. The remaining term is the right-hand side of (5.12). $\square$

On compact cylinders, derivatives of the cutoff produce convective, pressure, scale, and translation fluxes in the local energy inequality. Their exact form is recorded in Appendix C; these are the local terms placed explicitly in the moving error density (12.25).

The fixed- and moving-cutoff bounds follow from Propositions C.2 and C.3 and the suitable local-energy inequality on the normalized windows.

Pressure remains the multiplier for incompressibility. On an inner ball use the decomposition

$$(5.13) \quad P = P_{\text{loc}} + H, \quad -\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j), \quad \Delta H = 0,$$

with a spatial mean fixing H . The harmonic term can depend on space and time.

The local Calderón–Zygmund estimate and the harmonic interior estimate, with constants uniform on the normalized compact orbit, are the required conclusion of Section C.2.

Proposition 5.3 (Same-current covariance). *Let two admissible scale–translation charts represent the same physical solution on a common cylinder. The fields, test weights, and Stokes pairings transform by the corresponding pullback, while the physical current*

$$(5.14) \quad J_\nu = -\nu A_S u$$

is unchanged. In particular, (5.11) is invariant after the time change. Optimizing a concentration probe over admissible charts changes the probe but does not change J_ν .

Proof. Both charts are obtained from (5.1) by composing translations and positive dilations. The tensorial change of variables gives (5.10)–(5.11) in each chart. Since both right-hand sides equal the same physical energy and physical dissipation, their values agree. The current (5.14) is determined by u before either chart is selected. $\square$

6. MONOTONE CHARTS AND RECURRENT PRESSURE CANCELLATION

The scale and translation terms in (5.4) have two different roles. Their oscillation along a particular normalization is a choice of coordinates, whereas their limiting value on a recurrent compact class is part of the equation. The first result below removes artificial positive scale motion. The second identifies the recurrent equality class without estimating pressure on individual windows.

6.1. A monotone interpolation through ordered active cylinders.

Proposition 6.1 (Monotone scale interpolation). *Let*

$$J_j = [s_j, t_j], \quad t_j < s_{j+1}, \quad |J_j| = r_j^2, \quad r_{j+1} < r_j,$$

be an infinite ordered family of physical time intervals, and let $x_j \in \mathbb{R}^3$. There are locally absolutely continuous functions

$$\widehat{\Lambda} : (s_1, T) \rightarrow (0, \infty), \quad \widehat{X} : (s_1, T) \rightarrow \mathbb{R}^3,$$

where $T := \lim_j t_j$, with the following properties:

- (i) $\widehat{\Lambda} = r_j$ and $\widehat{X} = x_j$ on J_j ;
- (ii) $\widehat{\Lambda}$ is nonincreasing;
- (iii) after the time change

$$(6.1) \quad \widehat{\tau}(t) := \int_{s_1}^t \widehat{\Lambda}(q)^{-2} dq,$$

the coefficients of the corresponding chart satisfy

$$(6.2) \quad \widehat{a} := \frac{d}{d\widehat{\tau}} \log \widehat{\Lambda} \leq 0 \quad \text{a.e.}, \quad \widehat{b} := \frac{d\widehat{X}/d\widehat{\tau}}{\widehat{\Lambda}} \in L_{\text{loc}}^1;$$

- (iv) every J_j is a unit interval in the $\widehat{\tau}$ -clock and $\widehat{\tau}(t) \rightarrow \infty$ as $t \uparrow T$.

On each active interval the represented fields are exactly the fixed-scale parabolic fields

$$(6.3) \quad \widehat{V}_j(y, \sigma) = r_j u(x_j + r_j y, s_j + r_j^2 \sigma), \quad 0 \leq \sigma \leq 1,$$

and therefore satisfy (5.4) with $\widehat{a} = \widehat{b} = 0$ there.

Proof. Set $\widehat{\Lambda} = r_j$ and $\widehat{X} = x_j$ on J_j . On each gap $[t_j, s_{j+1}]$, interpolate $\log \widehat{\Lambda}$ affinely from $\log r_j$ to $\log r_{j+1}$, and interpolate $\widehat{X}$ affinely from x_j to x_{j+1} . The resulting functions are locally absolutely continuous. Since $r_{j+1} < r_j$, the scale is nonincreasing.

The map $t \mapsto \widehat{\tau}(t)$ is strictly increasing and locally absolutely continuous. Its inverse satisfies

$$\frac{dt}{d\widehat{\tau}} = \widehat{\Lambda}^2.$$

Consequently

$$\widehat{a} = \widehat{\Lambda} \frac{d\widehat{\Lambda}}{dt} \leq 0, \quad \widehat{b} = \widehat{\Lambda} \frac{d\widehat{X}}{dt},$$

which proves (6.2). Both coefficients are locally integrable because the interpolants are absolutely continuous and the scale is bounded above and below on every compact subinterval of (s_1, T) .

On J_j , the scale and center are constant, so

$$\widehat{\tau}(t) - \widehat{\tau}(s_j) = \frac{t - s_j}{r_j^2}.$$

Thus J_j has represented length one and the change of variables is exactly (6.3). Every active interval contributes one to (6.1). Since the intervals are disjoint and infinite in number, $\widehat{\tau}(t) \rightarrow \infty$ as $t \uparrow T$. Finally, Proposition 5.1 gives the represented equation. On J_j , both chart derivatives vanish, so $\widehat{a} = \widehat{b} = 0$. $\square$

Remark 6.2 (What monotone interpolation removes). Once the singular extraction has produced chronologically ordered active cylinders with decreasing scales, positive scale drift is unnecessary. It arises only if one insists on continuing a moment normalization through the inactive gaps. The monotone chart preserves the normalized fields on every active cylinder and represents the same physical solution in the gaps, but it does not by itself prove compactness of those gap segments or persistence of the target away from the active cylinders. Those are concentration–compactness questions, not modulation estimates.

6.2. The invariant mean identity. Fix constants $a_* \in \mathbb{R}$ and $b_* \in \mathbb{R}^3$, and consider entire smooth solutions of

$$(6.4) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a_*(V + y \cdot \nabla V) + b_* \cdot \nabla V, \quad \operatorname{div} V = 0.$$

For $c \in \mathbb{R}^3$ and $s \in \mathbb{R}$, define

$$(6.5) \quad (T_c V)(y, \tau) := V(y + e^{-a_* \tau} c, \tau), \quad (T_c P)(y, \tau) := P(y + e^{-a_* \tau} c, \tau),$$

$$(6.6) \quad (\Theta_s V)(y, \tau) := V(y, \tau + s), \quad (\Theta_s P)(y, \tau) := P(y, \tau + s).$$

Direct substitution shows that both maps preserve (6.4) and that

$$(6.7) \quad \Theta_s T_c \Theta_{-s} = T_{e^{-a_* s} c}.$$

They therefore generate the solvable, and hence amenable, group $G_{a_*} = \mathbb{R}^3 \rtimes \mathbb{R}$.

Lemma 6.3 (Pressure does no work outside the spatial zero mode). *Let $\mathcal{M}$ be a compact G_{a_*} -invariant family of bounded smooth entire solutions of (6.4), compact in a local topology in which the velocity and pressure-gradient evaluations below are continuous. Let μ be a G_{a_*} -invariant Borel probability measure on $\mathcal{M}$. In $L^2(\mathcal{M}, \mu)$, put*

$$u_i(V) := V_i(0, 0), \quad p_i(V) := \partial_i P(0, 0),$$

and let Π_0 be the orthogonal projection onto the subspace invariant under all T_c . If $u_i^0 := \Pi_0 u_i$ and $p_i^0 := \Pi_0 p_i$, then

$$(6.8) \quad \int_{\mathcal{M}} u_i p_i \, d\mu = \int_{\mathcal{M}} u_i^0 p_i^0 \, d\mu.$$

Proof. Let D_k be the skew-adjoint generator of the unitary action induced by T_{he_k} . Since the translation in (6.5) equals the ordinary translation by he_k at $\tau = 0$,

$$D_k u_i = \partial_k V_i(0, 0), \quad D_i u_i = 0.$$

Taking the divergence of (6.4) gives

$$(6.9) \quad -\Delta P = \partial_i \partial_j (V_i V_j).$$

The scale and translation terms have zero divergence. Hence, with $q_i := (I - \Pi_0)p_i$ and $A := -\sum_k D_k^2$, the commuting spectral resolution of the spatial translation action gives

$$q_k = D_k A^{-1} (I - \Pi_0) D_i D_j (u_i u_j)$$

on the nonzero spatial spectral subspace. More precisely, the right side is the L^2 -limit as $\varepsilon \downarrow 0$ of

$$q_{k,\varepsilon} := D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j (u_i u_j).$$

The pressure curl relation determines the vector from its divergence and excludes an additional nonzero-frequency harmonic part. Skew-adjointness and incompressibility give

$$\int_{\mathcal{M}} u_k q_{k,\varepsilon} \, d\mu = - \int_{\mathcal{M}} (D_k u_k) (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j (u_i u_j) \, d\mu = 0.$$

Pass to the L^2 -limit. Since $u_i - u_i^0 \perp \text{ran } \Pi_0$, one also has $\int (u_i - u_i^0) p_i^0 d\mu = 0$. These two identities prove (6.8). Only ∇P occurs, so the conclusion is independent of the additive pressure gauge. $\square$

Lemma 6.4 (Pressure work under activity-biased rooting). *Assume the hypotheses and notation of Lemma 6.3. On the nonzero spatial spectral subspace write*

$$(6.10) \quad q_k := (I - \Pi_0)p_k = D_k h, \quad h := A^{-1}(I - \Pi_0)D_i D_j(u_i u_j), \quad A = -\sum_k D_k^2,$$

with the identity understood through the resolvent approximation used in the proof of Lemma 6.3. Let $w \geq 0$ be bounded, belong to the form domains of the spatial generators, and satisfy

$$(6.11) \quad Z_w := \int_{\mathcal{M}} w d\mu > 0, \quad h_\varepsilon u_k D_k w \longrightarrow h u_k D_k w \quad \text{in } L^1(\mu),$$

For the rooted probability $d\mu_w = Z_w^{-1} w d\mu$,

$$(6.12) \quad \sum_k \int_{\mathcal{M}} u_k q_k d\mu_w = -\frac{1}{Z_w} \int_{\mathcal{M}} h u_k D_k w d\mu.$$

Thus activity-biased rooting cancels the nonzero pressure modes precisely when the root-current pairing on the right of (6.12) vanishes.

Proof. Use first the regularized potentials

$$h_\varepsilon := (\varepsilon + A)^{-1}(I - \Pi_0)D_i D_j(u_i u_j), \quad q_{k,\varepsilon} = D_k h_\varepsilon.$$

Skew-adjointness of D_k , the generator product rule, and incompressibility give

$$\begin{aligned} \sum_k \int w u_k D_k h_\varepsilon d\mu &= -\sum_k \int h_\varepsilon D_k(w u_k) d\mu \\ &= -\sum_k \int h_\varepsilon u_k D_k w d\mu - \int h_\varepsilon w \sum_k D_k u_k d\mu \\ &= -\sum_k \int h_\varepsilon u_k D_k w d\mu. \end{aligned}$$

The resolvent convergence in Lemma 6.3 gives $q_{k,\varepsilon} \rightarrow q_k$ in $L^2(\mu)$. Approximation of w in the common generator form domain passes the left side to its limit, while (6.11) passes the right side to its stated L^1 limit. Divide by Z_w to obtain (6.12). $\square$

Remark 6.5 (The retained root current). The right side of (6.12) is absent from the unweighted invariant calculation and need not have a sign. In an event-indexed spacetime decomposition it is represented by pressure and transport across the moving boundary of the rooted cell. A nonzero value is therefore a transport or boundary coordinate in (11.13); setting it to zero without proving the pairing or the corresponding oriented boundary trace vanishes would lose a possible successor.

Theorem 6.6 (Rigidity of a recurrent nonexpanding covariant class). *Let $\mathcal{M}$ be a nonempty compact minimal G_{a_*} -invariant family satisfying the hypotheses of Lemma 6.3. If $a_* \leq 0$, every element of $\mathcal{M}$ is spatially constant. Equivalently, every continuous Galilean-invariant local oscillation observable vanishes on $\mathcal{M}$.*

Proof. Amenability of G_{a_*} gives an invariant probability measure μ on $\mathcal{M}$. Its support is a nonempty closed invariant subset, so minimality makes $\text{supp } \mu = \mathcal{M}$.

The spatially invariant subspace is preserved by time translation because the spatial subgroup is normal by (6.7); hence Π_0 commutes with the time-translation group. Evaluate (6.4) at $(0, 0)$,

move every term to the left, and project onto that subspace. Spatial derivatives, including the nonlinearity written as $D_j(u_j u_i)$, vanish after projection. The term $y \cdot \nabla V$ vanishes at $y = 0$. Thus

$$(6.13) \quad \partial_\tau u_i^0 - a_* u_i^0 + p_i^0 = 0 \quad \text{in } L^2(\mathcal{M}, \mu).$$

Pairing with u_i^0 , summing in i , and using invariance under Θ_s gives

$$(6.14) \quad \int_{\mathcal{M}} u_i^0 p_i^0 \, d\mu = a_* \int_{\mathcal{M}} |u^0|^2 \, d\mu.$$

Next take the scalar product of the equation at $(0, 0)$ with u and average. The time derivative has zero mean. Integration by parts with the spatial generators gives

$$-\nu \int u \cdot \Delta V(0, 0) \, d\mu = \nu \int |\nabla V(0, 0)|^2 \, d\mu.$$

The nonlinear and constant-translation terms are spatial divergences and have zero mean. Using Lemma 6.3 and Eq. (6.14) for the pressure term yields

$$(6.15) \quad \nu \int_{\mathcal{M}} |\nabla V(0, 0)|^2 \, d\mu - a_* \int_{\mathcal{M}} |u - u^0|^2 \, d\mu = 0.$$

Here we used that multiplication by the spatially invariant constant a_* commutes with Π_0 . Both terms in (6.15) are nonnegative when $a_* \leq 0$; hence the first vanishes. Full support of μ and continuity of the gradient evaluation imply $\nabla V(0, 0) = 0$ for every $V \in \mathcal{M}$. The group action moves $(0, 0)$ to every point of spacetime, so $\nabla V \equiv 0$ for every element of $\mathcal{M}$. Such velocities are spatially constant, and every mean-subtracted local oscillation is zero. $\square$

The constant-coefficient formulation is convenient for minimal autonomous hulls, but a monotone scale interpolation naturally produces a stationary law of coefficient paths rather than a single constant. The same identity holds at that level.

Definition 6.7 (Stationary monotone covariant ensemble). A stationary monotone covariant ensemble is a Borel probability space $(\mathcal{M}, \mu)$ of entire smooth represented solutions (V, P, a, b) of (5.4) with the following properties.

- (i) The local velocity, pressure-gradient, and coefficient observables used below are integrable, and the velocity observables are in the domains of the spatial and time generators. This holds, for example, on a compact locally smooth class with a uniform local coefficient bound.
- (ii) The law is invariant under time translation of the four coefficient paths.
- (iii) For $c \in \mathbb{R}^3$, let q_c solve

$$(6.16) \quad q'_c(\tau) = -a(\tau)q_c(\tau), \quad q_c(0) = c,$$

and set

$$T_c(V, P, a, b) := (V(\cdot + q_c(\tau), \tau), P(\cdot + q_c(\tau), \tau), a, b).$$

The class and the law are invariant under every T_c .

- (iv) The scale coefficient satisfies $a \leq 0$ almost everywhere for μ -almost every path.

The integrability in item (i) includes

$$\int_{\mathcal{M}} (|\nabla V(0, 0)|^2 + |a(0)| |V(0, 0)|^2) \, d\mu < \infty,$$

with time-zero values understood through the stationary measurable representative. Equivalently, one may formulate every identity below after Steklov averaging in time and then pass to almost every time.

Theorem 6.8 (Rigidity of a stationary monotone covariant ensemble). *Let $(\mathcal{M}, \mu)$ be a stationary monotone covariant ensemble. Let $u_i = V_i(0, 0)$, and let u_i^0 be its orthogonal projection in $L^2(\mathcal{M}, \mu)$ onto the spatially invariant observables. Then*

$$(6.17) \quad \nu \int_{\mathcal{M}} |\nabla V(0, 0)|^2 d\mu + \int_{\mathcal{M}} (-a(0)) |u - u^0|^2 d\mu = 0.$$

Consequently $\nabla V = 0$ at every spacetime point for μ -almost every represented solution. In particular, every mean-subtracted local oscillation observable vanishes μ -almost surely.

Proof. Equation (6.16) is precisely the condition that makes T_c preserve (5.4). At $\tau = 0$, its infinitesimal generators are the ordinary spatial derivatives. The pressure Poisson equation remains (6.9), because both modulation fields are divergence free. The spectral proof of Lemma 6.3 therefore gives

$$(6.18) \quad \int u_i p_i d\mu = \int u_i^0 p_i^0 d\mu.$$

Although time translation no longer conjugates T_c by a constant parameter, it preserves every spatial orbit. Indeed,

$$\Theta_s T_c(V, P, a, b) = T_{c_s} \Theta_s(V, P, a, b), \quad c_s = c \exp\left(-\int_0^s a(q) dq\right).$$

Thus the spatially invariant sigma-algebra is time invariant, and its orthogonal projection commutes with time translation. The functions $a(0)$ and $b(0)$ are themselves spatially invariant.

Projecting the equation at $(0, 0)$ onto the spatial zero mode now gives

$$\partial_\tau u_i^0 - a(0) u_i^0 + p_i^0 = 0.$$

Pair with u_i^0 , sum in i , and use stationarity to obtain

$$(6.19) \quad \int u_i^0 p_i^0 d\mu = \int a(0) |u^0|^2 d\mu.$$

Testing the full equation at the origin against u , averaging, and using spatial and time invariance removes the time derivative, convection, and the b -transport term. Diffusion gives $\nu \int |\nabla V(0, 0)|^2 d\mu$. Combining (6.18) and (6.19) gives

$$\nu \int |\nabla V(0, 0)|^2 d\mu - \int a(0) (|u|^2 - |u^0|^2) d\mu = 0.$$

Multiplication by the spatially invariant observable $a(0)$ preserves the zero-mode subspace. Orthogonality therefore gives

$$\int a(0) (|u|^2 - |u^0|^2) d\mu = \int a(0) |u - u^0|^2 d\mu,$$

which proves (6.17). Both terms are nonnegative. Hence the gradient vanishes almost surely at the origin. Time and covariant spatial invariance propagate this conclusion to every spacetime point. The Steklov formulation mentioned in Definition 6.7 justifies the same calculation when the coefficient paths are merely locally integrable. $\square$

Corollary 6.9 (No target-retaining recurrent limit). *Let $\mathcal{K}$ be a compact invariant class of autonomous limits with $a_* \leq 0$, and let $\mathcal{O}$ be a continuous nonnegative observable on $\mathcal{K}$ which vanishes on spatially constant velocities. If every minimal invariant component of $\mathcal{K}$ retains $\mathcal{O} \geq \eta > 0$, then $\mathcal{K}$ is empty.*

Proof. Every nonempty compact invariant class contains a compact minimal invariant component. Apply Theorem 6.6 on that component. It gives $\mathcal{O} = 0$, contrary to the retained lower bound. $\square$

The preceding rigidity statement becomes quantitative on a compact successor class through the following elementary ergodic lemma. This is the point at which recurrence replaces a list of increasingly sharp estimates.

Lemma 6.10 (Uniform block deficit or a saturation law). *Let K be a compact metric space, let $S : K \rightarrow K$ be continuous, and let $h \in C(K)$. Write $\mathcal{P}_S(K)$ for the set of S -invariant Borel probability measures and put*

$$(6.20) \quad \beta := \max_{\mu \in \mathcal{P}_S(K)} \int_K h \, d\mu.$$

Then exactly one of the following alternatives holds.

(i) $\beta < 0$, and there exist $N \geq 1$ and $\delta > 0$ such that

$$(6.21) \quad \sum_{k=0}^{N-1} h(S^k w) \leq -\delta \quad (w \in K);$$

(ii) $\beta \geq 0$, and an invariant probability measure μ_* attains β .

In particular, if a PDE balance gives $\int h \, d\mu \leq 0$ for every invariant measure and every equality law is target-null, while a continuous target observable satisfies $\mathcal{O} \geq \eta > 0$ on K , then alternative (i) holds.

Proof. The set $\mathcal{P}_S(K)$ is nonempty and weak-* compact, and the map $\mu \mapsto \int h \, d\mu$ is continuous. Hence the maximum in (6.20) is attained.

Assume $\beta < 0$ and that (6.21) fails for every N . Then for each n there is $w_n \in K$ such that

$$\sum_{k=0}^{n-1} h(S^k w_n) > -n^{-1}.$$

For the empirical measures

$$\mu_n := \frac{1}{n} \sum_{k=0}^{n-1} \delta_{S^k w_n}$$

one consequently has $\liminf_n \int h \, d\mu_n \geq 0$. Pass to a weak-* convergent subsequence. For every $f \in C(K)$,

$$\int (f \circ S - f) \, d\mu_n = \frac{f(S^n w_n) - f(w_n)}{n} \rightarrow 0,$$

so the limit μ is S -invariant. It satisfies $\int h \, d\mu \geq 0$, contradicting $\beta < 0$. Thus a block length N exists; compactness of K makes the negative maximum of the continuous block sum a number $-\delta < 0$, proving (6.21).

For the final assertion, an invariant measure with zero average would be supported on the target-null equality class by the stated PDE rigidity, while $\mathcal{O} \geq \eta$ on all of K . Hence equality is impossible and the maximal invariant average is strictly negative. $\square$

Corollary 6.11 (Recurrent equality is the only zero-gap obstruction). *Let a compact target-retaining successor class have a continuous signed local-energy excess h , and suppose its finite block sums are the exact aggregate inward feeding minus viscous and suitable-defect production, up to bounded endpoint storage and true exterior input of sublinear block growth. If every invariant saturation law is realized either as a stationary monotone covariant ensemble in the sense of Definition 6.7 or as a zero-Dirichlet ordinary window, then the class has a uniform strictly dissipative block deficit.*

Proof. Signed additivity removes the internal currents before the invariant average is taken. The endpoint storage and exterior input vanish after division by the block length, so every invariant law has nonpositive mean excess. In the equality case, [Theorem 6.8](#) makes the first realization spatially constant, while [Theorem 12.6](#) makes the second oscillation-removable. Both contradict the retained target floor. Apply [Lemma 6.10](#). $\square$

Remark 6.12 (Interface with noncompact extraction). The theorem closes an *ordinary compact recurrent* scale-collapse or scale-rigid limit, including the limiting case $a_* = 0$. Rough, exterior, diffuse, and noncompact sequences are retained at their first failed coordinate by [Theorem 11.13](#); repeated active recentering is treated in the marked path space and does not require a covariant hull retaining a fixed root.

7. SCALE-CRITICAL QUOTIENT GEOMETRY

The physical energy identity and the scale-critical concentration problem live at different levels. This section separates them and fixes the quotient structure used by the critical orbit.

7.1. Physical action and critical action. The Navier–Stokes scaling based at (x_0, t_0) is

$$(7.1) \quad u^{(\lambda)}(y, s) = \lambda u(x_0 + \lambda y, t_0 + \lambda^2 s), \quad p^{(\lambda)}(y, s) = \lambda^2 p(x_0 + \lambda y, t_0 + \lambda^2 s).$$

The quantities C, D are invariant under (7.1). The localized Dirichlet quantity is invariant after the parabolic factor is included:

$$(7.2) \quad r^{-1} \int_{Q_r(z_0)} |\nabla u|^2 dx dt = \int_{Q_1} |\nabla u^{(r)}|^2 dy ds.$$

This is the dissipation seen by a normalized unit window.

By contrast, the physical Stokes action transforms according to (5.11) and carries the factor Λ :

$$(7.3) \quad \int_{t(\tau_1)}^{t(\tau_2)} \nu \|\nabla u(t)\|_2^2 dt = \int_{\tau_1}^{\tau_2} \Lambda(\tau) \nu \|\nabla V(\tau)\|_2^2 d\tau.$$

Consequently, finite physical action does not control

$$(7.4) \quad \int_{\tau_1}^{\tau_2} \nu \int_{B_R} |\nabla V|^2 dy d\tau$$

uniformly along a collapsing orbit. For example, in centered Type I coordinates $\Lambda = e^{-\tau/2}$. A represented orbit with order-one Dirichlet dissipation on every unit interval has finite contribution to (7.3) but infinite contribution to (7.4). The global energy inequality alone therefore cannot supply the required contradiction.

The corrected moving-corrected localized energy is designed precisely at the level of (7.4). Its lower bound replaces the missing unweighted global bound in represented time. The production gap then says that a retained core must spend a fixed amount of this critical action, or of the other covariant production terms, on every target-retaining window.

7.2. Galilean and pressure quotients. For a constant vector $c \in \mathbb{R}^3$, the Galilean transform

$$(7.5) \quad u^c(x, t) := u(x - ct, t) + c, \quad p^c(x, t) := p(x - ct, t)$$

preserves (1.1). Addition of a function $\pi(t)$ to the pressure also preserves the equation. These modes have no bearing on regularity but can change an uncentered local kinetic integral. The variance (9.6) is invariant under the induced constant-velocity action:

$$(7.6) \quad \frac{1}{2} \int m |(V + c) - \overline{(V + c)}_m|^2 = \frac{1}{2} \int m |V - \bar{V}_m|^2.$$

For $c \neq 0$, (7.5) does not preserve the global finite-energy class on $\mathbb{R}^3$. It is used only as a symmetry of local profiles and observables; the physical finite-energy solution and its Stokes current are never replaced by u^c . Likewise, D and the pressure atlas depend only on the pressure class modulo functions of time.

Fix nonzero nonnegative radial cutoffs $\zeta_0, \zeta_1 \in C_c^\infty(B_R)$, a level $\Theta_0 > 0$, and write $V_{\lambda,q}(y) = \lambda V(q + \lambda y)$. For a nonzero cutoff ξ , set

$$(7.7) \quad \langle W \rangle_\xi := \frac{\int \xi W \, dy}{\int \xi \, dy}, \quad W_\xi^\circ := W - \langle W \rangle_\xi.$$

Translation and dilation are fixed by the four-component, Galilean-invariant gauge map

$$(7.8) \quad \begin{aligned} \mathfrak{F}_0(\lambda, q; V) &:= \int \zeta_0 |(V_{\lambda,q})_{\zeta_0}^\circ|^3 \, dy - \Theta_0, \\ \mathfrak{F}_j(\lambda, q; V) &:= \int y_j \zeta_1 |(V_{\lambda,q})_{\zeta_1}^\circ|^2 \, dy, \quad 1 \leq j \leq 3. \end{aligned}$$

Indeed, replacing V by $V + c$ adds the constant λc to $V_{\lambda,q}$, which is removed by both means in (7.7). Thus the gauge selects only scale and spatial center; it does not select a velocity representative. At a zero for which

$$(7.9) \quad D_{(\lambda,q)} \mathfrak{F}(1, 0; V) \in \mathbb{R}^{4 \times 4} \quad \text{is invertible,}$$

the implicit-function theorem selects the scale and center locally. Differentiating the four constraints produces (a, b) . Degeneration of (7.9) is tested during canonical-observer construction; no smooth continuation through a degenerate point is assumed.

7.3. The quotient topology. On a fixed represented cylinder $B_R \times I$, put

$$(7.10) \quad \mathcal{X}_R(I) := (L_I^\infty L^2(B_R) \cap L_I^2 H^1(B_R) \cap L^3(B_R \times I)) \times L^{3/2}(B_R \times I) \times L^1(I; \mathbb{R}^4),$$

where the factors represent V , P , and (a, b) . The compactness topology consists of strong local L^3 convergence for velocity, weak local $L^2 H^1$ convergence for its gradient, strong local $L^{3/2}$ convergence for the normalized pressure, and L^1 convergence of modulation coefficients on the selected windows. Adjoint densities and cutoffs are added with the topology in (A.7) and (A.6).

Let $\mathcal{G}_{\text{ch}}$ denote the local chart action generated by translations, positive dilations, and pressure gauge. Near a zero satisfying (7.9), the normalized slice (10.3), together with (10.4), intersects the corresponding $\mathcal{G}_{\text{ch}}$ -orbit locally and transversely. Global continuation and consistency of these local slices are part of Theorem 10.2. Constant-velocity Galilean symmetry is not fixed by a time-dependent mean condition; instead, every scalar used below is invariant under that symmetry by construction. A compact family of normalized windows may therefore be studied on the chart slice and then quotiented by the residual Galilean action.

Proposition 7.1 (Covariance of the reduced data). *On the nondegenerate normalized slice, the following objects depend only on the quotient state:*

- (i) *the oscillatory kinetic energy $\mathcal{E}_\Gamma^\circ$ and concentration $\mathcal{C}_\Gamma^\circ$;*
- (ii) *the physical Stokes current and its inverse-metric norm;*
- (iii) *the ray distance after the target probe is pulled back with the chart;*
- (iv) *the normalized pressure class and, for a fixed padded terminal problem, the localized adjoint probability measure $m \, dy$.*

Proof. The first item follows from change of variables and (7.6). The second and third follow from Proposition 5.3 and covariance of orthogonal projection in the Stokes metric. Pressure means are removed by (10.4). For a fixed padded terminal problem, the normalized adjoint equation pulls back covariantly when its domain, cutoff, and terminal datum are pulled back with

the chart, and normalization fixes the mass. No comparison between distinct terminal problems is asserted or needed for the moving-corrected localized energy. $\square$

Compactness and coherent continuation of the observer are supplied by [Theorem 10.2](#). Moving-cutoff errors are retained in the signed current rather than imposed as a separate global condition.

8. BACKWARD ADJOINT LOCALIZATIONS AND ENTROPY

The adjoint density supplies scale-relative localization. Its entropy is kept separate from the kinetic-energy gradient identity; this separation prevents pressure and critical norms from acquiring artificial variational equations.

8.1. The normalized finite-horizon adjoint problem. Fix a repaired cylinder $B_R \times (\tau_0, \sigma)$, let

$$(8.1) \quad c(y, \tau) := V(y, \tau) - a(\tau)y - b(\tau), \quad \operatorname{div} c = -3a,$$

and choose $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, together with a nonnegative terminal datum $h \in C_c^\infty(B_R)$, $h \not\equiv 0$. First solve the linear terminal-boundary problem

$$(8.2) \quad -\partial_\tau \tilde{\rho} - \nu \Delta \tilde{\rho} - \operatorname{div}(c \tilde{\rho}) = 0, \quad \tilde{\rho}|_{\partial B_R} = 0, \quad \tilde{\rho}(\sigma) = h.$$

Define

$$(8.3) \quad Z(\tau) := \int_{B_R} \chi^2 \tilde{\rho} \, dy, \quad \rho := \frac{\tilde{\rho}}{Z}, \quad m := \chi^2 \rho.$$

We work on intervals on which $Z > 0$. Then $m \geq 0$ and

$$(8.4) \quad \int_{B_R} m(y, \tau) \, dy = 1.$$

For a retained unit window $I = [s, s+1]$, the adjoint problem is always solved on a padded interval ending at

$$(8.5) \quad \sigma_I = s + 1 + \kappa_*, \quad \kappa_* > 0.$$

Consequently

$$(8.6) \quad \kappa_* e^{-2M_*} \leq \theta_I(\tau) \leq (1 + \kappa_*) e^{2M_*} \quad (\tau \in I).$$

Here M_* is a uniform bound for $\int_s^{\sigma_I} |a(r)| \, dr$ on the normalized padded windows, and θ_I is the unique solution of the covariant terminal-value problem

$$(8.7) \quad \theta'_I + 2a\theta_I = -1, \quad \theta_I(\sigma_I) = 0.$$

Indeed,

$$(8.8) \quad \theta_I(\tau) = \int_\tau^{\sigma_I} \exp\left(2 \int_\tau^r a(q) \, dq\right) \, dr,$$

which proves (8.6). The Gaussian entropy is evaluated only on the retained window, where $\theta_I > 0$, not at the terminal time. It is a covariant diagnostic on each padded window; the global monotonicity balance constructed below does not compare or reset finite-horizon adjoint densities.

Lemma 8.1 (Normalized adjoint equation). *Set*

$$(8.9) \quad \kappa(\tau) := \frac{Z'(\tau)}{Z(\tau)} = \int_{B_R} (-\nu \Delta \chi^2 + c \cdot \nabla \chi^2) \rho \, dy.$$

Then

$$(8.10) \quad -\partial_\tau \rho - \nu \Delta \rho - \operatorname{div}(c \rho) = \kappa \rho$$

in distributions, and (8.4) holds at every time at which the weak solution is represented continuously in L^2 .

Proof. Differentiate Z using (8.2) and integrate by parts. The support of χ is contained in B_R , so

$$Z' = \int_{B_R} (-\nu \Delta \chi^2 + c \cdot \nabla \chi^2) \tilde{\rho} \, dy.$$

Division by Z gives (8.9). Substituting $\tilde{\rho} = Z\rho$ into (8.2) gives (8.10). The last statement follows from the definition of Z . $\square$

This construction solves the linear equation before normalization. The coefficient κ is therefore determined by the solution and does not enter the linear terminal problem.

Lemma 8.2 (Weak adjoint construction). *Assume*

$$(8.11) \quad V \in L_\tau^2 H_y^1(B_R), \quad a, b \in L_\tau^1, \quad \operatorname{div} c = -3a.$$

Then (8.2) has a nonnegative weak solution with

$$\tilde{\rho} \in L_\tau^\infty L_y^2(B_R) \cap L_\tau^2 H_{0,y}^1(B_R).$$

It is obtained as the weak limit of solutions with smooth drifts c_ε . On every compact subinterval where $Z \geq z_0 > 0$, the normalized densities satisfy the same bounds, and $\sqrt{\rho} \in L_\tau^2 H_{\operatorname{loc},y}^1$ whenever their Fisher information is uniformly bounded.

Proof. Reverse time in (8.2) and approximate V, a, b by smooth functions preserving $\operatorname{div} c_\varepsilon = -3a_\varepsilon$. Galerkin approximation gives smooth weak solutions. Testing by $\tilde{\rho}_\varepsilon$ makes the transport contribution $-\frac{1}{2} \int (\operatorname{div} c_\varepsilon) |\tilde{\rho}_\varepsilon|^2$, so Gronwall's inequality uses only $a_\varepsilon \in L^1$. This yields uniform $L^\infty L^2 \cap L^2 H_0^1$ bounds. Choose the approximating drifts to converge strongly in $L^2 L^6 + L^1 L^\infty$. The first summand follows from $V \in L^2 H^1$, and the affine part belongs to the second summand on B_R . The bounds on $\tilde{\rho}_\varepsilon$ then allow $c_\varepsilon \tilde{\rho}_\varepsilon$ to pass to the limit against every smooth test function. Weak compactness gives a solution of (8.2); testing the negative part proves nonnegativity. Division by $Z \geq z_0$ preserves the estimates. Finally,

$$\int \frac{|\nabla \rho|^2}{\rho + \varepsilon} \, dy = 4 \int |\nabla \sqrt{\rho + \varepsilon}|^2 \, dy,$$

and weak lower semicontinuity gives the last assertion as $\varepsilon \downarrow 0$. $\square$

8.2. Transport identity and the Gaussian equality model. For smooth positive ρ , write

$$(8.12) \quad \rho = (4\pi\nu\theta)^{-3/2} e^{-f}, \quad \theta > 0.$$

Equation (8.10) gives

$$(8.13) \quad \partial_\tau f + \nu \Delta f - \nu |\nabla f|^2 + c \cdot \nabla f = \kappa - \frac{3\theta'}{2\theta} + \operatorname{div} c.$$

Lemma 8.3 (Localized adjoint transport identity). *For every smooth scalar F ,*

$$(8.14) \quad \begin{aligned} \frac{d}{d\tau} \int_{B_R} \chi^2 F \rho \, dy &= \int_{B_R} \chi^2 (\partial_\tau F - \nu \Delta F + c \cdot \nabla F) \rho \, dy \\ &\quad + \int_{B_R} [F(-\nu \Delta \chi^2 + c \cdot \nabla \chi^2) - 2\nu \nabla \chi^2 \cdot \nabla F] \rho \, dy \\ &\quad - \kappa \int_{B_R} \chi^2 F \rho \, dy. \end{aligned}$$

Proof. Differentiate the integral, substitute (8.10), and integrate the Laplacian and divergence terms by parts. All spatial boundary terms vanish because $\chi \in B_R$. $\square$

The three lines of (8.14) separate interior evolution, cutoff transport, and normalization. Taking

$$(8.15) \quad W_f := \nu\theta|\nabla f|^2 + f - 3, \quad \mathcal{W}_{\text{ad},\chi}(\tau) := \int_{B_R} \chi^2 W_f \rho \, dy,$$

gives the exact localized derivative by setting $F = W_f$ in (8.14) and using (8.13). In particular, derivatives of χ and the scalar κ remain visible in the last two lines. The velocity drift contributes through ∇c , and its compression contributes through $\text{div } c = -3a$. No sign is assigned to those local terms without an additional estimate.

The equality model is most transparent without localization.

Proposition 8.4 (Adjoint entropy production with a variable heat scale). *Let ρ be a smooth positive probability density on $\mathbb{R}^3$ satisfying*

$$-\partial_\tau \rho - \nu \Delta \rho - \text{div}(c\rho) = 0$$

with sufficient decay. With f as in (8.12), for every positive absolutely continuous θ one has

$$(8.16) \quad \begin{aligned} \frac{d}{d\tau} \int_{\mathbb{R}^3} (\nu\theta|\nabla f|^2 + f - 3)\rho \, dy &= 2\nu^2\theta \int_{\mathbb{R}^3} \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy \\ &\quad + \int_{\mathbb{R}^3} (\text{div } c)\rho \, dy \\ &\quad + 2\nu\theta \int_{\mathbb{R}^3} (\nabla f \cdot \nabla \text{div } c - (\nabla f)^T (\nabla c) \nabla f) \rho \, dy \\ &\quad + (\theta' + 1) \left(\nu \int_{\mathbb{R}^3} |\nabla f|^2 \rho \, dy - \frac{3}{2\theta} \right). \end{aligned}$$

For $c = 0$ and $\theta' = -1$, the derivative is nonnegative and vanishes precisely when $\nabla^2 f = (2\nu\theta)^{-1}I$ on the support of ρ .

Proof. Let $H(\rho) = -\int \rho \log \rho$ and $I(\rho) = \int |\nabla \log \rho|^2 \rho$. Direct differentiation with the backward adjoint equation and integration by parts give

$$\frac{dH}{d\tau} = -\nu I + \int (\text{div } c)\rho$$

and

$$\frac{dI}{d\tau} = 2\nu \int |\nabla^2 \log \rho|^2 \rho - 2 \int (\nabla \log \rho \cdot \nabla \text{div } c + (\nabla \log \rho)^T (\nabla c) \nabla \log \rho) \rho.$$

Differentiate

$$\nu\theta I + H - \frac{3}{2} \log(4\pi\nu\theta) - 3$$

in τ , use $\nabla \log \rho = -\nabla f$, and first collect the terms corresponding to $\theta' = -1$. Completing their square gives the first three lines of (8.16). The difference between general θ' and -1 is exactly its last line. $\square$

For the repaired Navier–Stokes drift $c = V - ay - b$, one has $\nabla \operatorname{div} c = 0$ and $\nabla c = \nabla V - aI$. Hence the choice (8.7) cancels every scalar scale term in (8.16) and gives

$$(8.17) \quad \begin{aligned} \frac{d}{d\tau} \mathcal{W}_{\text{ad}} &= 2\nu^2\theta \int_{\mathbb{R}^3} \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy \\ &\quad - 2\nu\theta \int_{\mathbb{R}^3} (\nabla f)^T (\nabla V)_{\text{sym}} \nabla f \rho \, dy. \end{aligned}$$

Here $\mathcal{W}_{\text{ad}}$ denotes the whole-space entropy in Proposition 8.4, and $(\nabla V)_{\text{sym}} := \frac{1}{2}(\nabla V + (\nabla V)^T)$. Formula (8.17) isolates the positive Gaussian square from velocity strain. The translation coefficient b drops out because it has no spatial derivative, while a drops out because the heat scale is transported by (8.7). The strain term has no sign by itself, but it completes a square with viscous production below.

In the unmodulated case $a = b = 0$, for which $\theta' = -1$, the Euclidean heat kernel is

$$(8.18) \quad \rho_*(y, \tau) = (4\pi\nu\theta)^{-3/2} \exp\left(-\frac{|y|^2}{4\nu\theta}\right), \quad f_* = \frac{|y|^2}{4\nu\theta},$$

and satisfies

$$\nabla^2 f_* = \frac{1}{2\nu\theta} I, \quad \int_{\mathbb{R}^3} (\nu\theta |\nabla f_*|^2 + f_* - 3) \rho_* \, dy = 0.$$

Hence both the entropy and its production vanish. This is the Euclidean normalization of Perelman’s $\mathcal{W}$ -entropy [9].

For weak densities, the Fisher term is defined by

$$(8.19) \quad \int \chi^2 \frac{|\nabla \rho|^2}{\rho} \, dy := 4 \int \chi^2 |\nabla \sqrt{\rho}|^2 \, dy,$$

with value $+\infty$ when $\sqrt{\rho} \notin H_{\text{loc}}^1$. The entropy uses $0 \log 0 = 0$. At weak regularity the Fisher information has its usual lower-semicontinuous extension. The Shannon term in $\mathcal{W}_{\text{ad}, \chi}$ has the opposite semicontinuity orientation, so its passage to a limit requires an explicit strong-convergence and uniform-integrability argument, as recorded in Section C.3; no blanket lower-semicontinuity claim is made for the full entropy. No pointwise equation for $f = -\log \rho + \text{constant}$ is asserted on the zero set of ρ . Smooth positive adjoint approximations justify the localized transport formulae needed below.

9. STOKES-METRIC PROJECTION ONTO LOCALIZED KINETIC VARIATIONS

9.1. Scale-critical kinetic probes. On a compact represented-time interval I_τ , an admissible probe consists of a repaired scale–translation chart (X, Λ) and a nonnegative function

$$(9.1) \quad \begin{aligned} m &\in W^{1,1}(I_\tau; W^{1,\infty}(B_R)) \cap L^\infty(I_\tau; W^{2,\infty}(B_R)), \\ \operatorname{supp} m(\cdot, \tau) &\subset B_{R_0} \quad (0 < R_0 < R), \quad \int_{B_R} m(y, \tau) \, dy = 1. \end{aligned}$$

The smooth adjoint weights $m = \chi^2 \rho$ from Section 8, and regularized weak adjoint weights with the same normalization, are the principal examples. The physical multiplier is

$$(9.2) \quad w_\Gamma(x, t(\tau)) := \Lambda(\tau)^{-1} m\left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau\right).$$

Its physical mass and weighted velocity mean are

$$(9.3) \quad M_\Gamma(t) := \int_{\mathbb{R}^3} w_\Gamma \, dx = \Lambda(\tau)^2, \quad \bar{u}_\Gamma(t) := M_\Gamma(t)^{-1} \int_{\mathbb{R}^3} w_\Gamma u \, dx.$$

The Galilean-reduced localized kinetic probe is

$$(9.4) \quad F_\Gamma(u(t)) := \frac{1}{2} \int_{\mathbb{R}^3} w_\Gamma(x, t) |u(x, t) - \bar{u}_\Gamma(t)|^2 dx.$$

On the represented cylinder,

$$(9.5) \quad F_\Gamma(u(t(\tau))) = \frac{1}{2} \int_{B_R} m(y, \tau) |V(y, \tau) - \bar{V}_m(\tau)|^2 dy, \quad \bar{V}_m := \int_{B_R} mV dy.$$

Thus F_Γ is invariant under the Navier–Stokes scaling and is a Galilean-invariant version of the local quantity A . Denote its chart expression by

$$(9.6) \quad \mathcal{E}_\Gamma^\circ(V(\tau)) := \frac{1}{2} \int_{B_R} m |V - \bar{V}_m|^2 dy.$$

For variations of V with m fixed,

$$(9.7) \quad D\mathcal{E}_\Gamma^\circ(V)[\varphi] = \int_{B_R} m(V - \bar{V}_m) \cdot \varphi dy.$$

Indeed, the variation of $\bar{V}_m$ pairs with $\int m(V - \bar{V}_m) = 0$. Thus the target gradient below is the Stokes gradient of the Galilean-reduced variance in every representative. No time-dependent velocity-centering condition is imposed on the chart.

Let

$$(9.8) \quad q_\Gamma := \mathbb{P}(w_\Gamma(u - \bar{u}_\Gamma)).$$

For variations $\varphi \in \mathcal{H}_\sigma$,

$$DF_\Gamma(u)[\varphi] = (q_\Gamma, \varphi)_{L^2}.$$

Since $q_\Gamma \in \mathcal{V}_\sigma$ for almost every time, this derivative extends to currents $J \in \mathcal{V}_\sigma^*$ by $DF_\Gamma(u)[J] := \langle J, q_\Gamma \rangle$. The Stokes gradient current of the probe is

$$(9.9) \quad G_\Gamma := Kq_\Gamma.$$

Define the viscous action, the Stokes norm of the localized variation, and the signed localized power by

$$(9.10) \quad \mathcal{A}_\nu(u) := \|J_\nu\|_{K^{-1}}^2 = \nu \|\nabla u\|_2^2,$$

$$(9.11) \quad \mathcal{G}_\Gamma(u) := \|G_\Gamma\|_{K^{-1}}^2 = \nu \|\nabla q_\Gamma\|_2^2,$$

$$(9.12) \quad \mathcal{P}_\Gamma(u) := (J_\nu, G_\Gamma)_{K^{-1}} = DF_\Gamma(u)[J_\nu].$$

Lemma 9.1 (Explicit viscous power). *For smooth u and w_Γ ,*

$$(9.13) \quad \mathcal{P}_\Gamma = -\nu \int_{\mathbb{R}^3} w_\Gamma |\nabla u|^2 dx + \frac{\nu}{2} \int_{\mathbb{R}^3} |u - \bar{u}_\Gamma|^2 \Delta w_\Gamma dx.$$

The first term is local viscous dissipation. The second is the diffusive flux across the spatial variation of the probe.

Proof. Since $A_\Sigma u$ is solenoidal and $\mathbb{P}$ is self-adjoint,

$$\mathcal{P}_\Gamma = -\nu \int \nabla u : \nabla q_\Gamma = -\nu \int \nabla u : \nabla (w_\Gamma(u - \bar{u}_\Gamma)).$$

Expand the last gradient and integrate $-\nu \int \partial_j u_i (u_i - \bar{u}_{\Gamma,i}) \partial_j w_\Gamma$ by parts. This gives (9.13). $\square$

In the represented variables, the power measure has the scale-free form
(9.14)

$$\mathcal{P}_\Gamma(u(t(\tau))) \, dt = \widehat{\mathcal{P}}_\Gamma(V(\tau)) \, d\tau, \quad \widehat{\mathcal{P}}_\Gamma(V) := -\nu \int_{B_R} m |\nabla V|^2 \, dy + \frac{\nu}{2} \int_{B_R} |V - \bar{V}_m|^2 \Delta m \, dy.$$

For smooth solutions the total derivative of the probe splits as

$$(9.15) \quad \frac{d}{dt} F_\Gamma(u(t)) = \mathcal{T}_\Gamma(u, p) + \mathcal{P}_\Gamma(u),$$

where

$$(9.16) \quad \mathcal{T}_\Gamma(u, p) := \frac{1}{2} \int |u - \bar{u}_\Gamma|^2 (\partial_t w_\Gamma + u \cdot \nabla w_\Gamma) \, dx + \int p (u - \bar{u}_\Gamma) \cdot \nabla w_\Gamma \, dx.$$

The term $\mathcal{T}_\Gamma$ contains conservative transport, pressure flux, and the motion of the scale–center coordinates. The term $\mathcal{P}_\Gamma$ contains viscosity alone. For suitable weak solutions, define

$$(9.17) \quad \mu_{\text{loc}} := - \left[\partial_t \frac{|u|^2}{2} + \operatorname{div} \left(\left(\frac{|u|^2}{2} + p \right) u \right) - \nu \Delta \frac{|u|^2}{2} + \nu |\nabla u|^2 \right].$$

The local energy inequality says precisely that this distribution is a nonnegative Radon measure. Combining it with the weak momentum identity for the weighted mean gives

$$(9.18) \quad F_\Gamma(t_2) - F_\Gamma(t_1) = \int_{t_1}^{t_2} (\mathcal{T}_\Gamma + \mathcal{P}_\Gamma) \, dt - \mu_{\text{loc}}(w_\Gamma; [t_1, t_2]),$$

where $\mu_{\text{loc}}(w_\Gamma; [t_1, t_2])$ denotes the measure in (9.17) paired with the nonnegative test weight on the indicated time interval. The identity holds at the corresponding local energy times; arbitrary compact subintervals follow by one-sided representatives.

9.2. The Stokes-metric projection inequality.

Definition 9.2 (Viscous alignment). Set

$$(9.19) \quad \operatorname{Align}_\Gamma(u) := \begin{cases} \frac{(\mathcal{P}_\Gamma(u)_+)^2}{\mathcal{A}_\nu(u) \mathcal{G}_\Gamma(u)}, & \mathcal{A}_\nu(u) \mathcal{G}_\Gamma(u) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Positive power means that the viscous current points toward increasing the localized kinetic probe. Negative power means that viscosity lowers that probe.

Theorem 9.3 (Stokes-metric projection inequality). *For almost every time,*

$$(9.20) \quad (\mathcal{P}_\Gamma(u)_+)^2 \leq \mathcal{A}_\nu(u) \mathcal{G}_\Gamma(u), \quad 0 \leq \operatorname{Align}_\Gamma(u) \leq 1.$$

For every interval $I \Subset (0, T)$,

$$(9.21) \quad \int_I \mathcal{P}_\Gamma(u(t))_+ \, dt \leq \left(\int_I \mathcal{A}_\nu(u(t)) \, dt \right)^{1/2} \left(\int_I \mathcal{G}_\Gamma(u(t)) \, dt \right)^{1/2}.$$

The first factor is at most $\mathcal{E}(u(0))^{1/2}$.

Proof. The definitions give

$$\mathcal{P}_\Gamma = (J_\nu, G_\Gamma)_{K^{-1}}.$$

Cauchy–Schwarz in the K^{-1} -metric proves (9.20). A second application of Cauchy–Schwarz in time proves (9.21). The energy bound is (3.12). $\square$

The inequality separates the amount of irreversible motion from its orientation. Equal action can produce zero or maximal positive power, depending on the angle with the target gradient.

Definition 9.4 (Covariant ray-alignment defect). For a sufficiently regular divergence-free comparison curve v , set

$$(9.22) \quad R(v) := \mathfrak{D}_t^E v + K v.$$

For an admissible probe Γ and interval I , define

$$(9.23) \quad \begin{aligned} \mathcal{D}_{\Gamma, I}^{\text{ray}}(v) &:= \frac{1}{2} \int_I \|R(v)\|_{K^{-1}}^2 dt \\ &+ \frac{1}{2} \int_I \left(\mathcal{A}_\nu(v) - \frac{(\mathcal{P}_\Gamma(v)_+)^2}{\mathcal{G}_\Gamma(v)} \right) dt, \end{aligned}$$

where the quotient is defined as zero when $\mathcal{G}_\Gamma = 0$.

Proposition 9.5 (Metric meaning of the ray defect). *The functional $\mathcal{D}_{\Gamma, I}^{\text{ray}}$ is nonnegative. Pointwise in time,*

$$(9.24) \quad \mathcal{A}_\nu - \frac{(\mathcal{P}_{\Gamma+})^2}{\mathcal{G}_\Gamma} = \text{dist}_{K^{-1}}^2(J_\nu, \{\alpha G_\Gamma : \alpha \geq 0\}).$$

For an exact Navier–Stokes solution, the equation-defect term in (9.23) vanishes. Equality $\mathcal{D}_{\Gamma, I}^{\text{ray}}(u) = 0$ then means that the actual viscous current lies on the nonnegative target-gradient ray almost everywhere on I . Moreover,

$$(9.25) \quad 2\mathcal{D}_{\Gamma, I}^{\text{ray}}(u) = \int_I \mathcal{A}_\nu(u(t))(1 - \text{Align}_\Gamma(u(t))) dt.$$

Proof. Orthogonal projection of a vector J onto the closed ray generated by a nonzero vector G has coefficient $(J, G)_+ / \|G\|^2$. Its squared residual norm is $\|J\|^2 - (J, G)_+^2 / \|G\|^2$. Apply this identity in the K^{-1} -metric. When $G = 0$, the distance is $\|J\|$, in agreement with the convention in (9.23). The equation-defect statement is (3.7). Substitution of (9.19) gives (9.25). $\square$

Let $\mathfrak{G}_{\text{adm}}(I)$ denote repaired probes with scale-relative cutoff bounds and normalized adjoint weights as in Section 8. Let $\mathfrak{G}_{\text{adm}}(t)$ be their time- t slices. Define

$$(9.26) \quad \text{Align}_*(t) := \sup_{\Gamma \in \mathfrak{G}_{\text{adm}}(t)} \text{Align}_\Gamma(t), \quad \mathcal{D}_*^{\text{ray}}(I) := \inf_{\Gamma \in \mathfrak{G}_{\text{adm}}(I)} \mathcal{D}_{\Gamma, I}^{\text{ray}}(u).$$

Suprema and infima are understood through approximating sequences; no attainment is required. The Stokes mobility K and current J_ν are fixed throughout these optimizations.

Theorem 9.6 (Local energy identities and projection bound). *For every admissible probe and interval $[t_1, t_2] \subseteq (0, T)$, the following three statements hold simultaneously:*

- (i) *global energy decreases by viscous action and the nonnegative energy defect according to (3.11);*
- (ii) *the localized probe satisfies (9.18), separating coordinate/transport flux from viscous target power;*
- (iii) *positive viscous target power satisfies (9.21) and is therefore controlled by the finite global action.*

These conclusions remain valid along an optimizing sequence for (9.26).

Proof. Items (i)–(iii) are Theorem 3.3, the suitable local energy inequality tested by w_Γ , and Theorem 9.3, respectively. Each inequality holds for every admissible probe, so it holds termwise along any approximating sequence in (9.26). $\square$

9.3. Alignment, coercivity, and coherent probe selection. The alignment is an angle in the Stokes metric. It contains no information about the amount of action unless it is paired with $\mathcal{A}_\nu$, and it contains no compactness information unless the probe belongs to a normalized family. The following elementary observation fixes the equality orientation.

Lemma 9.7 (Positive-ray equality for a kinetic probe). *Let $w \geq 0$ have positive finite mass, put $\bar{u}_w = (\int w)^{-1} \int wu$, let $q = \mathbb{P}(w(u - \bar{u}_w))$, and suppose $u \in L^2_\sigma \cap \dot{H}^1_\sigma$. If*

$$(9.27) \quad -Ku = \alpha Kq \quad \text{for some } \alpha \geq 0,$$

then $u = 0$.

Proof. Apply K^{-1} in the solenoidal Gelfand triple and pair the resulting identity $-u = \alpha q$ with u . Since u is solenoidal,

$$-\|u\|_2^2 = \alpha(q, u)_{L^2} = \alpha \int w|u - \bar{u}_w|^2 dx \geq 0.$$

Both sides vanish, so $u = 0$. □

For a finite-energy whole-space state, vanishing ray defect is therefore a rigid equality. Along a singular rescaling, however, the physical ray defect is weighted by Λ exactly as in (7.3). It may have finite physical integral even when its scale-critical chart density remains positive on every unit window. This explains why the ray defect is only a covariant diagnostic; the exclusion proof instead controls the local Dirichlet chart density through the moving-cutoff energy.

Probe optimization also has a precise order. The physical solution first determines J_ν and the canonical chart. A fixed terminal-data rule then selects the finite-horizon adjoint problem on the padded window (8.5). Only after these choices are fixed may one take a supremum in Align or an infimum in $\mathcal{D}^{\text{ray}}$. Replacing the current by the gradient of an optimizing probe would change the equation and is not an admissible variation.

Definition 9.8 (Compatible family of localized probes). A compatible family of localized probes on a represented orbit is a family $\{\Gamma_j\}_{j \geq 1}$ on padded retained windows such that:

- (i) every Γ_j is constructed from the same physical solution by a fixed terminal-data rule and has the buffer (8.6);
- (ii) chart transitions satisfy the composition law in Theorem 10.2;
- (iii) the current paired with every target is the pullback of $J_\nu = -\nu A_S u$;
- (iv) approximating probes converge in the compact topology required by the local adjoint construction.

The terminal adjoint equation is a testing device. Coherence here means that all probes test the same physical current. It does not identify distinct finite-horizon adjoint densities with one global density, and no such identification is used by the corrected localized energy below.

Proposition 9.9 (Scale-critical window form of the ray defect). *Let I_τ be a represented interval and I_t its physical image. For an exact Navier–Stokes solution, the pullback of the physical ray defect has the form*

$$(9.28) \quad 2\mathcal{D}_{\Gamma, I_t}^{\text{ray}}(u) = \int_{I_\tau} \Lambda(\tau) \widehat{\mathfrak{D}}_{\text{ray}, \Gamma}(V(\tau)) d\tau,$$

where $\widehat{\mathfrak{D}}_{\text{ray}, \Gamma}$ is invariant under a constant parabolic rescaling of the represented window. Equivalently, it is defined by the Radon–Nikodym relation

$$(9.29) \quad \widehat{\mathfrak{D}}_{\text{ray}, \Gamma}(\tau) d\tau := \Lambda(\tau)^{-1} \text{dist}_{K^{-1}}^2(J_\nu, \mathbb{R}_+ G_\Gamma) dt.$$

Proof. Both vectors in the Stokes pairing transform by the pullback of the same chart. Their squared inverse-metric distance has the scaling of viscous action. Formula (5.11) shows that its physical measure is Λ times the dimensionless represented measure, which is exactly (9.29). $\square$

The hatted density exhibits the same physical-versus-critical distinction as (7.3) and (7.4). It is retained below as a covariant diagnostic, but the local coercive floor does not use it.

10. CANONICAL CRITICAL ORBITS

The covariant argument is carried by one normalized orbit rather than a list of asymptotic regimes. This section specifies the state space and the exact compactness conclusion required by the entropy proof.

10.1. Symmetry reduction and target-retaining windows. Fix a normalized probe radius $R_* > 1$ and a nonnegative inner cutoff $\zeta \in C_c^\infty(B_{R_*})$, $0 \leq \zeta \leq 1$, $\int \zeta^3 > 0$, for which $\{\zeta > 0\}$ is connected and whose support lies where the adjoint cutoff equals one. Put

$$(10.1) \quad \bar{V}_\zeta(\tau) := \frac{\int \zeta^3 V(y, \tau) dy}{\int \zeta^3 dy}.$$

For a unit represented interval $I = [s, s + 1]$, define

$$(10.2) \quad \mathcal{C}_\Gamma^\circ(V; I) := \int_I \int_{B_{R_*}} \zeta^3 |V - \bar{V}_\zeta|^3 dy d\tau.$$

This observable is independent of the adjoint terminal datum, continuous under strong local L^3 convergence, and invariant under constant velocity shifts.

The relevant symmetries are spatial translation, positive parabolic dilation, constant-velocity Galilean transformation, and addition of a function of time to the pressure. The repaired variables (X, Λ, a, b) encode scale and center. We fix those representatives by

$$(10.3) \quad \mathfrak{F}(1, 0; V(\tau)) = 0, \quad \det D_{(\lambda, q)} \mathfrak{F}(1, 0; V(\tau)) \neq 0$$

at almost every active time, with absolutely continuous continuation supplied by the canonical-observer construction. Galilean symmetry is not fixed by (10.3); the gauge itself is invariant under it, and the scalar observables remove it through (10.1) and (9.3). The pressure gauge is

$$(10.4) \quad \int_{B_1} P(y, \tau) dy = 0.$$

The nondegeneracy and absolute continuity of this gauge are conclusions of the canonical-observer theorem, not extra conditions imposed on an arbitrary weak solution.

Definition 10.1 (Normalized target-retaining window). Fix $\eta_0 > 0$ and a collection of compact-cylinder bounds $\mathbf{M} = (M_R)_{R \geq 1}$. A represented unit window $\mathfrak{w} = (V, P, a, b; I)$ is an $(\eta_0, \mathbf{M})$ -normalized target-retaining window when:

- (i) (V, P) is a suitable solution of (5.4) on a neighborhood of $B_R \times I$ for every fixed R ;
- (ii) the gauges (10.3) and (10.4) hold, and the inverse of the gauge Jacobian in (7.9) is uniformly bounded on the window;
- (iii) the local suitable norms satisfy, on each fixed $B_R \times I$,

$$(10.5) \quad \|V\|_{L_I^\infty L^2(B_R)} + \|V\|_{L_I^2 H^1(B_R)} + \|V\|_{L^3(B_R \times I)} + \|P\|_{L^{3/2}(B_R \times I)} + \|a\|_{L^1(I)} + \|b\|_{L^1(I)} \leq M_R;$$

- (iv) the oscillatory floor

$$(10.6) \quad \mathcal{C}_\Gamma^\circ(V; I) \geq \eta_0$$

holds for the selected inner probe.

The subtraction in (10.2) is essential. A spatially constant represented velocity can have positive uncentered L^3 mass, but it has no spatial oscillation and cannot by itself support a singular point; constant Galilean shifts are a special case. The canonical-observer theorem therefore retains concentration only after passing to this quotient.

10.2. Canonical observation.

Theorem 10.2 (Canonical observer at terminal loss). *Let u be a classical finite-energy solution of (1.1) on $[0, T)$, where $T < \infty$ is its maximal classical existence time, and let (u, p) denote any suitable continuation through T . Then there exist $\eta_0 = \eta_0(\nu) > 0$, compact bounds $\mathbf{M}$, an absolutely continuous repaired chart (X, Λ) , and a represented suitable orbit*

$$(V, P, a, b)(\tau), \quad \tau \in [0, \infty),$$

with unit intervals $I_j = [\tau_j, \tau_j + 1]$ having pairwise disjoint interiors and $\tau_j \rightarrow \infty$, such that:

- (i) after passing once to a subsequence and relabeling, with $x_j := X(\tau_j)$ and $r_j := \Lambda(\tau_j)$, one has $r_j \downarrow 0$, and the physical time images of the windows converge to T ; the centers may remain bounded or escape to spatial infinity;
- (ii) every restricted state $\mathfrak{w}_j = (V, P, a, b; I_j)$ is an $(\eta_0, \mathbf{M})$ -normalized target-retaining window;
- (iii) in the product topology specified in Section 7 and made explicit here, the sequential closure of the selected windows is sequentially compact. More explicitly, after translating each time interval to $[0, 1]$, every sequence in that closure has a subsequence for which

$$(10.7) \quad V_j \rightarrow V_\infty \text{ strongly in } L^3_{\text{loc}}, \quad \nabla V_j \rightharpoonup \nabla V_\infty \text{ in } L^2_{\text{loc}},$$

on every compact spatial cylinder in $\mathbb{R}^3 \times [0, 1]$, and in particular strongly in $L^3(\text{supp } \zeta \times [0, 1])$, with strong local $L^{3/2}$ convergence of normalized pressures and L^1 convergence of (a, b) ;

- (iv) chart changes obey the composition law and represent the same physical solution.

Proof. See Appendix B, in particular Theorem B.1 and Propositions B.4, B.52 and B.53. $\square$

Remark 10.3 (Why one orbit is required). The corrected energy decrease is accumulated along one scalar function of represented time. Independent rescaling on each target-retaining window would provide unrelated endpoint values and would not support the telescoping argument. Clause (iv) is therefore part of the construction rather than a convention made after extraction.

10.3. The two classical scale regimes. Let $\ell(t) = \sqrt{T - t}$ denote the parabolic terminal scale. Along the canonical orbit, the dimensionless ratio

$$(10.8) \quad \mathfrak{r}(\tau) := \frac{\Lambda(\tau)}{\ell(t(\tau))}$$

records the classical distinction. A Type I window has $\mathfrak{r}$ bounded above and below after normalization; in the centered choice $t = -e^{-\tau}$, $\Lambda = e^{-\tau/2}$, one has $a = -1/2$. A Type II sequence contains windows on which $\mathfrak{r} \rightarrow 0$, and its coefficient $a = \Lambda'/\Lambda$ must be controlled through the repaired gauge (7.8).

The covariant equation (5.6) is identical in the two regimes. The difference is carried entirely by (a, b) and by the asymptotic path of the normalized orbit. The moving-cutoff alternative is consequently formulated for locally integrable a, b , without fixing $a = -1/2$ or assuming convergence of a . A centered ancient orbit, a rapid scale collapse, and a recurrent repaired orbit are either reduced to the corresponding local estimates or tested by the same moving production measure. The gap argument uses only the coherent orbit, compactness, the oscillatory floor, and the summable moving-cutoff case. Thus the argument uses precisely the compactness in Theorem 10.2 and introduces no additional rescaled solution.

There remains one compatibility issue. A window selected at time τ_j must extend to the next selected window without replacing the underlying solution. Let $g_{j+1 \leftarrow j} \in \mathcal{G}_{\text{ch}}$ be the accumulated canonical transition from the terminal representative of I_j to the initial representative of I_{j+1} . Coherence means

$$(10.9) \quad g_{k \leftarrow j} = g_{k \leftarrow k-1} \circ \cdots \circ g_{j+1 \leftarrow j}, \quad j < k,$$

and every composition pulls back the same physical pair (u, p) . The backward adjoint localization and its cutoff are transported on this one orbit. There is no cross-window reset in the exclusion argument.

11. EXACT CRITICAL FINANCING

The material identities identify the mechanism but do not by themselves produce a scale-critical finite measure. The required statement is a dissipation-or-reduction alternative: signed transport between nested regions is internal, whereas viscous dissipation and true exterior leakage are counted once. Its precise interface with the compact orbit is derived below from the localized energy equation.

Let $I = [s, t]$ be a represented interval and let $\psi \in C^\infty(I; C_c^\infty(\mathbb{R}^3))$, $\psi \geq 0$. Set

$$(11.1) \quad K_\psi(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V(y, \tau)|^2 \psi(y, \tau) dy, \quad D_I(\psi) := \nu \int_I \int_{\mathbb{R}^3} |\nabla V|^2 \psi dy d\tau,$$

and define the signed localization current

$$(11.2) \quad J_I(\psi) := \int_I \left[\frac{1}{2} \int |V|^2 (\partial_\tau \psi + \nu \Delta \psi) + \int \left(\frac{1}{2} |V|^2 + P \right) V \cdot \nabla \psi - \frac{a}{2} \int |V|^2 (\psi + y \cdot \nabla \psi) - \frac{1}{2} b \cdot \int |V|^2 \nabla \psi \right] d\tau.$$

All spatial integrals in (11.2) are over $\mathbb{R}^3$. The pressure contribution is invariant under $P \mapsto P + c(\tau)$, because $\int V \cdot \nabla \psi = 0$.

Proposition 11.1 (Signed-current additivity). *For every represented suitable solution there is a nonnegative local energy-defect functional $M_I(\psi)$, linear on nonnegative tests, such that*

$$(11.3) \quad K_\psi(t) + D_I(\psi) + M_I(\psi) = K_\psi(s) + J_I(\psi).$$

For a smooth solution equality holds and $M_I(\psi) = 0$. If $\psi = \sum_{k=1}^N \psi_k$ with $\psi_k \geq 0$, then

$$(11.4) \quad K_\psi = \sum_{k=1}^N K_{\psi_k}, \quad D_I(\psi) = \sum_{k=1}^N D_I(\psi_k), \quad J_I(\psi) = \sum_{k=1}^N J_I(\psi_k), \quad M_I(\psi) = \sum_{k=1}^N M_I(\psi_k).$$

Consequently, for an exact parent-child partition $\psi_\alpha = \sum_{\beta \in \text{ch}(\alpha)} \psi_\beta$, the difference between the parent current and the sum of its child currents is identically zero. This is refinement invariance, not selective cancellation: adding the parent balance to all child balances double-counts every term, whereas subtracting the parent balance cancels every term, including D_I and M_I .

Proof. The suitable local energy inequality for the repaired equation defines a nonnegative defect distribution. Integrating it between s and t and placing that defect on the left gives (11.3); the terms generated by the repaired drift are the last two terms of (11.2). Every displayed term is linear in ψ and its derivatives, which proves (11.4). For smooth solutions the local energy identity holds with equality. $\square$

Theorem 11.2 (Exact critical financing identity). *Let $C_0, \dots, C_{N-1}$ be consecutive normalized spacetime cells along the canonical observer. The represented-time interval of C_j begins at the incoming face of the j -th selected window and ends at the incoming face of the next one; hence it contains both that active window and any following inactive interval. Choose compatible smooth localization tests and oriented endpoint traces. Write K_j^- and K_j^+ for the localized kinetic energy at the incoming and outgoing faces, D_j for all viscous dissipation in C_j , $M_j \geq 0$ for all suitable-energy defect in C_j , and H_j for the complete signed pressure, transport, dilation, translation, and cutoff current. Then*

$$(11.5) \quad K_j^+ + D_j + M_j = K_j^- + H_j \quad (0 \leq j < N),$$

and

$$(11.6) \quad \sum_{j=0}^{N-1} (D_j + M_j) = K_0^- - K_{N-1}^+ + \sum_{j=0}^{N-1} H_j.$$

In particular, D_j dominates the dissipation on the active unit window contained in C_j . The identity is invariant under addition of a function of time to the pressure.

Proof. Apply the local energy equality with defect measure to each cell. This is (11.3) with the nonnegative defect placed on the left; all terms in which a derivative hits the localization or the moving chart comprise H_j , giving (11.5). On a common face, the outgoing kinetic trace of C_j is the incoming trace of C_{j+1} . The two appear with opposite signs and cancel. Because each cell already includes the inactive interval preceding the next selected window, its dissipation and defect remain in $D_j + M_j$, rather than being reclassified as current. Summation yields (11.6). Finally, $\int V \cdot \nabla \psi = 0$ makes the pressure part invariant under $P \mapsto P + c(\tau)$. $\square$

11.1. Normalized shell decomposition. Let

$$(11.7) \quad \mathfrak{C}_{\text{NS}} := \{\text{mat}, \text{vis}, \text{int}, \text{gauge}, \text{ext}\}.$$

The labels denote material deformation, viscous motion, signed interior transport including pressure, chart and material-relabeling motion, and true exterior flux. This is a decomposition of the actual repaired Navier–Stokes current, not a decomposition of its absolute value.

Definition 11.3 (Normalized singular-target shells). Let $\widehat{\mathcal{X}}_{\text{sing}}$ be a Hausdorff state-or-germ space containing the regular orbit and the nonremovable singular germs. A normalized shell presentation consists of nested neighborhoods U_ε of the singular set and continuous functions $\mathcal{S}_\varepsilon : \widehat{\mathcal{X}}_{\text{sing}} \rightarrow [0, 1]$ such that

$$(11.8) \quad \mathcal{S}_\varepsilon = 0 \quad \text{outside } U_{2\varepsilon}, \quad \mathcal{S}_\varepsilon = 1 \quad \text{on } U_\varepsilon.$$

On every regular precontact segment on which it is used, $\mathcal{S}_\varepsilon(u(\cdot))$ is absolutely continuous and its chain rule has the exact form

$$(11.9) \quad \frac{d}{dt} \mathcal{S}_\varepsilon(u(t)) = \sum_{c \in \mathfrak{C}_{\text{NS}}} q_\varepsilon^c(t) \quad \text{for a.e. } t.$$

A shell-crossing record is an interval $J_\varepsilon = [t_\varepsilon^-, t_\varepsilon^+]$ lying strictly before contact such that the endpoint values in (11.8) are zero and one, respectively.

Theorem 11.4 (Exact shell-crossing identity). *Every shell-crossing record for which the five functions in (11.9) are integrable satisfies*

$$(11.10) \quad 1 = \sum_{c \in \mathfrak{C}_{\text{NS}}} Q_\varepsilon^c, \quad Q_\varepsilon^c := \int_{J_\varepsilon} q_\varepsilon^c(t) dt.$$

Consequently

$$(11.11) \quad \max_{c \in \mathfrak{C}_{\text{NS}}} (Q_\varepsilon^c)_+ \geq \frac{1}{5}.$$

Neither conclusion uses an energy or entropy bound.

Proof. Integrate (11.9) over the crossing interval and use the endpoint values. Since the sum of five real numbers is one, the sum of their positive parts is at least one and one positive part is at least $1/5$. $\square$

Proposition 11.5 (Persistent contributing component). *Suppose contact supplies infinitely many shell-crossing records on pairwise disjoint precontact events. After passage to an infinite subsequence, one component $c_* \in \mathfrak{C}_{\text{NS}}$ satisfies*

$$(11.12) \quad Q_j^{c_*} \geq \frac{1}{5} \quad \text{for every index in the subsequence.}$$

Proof. Choose on each record a component satisfying (11.11). One of the five labels is selected infinitely often. $\square$

11.2. Finite-depth concentration decomposition. The shell identity becomes useful only after every loss of compactness or current cancellation is retained as a new Navier–Stokes object. We use the following five PDE source families. They describe the origin of a term and do not introduce additional equations:

$$(11.13) \quad \mathfrak{F}_{\text{PDE}} := \{\text{vis}, \text{tr}, \text{quo}, \text{sc}, \text{bd}\}.$$

Here vis is the localized Stokes form and its Dirichlet production; tr is the signed convection–pressure–moving-frame current together with kinetic storage; quo consists of the Leray, pressure-gauge, and Galilean quotients; sc contains parabolic rebasing, weak-product defects, and scale current; and bd contains exterior, annular, terminal, and trace currents. The order in (11.13) is fixed whenever a first failure is selected.

Definition 11.6 (Finite-depth localized event record). A finite-depth localized event record of length N consists of exact parabolic cells $C_0, \dots, C_N$ cut from one physical suitable solution, their normalized velocity and pressure representatives, compatible additive pressure gauges, the scale–translation maps between consecutive cells, and the local-energy coordinates (B.19) on every common interface. It is *target retaining* at level $\eta > 0$ when

$$(11.14) \quad \mathcal{O}_{Q_1}(V_j) \geq \eta, \quad 0 \leq j \leq N.$$

Every weak limit used in the record carries its equation residual, Reynolds stress, normalized pressure, scale–translation parameters, suitable-energy defect, and exterior current. A limit is called ordinary only when all these coordinates converge in the topologies in which their local weak identities pass to the limit. Records of increasing depth are *compatible* when the restriction of a depth- $N + 1$ realization to its first N cells is a realization of the depth- N record with the same physical origin, pressure gauges, oriented traces, and composed observer. A terminal family is *cofinal* when the physical radius of its last cell tends to zero with the depth.

Theorem 11.7 (Finite-depth Navier–Stokes source alternative). *Let the record-point extraction in Theorem B.1, together with its exact descendants, generate a compatible cofinal family of target-retaining localized event records of arbitrarily large finite depth. After an ordered subsequence selection, the first applicable one of the following outputs occurs.*

- (F1) *The Galilean-reduced target is lost. The selected velocity is spatially homogeneous on the retained component.*

- (F2) *One nonnegative target-aligned charge is bounded below by a fixed positive number on a pairwise disjoint family of events and is dominated by one finite physical Radon measure.*
- (F3) *There is a first nonordinary coordinate. In the order (11.13) it is a viscous-domain defect, a convection or Reynolds defect, a non-affine pressure-quotient defect, a parabolic scale or realization defect, or an exterior, annular, terminal, or trace defect. The coordinate and the shell covector detecting it are retained at the same physical origin and scale.*
- (F4) *Every finite coordinate is ordinary, all local identities pass to the limit, and the inverse system of records gives an actual same-origin scale-collapsing Navier–Stokes path satisfying (11.14) at every level.*

No pressure tail, diffuse velocity remainder, or escaping observer is identified with an ordinary profile in this alternative.

Proof. Enumerate the countable compact-cylinder exhaustion, the tests in $\mathcal{T}$, and the coordinates in Definition 11.6. First test the quotient target. Its loss gives (F1) by the connected-overlap argument in Proposition B.3. On the retained target, test the nonnegative charges admitted by the physical local-energy measure. A uniform positive value on disjoint events gives (F2). A normalized Dirichlet value carrying an additional scale factor is not admitted at this step and remains in the scale-current test below.

For the remaining coordinates, select the first index in the fixed source order and countable exhaustion at which ordinary convergence fails. The local pressure decomposition identifies failure of pressure compactness with a non-affine harmonic tail; Proposition B.5 retains its exterior ancestry. Failure of velocity tightness is subjected to the first-bad-scale selection of Proposition B.4; hence it gives an actual rescaled velocity profile, a homogeneous quotient, or a retained pressure tail. Failure of an observer, defect, current, or realization coordinate is retained in the corresponding coordinate of the finite-depth record. These are precisely the alternatives in (F3).

If no finite coordinate fails, diagonal compactness gives compatible limits on every member of the exhaustion. The inverse-limit support coordinate supplies simultaneous realization by the original terminal sequence, and compatibility of the observer maps gives one same-origin path. Strong local L^3 convergence preserves (11.14), and cofinality gives scale collapse. This is (F4). The first-applicable convention makes the four outputs disjoint, and the countable coordinate exhaustion makes them exhaustive. $\square$

Proposition 11.8 (Closure of the compact nonexpanding output). *Suppose the ordinary output (F4) has a compact covariant closure on which the normalized scale coefficient is nonpositive and every minimal covariant component retains the target floor (11.14). Then (F4) is impossible. The same conclusion holds when the closure carries a stationary monotone covariant ensemble in the sense of Definition 6.7 whose law retains the target almost surely.*

Proof. In the autonomous case apply Corollary 6.9. In the variable-coefficient case Theorem 6.8 gives $\nabla V = 0$ almost surely. The quotient activity therefore vanishes, contrary to (11.14). $\square$

Proposition 11.9 (First unbounded activity scale). *Let U be the bounded ancient profile in Theorem B.1, and let z_j be points in the diffuse remainder of Proposition B.3. For a fixed $0 < \varepsilon_0 < \varepsilon_{\text{reg}}$, let ρ_j be the first radius at which $C_V^\circ(z_j, \rho_j) = \varepsilon_0$, after the homogeneous mode is removed. If $\rho_j \rightarrow \infty$, the stopping-time construction gives one of the following concrete outputs:*

- (T1) *an actual target-retaining parabolic descendant;*
- (T2) *a spatially homogeneous velocity with affine pressure, which is target-null;*
- (T3) *a non-affine harmonic pressure tail with exterior ancestry.*

The last output remains a typed member of (F3) until it is converted into an actual target-retaining successor or shown to be target-null.

Proof. Apply [Proposition B.3](#). Its concentrated output is (T1), and its homogeneous output is (T2). On the diffuse output, apply [Proposition B.4](#); alternatives (a) and (b) there give (T1), while alternative (c) gives (T3). Those alternatives are exhaustive under the hypotheses of that proposition. Accessibility of the pressure tail follows from [Proposition B.5](#). None of these steps replaces a tail coordinate by an unproved velocity limit. $\square$

Definition 11.10 (Collapsed same-origin residual). A collapsed same-origin residual is an output of (F4) which

- (J1) retains a fixed positive quotient activity on a cofinal sequence of shrinking exact parabolic cells;
- (J2) satisfies the Navier–Stokes equation, incompressibility, the compatible pressure equations, and the signed suitable local-energy identities on every finite block;
- (J3) has no terminal L^3 endpoint, no homogeneous quotient, and no positive-density compact covariant component retaining the target;
- (J4) retains every nonzero Reynolds, pressure, scale, exterior, trace, and realization coordinate as a first-failure successor; and
- (J5) is not a uniformly record-bounded separated marked cascade, since that case is excluded by [Theorem B.22](#); and
- (J6) remains in the normalized window class [Definition 10.1](#) with the fixed bounds $\mathbf{M}$, and on the record-based realization retains the Type I pointwise envelope (11.15) on every normalized domain used by the marked projective family. By [Lemma 11.11](#), loss of that envelope is a scale/Type II coordinate in (J4), while loss of any other normalized bound is an observer, pressure, or compactness coordinate; neither is a member of the residual.

Thus the residual consists precisely of amplitude-renormalizing scale collapse together with any current needed to preserve the target through its successive changes of observer.

Lemma 11.11 (Type I envelope versus scale escape). *Let $C_j = Q_{r_j}(z_j)$ be the exact cells in an ordinary same-origin path, and fix $\kappa > 1$ so that $Q_{\kappa r_j}(z_j)$ lies in the realizing domain. After passage to a subsequence, exactly one of the following holds:*

- (I1) *the scale-invariant pointwise envelope*

$$(11.15) \quad \sup_j r_j \|u\|_{L^\infty(Q_{\kappa r_j}(z_j))} < \infty$$

holds on every normalized domain used by the marked projective family;

- (I2) *there are cells for which*

$$(11.16) \quad r_j \|u\|_{L^\infty(Q_{\kappa r_j}(z_j))} \longrightarrow \infty.$$

The second alternative is the scale/Type II output in (F3). It is selected before entry into the collapsed residual.

Proof. Apply the bounded/unbounded dichotomy to the nonnegative sequence $r_j \|u\|_{L^\infty(Q_{\kappa r_j}(z_j))}$. In the bounded case, parabolic change of variables gives one common L^∞ bound on the normalized cells. In the unbounded case, select a subsequence tending to infinity and retain the point, scale, and realizing pressure atlas. This is precisely a failure of the scale envelope in the sc coordinate of (11.13), hence (F3). The ordered finite-depth alternative selects that failure before the ordinary output (F4). $\square$

Corollary 11.12 (Reduction to one explicit residual mechanism). *For the finite-depth event construction, target loss, the compact nonexpanding output, finite ordinary descendant chains, and uniformly record-bounded separated cascades are closed by [Proposition 11.8](#), [Corollary B.9](#),*

and [Theorem B.22](#). Every non-tight or nonordinary failure is retained by [Theorem 11.7](#) and [Proposition 11.9](#). Consequently the only output not closed by those results is the collapsed same-origin residual of [Definition 11.10](#), including any infinite chain of its typed first-failure successors. Its event-indexed closure is proved in [Theorem B.43](#).

Proof. Apply the alternatives in [Theorem 11.7](#). Alternative (F1) is target-null. Alternative (F2) is impossible by countable additivity of its finite dominating physical measure. Alternative (F3) is treated by [Proposition 11.9](#); it is not discarded. In (F4), compact nonexpanding recurrence is excluded by [Proposition 11.8](#), finite chains by [Corollary B.9](#), and the uniformly record-bounded separated value by [Theorem B.22](#). Apply [Lemma B.29](#) to the remaining realized active events. Unbounded nesting gives the longitudinal path in [Definition 11.10](#); large antichains are reduced by [Theorem B.31](#) to a first typed output or to a coalescent marked path governed by the same signed-current identity. Removing these disjoint subcases leaves exactly the properties in [Definition 11.10](#); a non-affine intermediate pressure source remains one of its typed successor coordinates. $\square$

The shell layer identifies the signed current which finances repeated progress toward the singular target. It is kept in full: pressure, convection, translation, dilation, cutoff motion, kinetic storage, dissipation, and the suitable-energy defect are never separated into unrelated estimates.

Theorem 11.13 (Persistent-current compactification). *Let the canonical orbit of [Theorem 10.2](#) contain infinitely many target-retaining unit windows, and let their complete signed currents be defined by [\(11.2\)](#). If the cumulative current is not sublinear in the number of selected windows, then, after one ordered subsequence selection, exactly one of the following occurs:*

- (i) *a first noncompact center, pressure, scale, exterior, trace, or weak-product coordinate produces a sequence-realized critical-tail datum;*
- (ii) *every finite coordinate is compact, the successor observers and their oriented currents converge as marks, and recurrence of the marked successor path produces a nonzero sequence-realized logarithmic critical tail.*

In both cases the tail is realized by exact translations, Galilean changes, and parabolic rescalings of the original solution. No diffuse measure or pressure remainder is declared to be an ordinary velocity profile.

Proof. Apply the finite-depth local-energy decomposition to longer and longer initial blocks of the canonical orbit. Target loss contradicts the retained oscillation floor, while infinitely many disjoint positive charges contradict the finiteness of the local dissipation and defect measures. If a compact coordinate first fails, its first detecting test and physical realization survive the diagonal selection and give alternative (i).

Otherwise the complete event records are compact. The nesting–antichain alternative reduces transverse families to a first failed coordinate or a coalescent marked path; longitudinal nesting gives the same marked path directly. The exact moving-root identity excludes sublinear current. For persistent current, the event-shift closure retains both activity and the oriented boundary current. Recurrence of this compact marked path gives the sequence-realized tail in alternative (ii). The complete derivation, including the sparse moving-root case, is given in [Lemmas B.23](#) and [B.29](#) and [Theorems B.31](#) and [B.42](#). $\square$

Theorem 11.14 (Critical-tail rigidity). *Every sequence-realized logarithmic critical tail produced by persistent signed current is target-null. This includes diffuse, homogeneous, non-Dirac defect, log-periodic, and aperiodic tails.*

Proof. Decompose the tail on logarithmic shells and retain every compact shell carrying positive quotient activity. Shell-support realization turns every support point into an actual translated-cylinder limit of the original solution. Failure of logarithmic tightness is the diffuse alternative; on a tight shell, local compactness either records the first pressure, observer, Young, or viscous-defect coordinate, or gives strong L^3 convergence. Every recorded coordinate is rescaled at its first visible parabolic scale and therefore returns to the same marked-current alternative, whose sparse case is closed by the exact critical-currency identity. If no coordinate survives, the tail is coherent. Bounded-origin decay then makes homogeneous, log-periodic, and aperiodic minimal coherent tails zero. In every case the quotient activity vanishes. The detailed exhaustion is proved in [Lemmas B.37 to B.39](#) and [B.41](#) and [Theorem B.42](#). $\square$

12. COVARIANT ADJOINT GEOMETRY AND THE MOVING-CUTOFF ENERGY

The adjoint entropy $\mathcal{W}_{\text{ad}}$, the kinetic variance $\mathcal{E}_\Gamma^\circ$, and the ray defect $\mathcal{D}^{\text{ray}}$ have distinct roles. The first controls the localization geometry, the second detects the retained velocity core, and the third measures the component of viscous motion transverse to the target-gradient ray. None of them separately has the monotonicity and coercivity needed below.

12.1. The explicit nonnegative components. For a smooth positive adjoint density, define the Gaussian production

$$(12.1) \quad \mathfrak{D}_G(\rho) := 2\nu^2\theta \int_{B_R} \chi^2 \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \, dy.$$

Define also the localized viscous production

$$(12.2) \quad \mathfrak{D}_{\nu,\Gamma}(V) := \nu \int_{B_R} m |\nabla V|^2 \, dy$$

and the instantaneous ray production

$$(12.3) \quad \mathfrak{D}_{\text{ray},\Gamma}(V) := \widehat{\mathfrak{D}}_{\text{ray},\Gamma}(V),$$

where the hatted, scale-critical density is defined by [\(9.29\)](#). At weak regularity $\mathfrak{D}_{\nu,\Gamma}$ is defined directly from the weighted Dirichlet integral. If the Gaussian or ray terms are retained in a weak production measure, their identification and compact stability require additional adjoint compactness beyond the localized energy argument; neither is inferred merely from local weak velocity convergence. The coercive floor used in the compactness argument is independent of the adjoint density:

$$(12.4) \quad \mathfrak{D}_{0,\zeta}(V) := \nu \int_{\mathbb{R}^3} \zeta^2 |\nabla V|^2 \, dy.$$

The Gaussian production remains a favorable nonnegative term in the smooth Perelman identity, but it may be discarded before taking a weak limit. The ray production is likewise an additional nonnegative covariant diagnostic on each finite-energy representative. Neither term is needed to separate retained compact limits from equality. Inserting the ray term into [\(12.4\)](#) would in addition require a localization theorem for the whole-space Helmholtz projection under merely local blow-up convergence.

Formula [\(8.17\)](#) contains a raw velocity-strain term with no separate sign. It becomes coercive after coupling it to the viscous and Gaussian terms in [Proposition 12.2](#). The moving-cutoff kinetic energy below handles pressure and cutoff transport directly and dominates [\(12.4\)](#) because $\Phi = 1$ on $\text{supp } \zeta$.

12.2. The uncoupled entropy balance. The whole-space smooth calculation identifies every term in the correction. Put

$$(12.5) \quad \mathcal{L}_c F := \partial_\tau F - \nu \Delta F + c \cdot \nabla F, \quad c = V - ay - b.$$

For the kinetic density $e = |V|^2/2$, the repaired equation gives

$$(12.6) \quad \mathcal{L}_c e = -\nu |\nabla V|^2 + 2ae - \operatorname{div}(PV).$$

Indeed, after moving the dilation and translation derivatives to the left, (5.4) becomes

$$\partial_\tau V + c \cdot \nabla V + \nabla P = \nu \Delta V + aV.$$

Taking its scalar product with V proves (12.6).

Proposition 12.1 (Scale-covariant kinetic–entropy balance). *Let (V, P) and a positive probability density ρ be smooth on $\mathbb{R}^3$, with sufficient decay, and suppose that ρ solves the unnormalized whole-space adjoint equation. Set*

$$\bar{V}_\rho(\tau) := \int_{\mathbb{R}^3} V \rho \, dy, \quad \mathcal{K}_\rho^\circ(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V - \bar{V}_\rho|^2 \rho \, dy.$$

Then

$$(12.7) \quad \frac{d}{d\tau} \mathcal{K}_\rho^\circ = -\nu \int |\nabla V|^2 \rho + 2a\mathcal{K}_\rho^\circ - \int P(V - \bar{V}_\rho) \cdot \nabla f \, \rho.$$

Let θ solve (8.7). For every $\beta > 0$, define

$$(12.8) \quad \mathcal{G}_\beta := \theta \mathcal{K}_\rho^\circ - \beta \mathcal{W}_{\text{ad}}.$$

Then

$$(12.9) \quad \begin{aligned} \frac{d}{d\tau} \mathcal{G}_\beta = & -\mathcal{K}_\rho^\circ - \theta \nu \int |\nabla V|^2 \rho - 2\beta \nu^2 \theta \int \left| \nabla^2 f - \frac{1}{2\nu\theta} I \right|^2 \rho \\ & - \theta \int P(V - \bar{V}_\rho) \cdot \nabla f \, \rho + 2\beta \nu \theta \int (\nabla f)^T (\nabla V)_{\text{sym}} \nabla f \, \rho. \end{aligned}$$

Proof. Apply the whole-space form of Lemma 8.3 componentwise to the represented velocity equation. Since $\mathcal{L}_c V = aV - \nabla P$,

$$(12.10) \quad \bar{V}_\rho' = a\bar{V}_\rho - \int_{\mathbb{R}^3} \nabla P \, \rho \, dy.$$

Applying the same transport identity to (12.6) and integrating by parts gives

$$-\int \operatorname{div}(PV) \rho = \int PV \cdot \nabla \rho = -\int PV \cdot \nabla f \, \rho.$$

Subtracting $\frac{1}{2}|\bar{V}_\rho|^2$ and using (12.10) proves (12.7). In particular,

$$(\theta \mathcal{K}_\rho^\circ)' = -\mathcal{K}_\rho^\circ - \theta \nu \int |\nabla V|^2 \rho - \theta \int P(V - \bar{V}_\rho) \cdot \nabla f \, \rho,$$

because $\theta' + 2a\theta = -1$. Subtracting β times (8.17) proves (12.9). $\square$

Thus the covariant heat scale removes the scalar modulation from the kinetic variance as well as from the adjoint entropy. No sign or integrability condition on a is used in this cancellation. The strain term in (12.9) combines exactly with the viscous and Gaussian terms.

Proposition 12.2 (Strain–Gaussian square completion). *Under the hypotheses of Proposition 12.1, set*

$$S := \frac{1}{2}(\nabla V + (\nabla V)^T), \quad \Omega := \frac{1}{2}(\nabla V - (\nabla V)^T), \quad A_f := \nabla^2 f - \frac{1}{2\nu\theta}I.$$

Then

$$(12.11) \quad \int_{\mathbb{R}^3} \rho S : (\nabla f \otimes \nabla f) dy = \int_{\mathbb{R}^3} \rho S : A_f dy.$$

Consequently the entropy balance has the exact form

$$(12.12) \quad \frac{d}{d\tau} \mathcal{G}_\beta + \mathcal{K}_\rho^\circ + \mathfrak{Q}_{\beta,\theta}(V, f) = -\theta \int_{\mathbb{R}^3} P(V - \bar{V}_\rho) \cdot \nabla f \rho dy,$$

where

$$(12.13) \quad \begin{aligned} \mathfrak{Q}_{\beta,\theta}(V, f) := & \theta\nu \int_{\mathbb{R}^3} |\Omega|^2 \rho dy + \theta \left(\nu - \frac{\beta}{2} \right) \int_{\mathbb{R}^3} |S|^2 \rho dy \\ & + 2\beta\nu^2\theta \int_{\mathbb{R}^3} \left| A_f - \frac{1}{2\nu}S \right|^2 \rho dy. \end{aligned}$$

If $0 < \beta \leq \nu$, then

$$(12.14) \quad \mathfrak{Q}_{\beta,\theta}(V, f) \geq \frac{\theta\nu}{2} \int_{\mathbb{R}^3} |\nabla V|^2 \rho dy.$$

In particular, the fixed choice

$$(12.15) \quad 0 < \beta \leq \nu$$

ensures (12.14). On every retained window, (8.6) further gives

$$(12.16) \quad \mathfrak{Q}_{\beta,\theta}(V, f) \geq \frac{\nu\kappa_* e^{-2M_*}}{2} \int_{\mathbb{R}^3} |\nabla V|^2 \rho dy.$$

Proof. Since $\rho = (4\pi\nu\theta)^{-3/2}e^{-f}$,

$$(12.17) \quad \nabla^2 \rho = \rho(\nabla f \otimes \nabla f - \nabla^2 f).$$

Incompressibility gives

$$\partial_i \partial_j S_{ij} = \Delta \operatorname{div} V = 0.$$

The decay hypotheses therefore justify two integrations by parts and give

$$\int_{\mathbb{R}^3} S : \nabla^2 \rho dy = \int_{\mathbb{R}^3} \rho \partial_i \partial_j S_{ij} dy = 0.$$

Using (12.17) and $\operatorname{tr} S = \operatorname{div} V = 0$ yields

$$\int \rho S : (\nabla f \otimes \nabla f) = \int \rho S : \nabla^2 f = \int \rho S : A_f.$$

This proves (12.11). Substitution into (12.9) gives (12.12). Finally,

$$\begin{aligned} & \theta\nu |\nabla V|^2 + 2\beta\nu^2\theta |A_f|^2 - 2\beta\nu\theta S : A_f \\ & = \theta\nu |\Omega|^2 + \theta \left(\nu - \frac{\beta}{2} \right) |S|^2 + 2\beta\nu^2\theta \left| A_f - \frac{S}{2\nu} \right|^2, \end{aligned}$$

because $|\nabla V|^2 = |S|^2 + |\Omega|^2$. This proves (12.13). When $\beta \leq \nu$, the first two terms on its right control $(\theta\nu/2)(|S|^2 + |\Omega|^2)$, proving (12.14). Combining this with the lower bound in (8.6) proves (12.16). $\square$

Lemma 12.3 (Localized strain transmutation). *Let V, ρ, f be smooth on B_R , with $\operatorname{div} V = 0$, let $\chi \in C_c^\infty(B_R)$, and assume $\rho = (4\pi\nu\theta)^{-3/2}e^{-f}$ and $\int_{B_R} \chi^2 \rho = 1$. With A_f as in [Proposition 12.2](#),*

$$(12.18) \quad \int_{B_R} \chi^2 \rho S : (\nabla f \otimes \nabla f) \, dy = \int_{B_R} \chi^2 \rho S : A_f \, dy + \mathfrak{E}_\chi(S, \rho),$$

$$(12.19) \quad \mathfrak{E}_\chi(S, \rho) := - \int_{B_R} \rho S : \nabla^2(\chi^2) \, dy - 2 \int_{B_R} S : (\nabla(\chi^2) \otimes \nabla \rho) \, dy.$$

Writing $\mathcal{A}_\chi := \operatorname{supp}(|\nabla \chi| + |\nabla^2 \chi|)$, one has for every $\hat{\chi} \in C_c^\infty(B_R)$ satisfying $\hat{\chi} = 1$ on a neighborhood of $\mathcal{A}_\chi$,

$$(12.20) \quad |\mathfrak{E}_\chi(S, \rho)| \leq \|\nabla^2(\chi^2)\|_\infty \int_{\mathcal{A}_\chi} \rho |S| \, dy + 2\|\nabla(\chi^2)\|_\infty \left(\int_{\mathcal{A}_\chi} \rho |S|^2 \, dy \right)^{1/2} \left(\int_{B_R} \hat{\chi}^2 \frac{|\nabla \rho|^2}{\rho} \, dy \right)^{1/2}$$

Moreover, with $m = \chi^2 \rho$, the localized quadratic form

$$(12.21) \quad \begin{aligned} \mathfrak{Q}_{\beta, \theta}^\chi(V, f) &:= \theta\nu \int_{B_R} m |\Omega|^2 \, dy + \theta \left(\nu - \frac{\beta}{2} \right) \int_{B_R} m |S|^2 \, dy \\ &\quad + 2\beta\nu^2\theta \int_{B_R} m \left| A_f - \frac{S}{2\nu} \right|^2 \, dy \end{aligned}$$

satisfies, whenever $0 < \beta \leq \nu$,

$$(12.22) \quad \mathfrak{Q}_{\beta, \theta}^\chi(V, f) \geq \frac{\theta\nu}{2} \int_{B_R} m |\nabla V|^2 \, dy = \frac{\theta}{2} \mathfrak{D}_{\nu, \Gamma}(V).$$

Proof. Identity (12.17) holds on B_R . Moreover $\partial_i \partial_j S_{ij} = 0$, as in the proof of [Proposition 12.2](#). Testing this distributional identity against the compactly supported function $\chi^2 \rho$ gives

$$0 = \int_{B_R} S : \nabla^2(\chi^2 \rho) \, dy.$$

Expanding the Hessian and using the symmetry of S yields

$$\int_{B_R} \chi^2 S : \nabla^2 \rho \, dy = - \int_{B_R} \rho S : \nabla^2(\chi^2) \, dy - 2 \int_{B_R} S : (\nabla(\chi^2) \otimes \nabla \rho) \, dy.$$

Substitute (12.17), then use $\operatorname{tr} S = 0$. This proves (12.18) and (12.19). The first term in (12.20) follows directly from support and the L^∞ norms of the cutoff derivatives. For the second, use

$$|S| |\nabla \rho| = (\sqrt{\rho} |S|) \left(\frac{|\nabla \rho|}{\sqrt{\rho}} \right)$$

and Cauchy–Schwarz on $\mathcal{A}_\chi$. The last integral over $\mathcal{A}_\chi$ is bounded by the displayed $\hat{\chi}$ -weighted Fisher information because $\hat{\chi} = 1$ there. The identity $|\nabla V|^2 = |S|^2 + |\Omega|^2$ and the same pointwise square completion as in [Proposition 12.2](#) prove (12.21)–(12.22). $\square$

Thus velocity strain requires no independent correction in the interior smooth identity. The only localized discrepancy is the annular commutator $\mathfrak{E}_\chi$. The scalar scale term has disappeared exactly from (12.12). The sole unsigned interior term in the whole-space scale-covariant identity is the pressure transfer

$$-\theta \int P(V - \bar{V}_\rho) \cdot \nabla f \, \rho.$$

For a localized adjoint density there are, in addition, the explicitly displayed cutoff and normalization terms from (8.14). We now compare this structural identity with the local kinetic-energy balance, where pressure occurs as a signed cutoff flux.

In particular, the adjoint pairing $-\theta \int P(V - \bar{V}_\rho) \cdot \nabla f \rho$ is not identified with the moving-cutoff flux $\int PV \cdot \nabla \Phi$. The pressure-tail estimates apply to the latter. The proof therefore discards the adjoint balance before invoking monotonicity; no cancellation between these two distinct pressure terms is claimed.

12.3. The moving-cutoff kinetic energy. Fix a radial nonincreasing $\phi \in C_c^\infty(B_2)$, with $0 \leq \phi \leq 1$ and $\phi = 1$ on B_1 . Let $R : [\tau_0, \infty) \rightarrow [R_*, \infty)$ be locally absolutely continuous and nondecreasing, and put

$$(12.23) \quad \Phi(\tau, y) := \phi(y/R(\tau)), \quad \mathcal{E}_R(\tau) := \frac{1}{2} \int_{\mathbb{R}^3} |V|^2 \Phi \, dy.$$

Write $a_+ = \max\{a, 0\}$, $a_- = \max\{-a, 0\}$, and define

$$(12.24) \quad \tilde{\mathfrak{D}}_R(\tau) := \nu \int |\nabla V|^2 \Phi \, dy + a_+(\tau) \int |V|^2 \Phi \, dy.$$

The full unsigned boundary-flux density is

$$(12.25) \quad \begin{aligned} B_{\text{mov}}(\tau) := & \frac{1}{2} \left| \int |V|^2 \partial_\tau \Phi \right| + \frac{\nu}{2} \left| \int |V|^2 \Delta \Phi \right| + \frac{1}{2} \left| \int |V|^2 V \cdot \nabla \Phi \right| + \left| \int PV \cdot \nabla \Phi \right| \\ & + \frac{1}{2} |b(\tau)| \left| \int |V|^2 \nabla \Phi \right| + \frac{1}{2} a_-(\tau) \int |V|^2 \Phi + \frac{1}{2} a_+(\tau) \int |V|^2 (-y \cdot \nabla \Phi). \end{aligned}$$

Every integral in (12.24)–(12.25) is over $\mathbb{R}^3$. The pressure term is unchanged by adding a function of time to P , because $\int V \cdot \nabla \Phi = 0$.

Theorem 12.4 (Corrected moving-cutoff monotonicity). *Let (V, P) be a represented suitable weak solution of (5.4). If $B_{\text{mov}} \in L^1(\tau_0, \infty)$, then*

$$(12.26) \quad \mu_{\text{NS}}(\tau) := \mathcal{E}_R(\tau) + \int_\tau^\infty B_{\text{mov}}(s) \, ds$$

has a nonincreasing representative and, in distributions,

$$(12.27) \quad \frac{d}{d\tau} \mu_{\text{NS}} + \frac{1}{2} \tilde{\mathfrak{D}}_R \leq 0.$$

In particular $\mu_{\text{NS}} \geq 0$, and for $\tau_0 \leq s < t$,

$$(12.28) \quad \mu_{\text{NS}}(t) + \mathfrak{D}_{\text{NS}}((s, t]) \leq \mu_{\text{NS}}(s), \quad \mathfrak{D}_{\text{NS}}((s, t]) := \frac{1}{2} \int_s^t \tilde{\mathfrak{D}}_R(\tau) \, d\tau.$$

Proof. Testing the suitable local energy inequality with Φ gives

$$\begin{aligned} \frac{d}{d\tau} \mathcal{E}_R \leq & -\nu \int |\nabla V|^2 \Phi + \frac{1}{2} \int |V|^2 \partial_\tau \Phi + \frac{\nu}{2} \int |V|^2 \Delta \Phi + \frac{1}{2} \int |V|^2 V \cdot \nabla \Phi \\ & + \int PV \cdot \nabla \Phi - \frac{a}{2} \int |V|^2 \Phi - \frac{a}{2} \int |V|^2 y \cdot \nabla \Phi - \frac{1}{2} b \cdot \int |V|^2 \nabla \Phi. \end{aligned}$$

Add $\tilde{\mathfrak{D}}_R/2$. The viscous term leaves $-\frac{\nu}{2} \int |\nabla V|^2 \Phi \leq 0$, while $-aM/2 + a_+M/2 = a_-M/2$. Since ϕ is radial nonincreasing, $y \cdot \nabla \Phi \leq 0$; hence the scale-shell term is bounded above by the last term of (12.25). Taking absolute values of the other fluxes proves

$$\frac{d}{d\tau} \mathcal{E}_R + \frac{1}{2} \tilde{\mathfrak{D}}_R \leq B_{\text{mov}}.$$

The suitable energy-defect measure has the favorable sign and was discarded in this inequality. Differentiating the finite tail in (12.26) subtracts B_{mov} , proving (12.27). Both terms in (12.26) are nonnegative. $\square$

12.4. Finite-block use of the moving cutoff. The proof uses Theorem 12.4 on the finite blocks selected by the canonical observer. No global summable-radius schedule is needed. The complete error B_{mov} on a block is part of its signed current H_j in Theorem 11.2; if these errors are sublinear, the critical price closes the path, and if they persist, their first noncompact coordinate or recurrent support is retained by Theorem 11.13. Thus the adjoint localization and the eventwise current are two expressions of the same financing identity.

12.5. Equality states.

Definition 12.5 (Oscillation-removable equality state). A represented state is oscillation-removable when its velocity is spatially constant on the connected retained region $\{\zeta > 0\}$ for almost every represented time. The constant may depend on time. Such a state has zero Galilean-reduced oscillatory concentration. No assertion that a localized Dirichlet adjoint density equals the whole-space Gaussian is included in this definition; the Gaussian remains the equality model only for the whole-space identity (8.16).

Theorem 12.6 (Inner-Dirichlet and zero-production rigidity). *Let $\mathfrak{w} = (V, P, a, b; [0, 1])$ be a normalized window or a compact limit of normalized windows. If*

$$(12.29) \quad \int_0^1 \mathfrak{D}_{0,\zeta}(V(\tau)) \, d\tau = 0,$$

then $\mathfrak{w}$ is oscillation-removable and

$$(12.30) \quad \mathcal{C}_\Gamma^\circ(V; [0, 1]) = 0.$$

In particular, zero moving-localized dissipation on the window implies (12.30) whenever $\Phi = 1$ on $\text{supp } \zeta$.

Proof. Since $\nu > 0$, (12.29) gives $\nabla V = 0$ almost everywhere on the connected set $\{\zeta > 0\}$ for almost every τ . Hence $V(\cdot, \tau) = c(\tau)$ there. Its ζ^3 -weighted mean is the same vector $c(\tau)$, so the integrand in (10.2) vanishes. This proves (12.30). The final assertion follows from the bound $\mathfrak{D}_{\text{NS}}(I) \geq \frac{1}{2} \int_I \mathfrak{D}_{0,\zeta}$ and nonnegativity. $\square$

13. THE POSITIVE CRITICAL PRICE

This section identifies the abstract covariant dissipation of Section 11 with the moving-localized dissipation. It is a concrete realization of the structural balance. No numerical estimate for the gap is needed: compactness upgrades the qualitative equality theorem to a uniform statement.

Lemma 13.1 (Compactness–rigidity gap principle). *Let $\mathcal{K}$ be a sequentially compact space, let $\mathcal{B} \subset \mathcal{K}$ be sequentially closed, and let $\mathfrak{d} : \mathcal{K} \rightarrow [0, \infty]$ be sequentially lower semicontinuous. If*

$$(13.1) \quad \{\mathfrak{d} = 0\} \cap \mathcal{B} = \emptyset,$$

then

$$(13.2) \quad \inf_{x \in \mathcal{B}} \mathfrak{d}(x) > 0.$$

Proof. If the infimum were zero, there would be $x_j \in \mathcal{B}$ with $\mathfrak{d}(x_j) \rightarrow 0$. Sequential compactness gives $x_j \rightarrow x_\infty$ after extraction. Closedness gives $x_\infty \in \mathcal{B}$, while lower semicontinuity gives $\mathfrak{d}(x_\infty) = 0$, contradicting (13.1). $\square$

For the Navier–Stokes application, $\mathcal{K}$ is the sequential closure of the selected windows from one canonical orbit in the full topology of clause (iii) of [Theorem 10.2](#); $\mathcal{B}$ is the subset satisfying (10.6); and

$$(13.3) \quad \mathfrak{d}_\nu(\mathfrak{w}) := \int_0^1 \mathfrak{D}_{0,\zeta}(V(\tau)) \, d\tau.$$

Strong local L^3 convergence makes $\mathcal{B}$ closed, while (A.8) makes $\mathfrak{d}_\nu$ lower semicontinuous. Both statements are proved in [Proposition A.2](#). The first assertion of [Theorem 12.6](#) supplies (13.1). Thus the compactness gap uses no semicontinuity assertion about the corrected localized energy, the Gaussian Hessian, or the nonlocal ray distance.

Theorem 13.2 (Positive critical dissipation price). *Fix the canonical orbit generated by one putative terminal loss. There exists $d_* = d_*(\text{orbit}) > 0$ such that every selected target-retaining window satisfies*

$$(13.4) \quad D_j^{\text{act}} := \nu \int_{I_j} \int_{\mathbb{R}^3} \zeta^2 |\nabla V|^2 \, dy \, d\tau \geq d_*.$$

Proof. Apply [Lemma 13.1](#) to the closure of the selected windows with $\mathfrak{d} = \mathfrak{d}_\nu$, as described above. It gives a number $\delta_\nu > 0$, depending on this orbit, such that $\mathfrak{d}_\nu(\mathfrak{w}_j) \geq \delta_\nu$ for every selected window. By (12.4), the left side of (13.4) is $\mathfrak{d}_\nu(\mathfrak{w}_j)$. Take $d_* = \delta_\nu$. $\square$

Corollary 13.3 (Infinite normalized expenditure). *For the infinite pairwise disjoint selected windows of the canonical orbit,*

$$(13.5) \quad \sum_{j=1}^{\infty} D_j^{\text{act}} = \infty.$$

More generally, any nonnegative countably additive measure μ satisfying (E.6) on those windows obeys $\sum_j \mu(I_j) = \infty$.

Proof. The first conclusion follows from (13.4). The general conclusion is [Corollary E.3](#). Only the component selected by the retained target is used; no estimate of target-null transport or material-frame motion enters the sum. $\square$

Corollary 13.4 (One-window entropy estimate). *If the moving-cutoff entropy satisfies $\mathfrak{D}_{\text{NS}}(I_j) \geq \frac{1}{2} D_j^{\text{act}}$, every selected window satisfies*

$$(13.6) \quad \mu_{\text{NS}}(\tau_j + 1) \leq \mu_{\text{NS}}(\tau_j) - \frac{1}{2} d_*.$$

Proof. Apply (12.28) on I_j and then use (13.4). $\square$

14. COVARIANT EXCLUSION OF SINGULARITIES

Proof of Theorem 1.1. Let T_* be the maximal classical existence time and suppose that $T_* < \infty$. Choose a global finite-energy suitable Leray solution. By weak–strong uniqueness it agrees with the classical solution on $[0, T_*)$ and gives a suitable continuation through T_* .

Canonical observation, [Theorem 10.2](#), supplies one coherent scale–center path and pairwise disjoint unit windows I_j , all satisfying

$$\mathcal{C}_\Gamma^\circ(V; I_j) \geq \eta_0.$$

By the positive-price theorem [Theorem 13.2](#),

$$D_j^{\text{act}} \geq d_* > 0 \quad \text{for every } j.$$

Apply the exact financing identity [Theorem 11.2](#) to the first N consecutive cells:

$$\sum_{j < N} (D_j + M_j) = K_0^- - K_{N-1}^+ + \sum_{j < N} H_j.$$

The kinetic endpoints are uniformly bounded. Since each full-cell dissipation satisfies $D_j \geq D_j^{\text{act}}$, one has $\sum_{j < N} (D_j + M_j) \geq Nd_*$, and the cumulative signed current cannot be sublinear.

The remaining possibility is persistent critical current. By [Theorem 11.13](#), it either loses a first center, pressure, scale, exterior, trace, or defect coordinate, or repeatedly returns through active centers. The first alternative produces a sequence-realized critical tail at its first visible parabolic scale. In the second, the observer and the complete current are retained as marks; recurrence of the marked successor path again produces a nonzero sequence-realized logarithmic critical tail. Both alternatives contradict [Theorem 11.14](#). Hence $T_* = \infty$.

Standard parabolic regularity then gives smoothness on every compact positive time interval, and weak–strong uniqueness identifies the Leray solution with this global classical solution. $\square$

This is the entire global argument. The appendices below prove its five structural theorems; all local configurations enter only through persistent-current compactification and critical-tail rigidity.

APPENDIX A. WEAK COVARIANT CALCULUS AND COMPACT STABILITY

This appendix records the standard weak covariant identities and proves the local stability statements used in the critical-price theorem.

A.1. Regularization of the equation. Let $S_h v(t) = h^{-1} \int_t^{t+h} v(s) ds$ be a Steklov average on an interior interval, and let v_ε denote spatial mollification. After a diagonal choice,

$$(A.1) \quad u_{\varepsilon,h} \rightarrow u \quad \text{strongly in } L_{\text{loc}}^2 H_{\text{loc}}^1,$$

with weak-* convergence in $L^\infty L^2$. The nonlinear commutator is

$$(A.2) \quad \mathcal{C}_\varepsilon := \operatorname{div}((u \otimes u)_\varepsilon - u_\varepsilon \otimes u_\varepsilon).$$

Since $u \in L_{\text{loc}}^{8/3} L_{\text{loc}}^4$,

$$(A.3) \quad \mathcal{C}_\varepsilon \rightarrow 0 \quad \text{in } L_{\text{loc}}^{4/3} H_{\text{loc}}^{-1}.$$

On a compact repaired interval, spatial mollification commutes with the translation derivative, while

$$(A.4) \quad a[y \cdot \nabla, \varphi_\varepsilon *]V \rightarrow 0 \quad \text{in } L_{\text{loc}}^1 L_{\text{loc}}^2.$$

The spatial commutator is uniformly L^2 -bounded and converges strongly to zero; dominated convergence uses $a \in L_{\text{loc}}^1$ and the local $L_\tau^\infty L_y^2$ bound. Moreover,

$$\|(b_1 - b_2) \cdot \nabla V\|_{H^{-1}(B_R)} \leq c_R |b_1 - b_2| \|V\|_{L^2(B_R)},$$

with the analogous estimate for $(a_1 - a_2)(V + y \cdot \nabla V)$.

Proposition A.1 (Weak validity of the covariant identities). *For every suitable weak solution:*

- (i) $R(u) = 0$ in distributions, and $\mathfrak{D}_\tau^{X,\Lambda} V + \nu A_S V = 0$ on every repaired compact cylinder;
- (ii) the global energy balance is (3.11);
- (iii) the localized identity is (9.18) for every admissible smooth compactly supported weight;
- (iv) the Stokes-metric projection inequalities hold for almost every weak time.

Proof. For item (i), write the mollified equation with residual $\mathcal{C}_\varepsilon$. The linear and pressure terms pass by distributional convergence, the nonlinear residual vanishes by (A.3), and the modulation terms pass by (A.4) and L^1 -approximation of a, b . Item (ii) is the Leray–Hopf energy inequality in the measure form proved in Theorem 3.3. For item (iii), test the suitable local energy inequality by w_Γ , and test the weak momentum equation componentwise by the same scalar weight. The latter identity gives the distributional derivative of $M_\Gamma^{-1} |\int w_\Gamma u|^2 / 2$; subtracting it from the weighted kinetic-energy relation gives exactly (9.18), including the same nonnegative local defect measure. Approximation of temporal indicators gives the asserted one-sided representatives. Item (iv) is Cauchy–Schwarz in the solenoidal Gelfand triple. $\square$

A.2. Compactness of the variational data. Suppose, on compact cylinders,

$$(A.5) \quad V_j \rightarrow V \quad \text{strongly in } L_{\text{loc}}^2 \text{ and } L_{\text{loc}}^3, \quad \nabla V_j \rightharpoonup \nabla V \quad \text{in } L_{\text{loc}}^2,$$

with the corresponding pressure and modulation convergence. For the additional adjoint-weighted conclusions, assume

$$(A.6) \quad m_j \rightarrow m \quad \text{strongly in } L^\infty(B_R \times I), \quad m_j \geq m_* > 0 \quad \text{on } \operatorname{supp} \zeta \times I,$$

and, for normalized adjoint densities,

$$(A.7) \quad \inf_j \inf_\tau Z_j(\tau) > 0, \quad \sup_j \|\tilde{\rho}_j\|_{L^\infty L^2 \cap L^2 H_0^1} < \infty, \quad \sup_j \int_I \int_{B_R} \chi_j^2 |\nabla \sqrt{\rho_j}|^2 dy d\tau < \infty.$$

Proposition A.2 (Stability of the local variational data). *Under (A.5):*

- (i) *the fixed-cutoff means $\bar{V}_{j,\zeta}$ converge strongly in $L^3(I)$, and $C_{\Gamma_j}^\circ(V_j; I) \rightarrow C_\Gamma^\circ(V; I)$;*
- (ii) *the adjoint-weighted kinetic variances converge in $L^1(I)$ whenever $m_j \rightarrow m$ strongly on the full support of the probe;*
- (iii) *the fixed inner Dirichlet floor is lower semicontinuous:*

$$(A.8) \quad \int_I \mathfrak{D}_{0,\zeta}(V) \, d\tau \leq \liminf_{j \rightarrow \infty} \int_I \mathfrak{D}_{0,\zeta}(V_j) \, d\tau;$$

if in addition (A.6)–(A.7) hold, then the adjoint-weighted viscous production is also lower semicontinuous:

$$(A.9) \quad \int_I \mathfrak{D}_{\nu,\Gamma}(V) \, d\tau \leq \liminf_{j \rightarrow \infty} \int_I \mathfrak{D}_{\nu,\Gamma_j}(V_j) \, d\tau;$$

- (iv) *under (A.6), the positivity bound for m on $\text{supp } \zeta$ passes to the limit.*

No lower-semicontinuity claim for the Gaussian Hessian square, the varying-ray distance, or the corrected production is made here, and none is used in the gap theorem. Retaining such terms in a stronger weak-production statement would require identification of the adjoint logarithmic Hessian and all associated localization terms; that refinement is not part of the moving-cutoff argument.

Proof. The first assertion follows from strong L^3 convergence and Hölder’s inequality on the fixed support of ζ . The same argument, together with strong convergence of the weights, gives the second assertion. For the third, $\zeta \nabla V_j \rightharpoonup \zeta \nabla V$ weakly in L^2 , which proves (A.8). Under the additional weight convergence, $m_j^{1/2} \nabla V_j \rightharpoonup m^{1/2} \nabla V$ weakly in L^2 ; weak lower semicontinuity of the L^2 -norm gives (A.9). Uniform convergence of the weights gives the final assertion. $\square$

APPENDIX B. SINGULAR-GERM EXTRACTION

This appendix derives [Theorem 10.2](#) directly from the suitable local-energy inequality, pressure reconstruction, and parabolic compactness.

B.1. Velocity concentration. The local concentration assertion was proved in [Theorem 2.1](#). For the global regularity problem one can start more directly from a maximal classical solution. The following record-point construction removes the existence of an ancient PDE profile from the list of assumptions.

Theorem B.1 (Record-point ancient-profile extraction). *Let u be a classical finite-energy solution of (1.1) on $[0, T)$, with divergence-free initial datum in $H^m(\mathbb{R}^3)$, $m > 5/2$, and suppose that $T < \infty$ is its maximal classical time. Then there are $t_j \uparrow T$, $x_j \in \mathbb{R}^3$, and $r_j \downarrow 0$ such that*

$$(B.1) \quad r_j^{-1} = M_j := \sup_{0 \leq t \leq t_j} \|u(t)\|_{L^\infty}, \quad |u(x_j, t_j)| \geq \frac{1}{2} M_j,$$

and the rescaled solutions

$$(B.2) \quad U_j(y, s) := r_j u(x_j + r_j y, t_j + r_j^2 s), \quad Q_j(y, s) := r_j^2 p(x_j + r_j y, t_j + r_j^2 s)$$

are defined on $\mathbb{R}^3 \times (-t_j/r_j^2, 0]$, satisfy

$$(B.3) \quad \|U_j\|_{L^\infty(\mathbb{R}^3 \times (-t_j/r_j^2, 0])} \leq 1, \quad |U_j(0, 0)| \geq \frac{1}{2},$$

and have a subsequence converging locally smoothly in velocity, and locally in pressure modulo functions of time, to a bounded mild ancient solution (U, Q) on $\mathbb{R}^3 \times (-\infty, 0]$. The limit satisfies

$$(B.4) \quad \|U\|_{L^\infty} \leq 1, \quad |U(0, 0)| \geq \frac{1}{2}.$$

Every further spacetime translation with translation time tending to $-\infty$ has, after pressure normalization and passage to a subsequence, a bounded mild entire Navier–Stokes limit.

Proof. The standard L^∞ continuation criterion implies $\limsup_{t \uparrow T} \|u(t)\|_\infty = \infty$; otherwise the mild solution can be continued a uniform positive time beyond T . Choose record times $t_j \uparrow T$ for which $M_j \rightarrow \infty$, and choose x_j as in (B.1). Such approximate maximizers are sufficient; for H^m -data the velocity is continuous and vanishes at spatial infinity at every time. Then $r_j = M_j^{-1} \downarrow 0$, and parabolic invariance gives the equation for (U_j, Q_j) . The record property gives (B.3). Moreover $t_j/r_j^2 = t_j M_j^2 \rightarrow \infty$.

Fix $R, L < \infty$. For all large j , the cylinder $B_{2R} \times [-2L, 0]$ lies in the domain of U_j . Bounded mild estimates, applied from a slightly earlier time, give uniform interior derivative bounds for the velocity on $B_R \times [-L, 0]$. Equivalently, one may combine the local energy inequality, the local Calderón–Zygmund/harmonic pressure decomposition, and interior parabolic regularity. Normalize Q_j by its spatial mean on each member of a countable nested cylinder exhaustion. The local pressure bounds and a diagonal Arzelà–Ascoli extraction give a smooth velocity limit and a compatible local pressure atlas. Passing to the mild formula proves that the limit is mild. The pointwise normalization passes to the limit and gives (B.4).

Finally, translate (U, Q) by times $s_j \rightarrow -\infty$. Every fixed compact time interval then lies in the translated domain. The same uniform bounded-mild estimates and diagonal argument give a solution on all of $\mathbb{R}^3 \times \mathbb{R}$. Spatial translations and constant parabolic rescalings commute with this extraction and preserve the assertion. $\square$

The nonzero value in (B.4) is not yet the Galilean-invariant target: a locally constant ancient limit can carry that normalization. The concentration–remainder selection below removes the homogeneous mode and chooses the first larger parabolic scale carrying fixed positive oscillation. Finite energy excludes a nonzero constant physical representative on all of $\mathbb{R}^3$.

B.2. Compactness selection. The endpoint of a finite descendant chain is controlled by an existing ancient-solution Liouville theorem. We record its exact form because it is the reason that only infinite or diffuse successor chains require the new covariant argument.

Theorem B.2 (Endpoint ancient Liouville theorem). *Let U be a mild ancient solution of the three-dimensional Navier–Stokes equations on $\mathbb{R}^3 \times (-\infty, 0)$. If there are times $s_k \downarrow -\infty$ such that*

$$(B.5) \quad \sup_k \|U(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $U \equiv 0$.

Proof. This is the sequence- L^3 Liouville theorem of Albritton and Barker [1, Theorem 1.2]. Their statement requires mildness but no terminal L^3 hypothesis. $\square$

For a bounded ancient velocity define the unit quotient activity

$$(B.6) \quad \mathfrak{a}_U(z) := \inf_{c \in \mathbb{R}^3} \int_{Q_1(z)} |U - c|^3 dx dt, \quad z = (x, t), \quad t < -1.$$

The pressure is not part of this target. On a bounded mild family it is reconstructed after subtracting spatial means, and ordinary spacetime translations are compact on every fixed cylinder whose translated times tend to $-\infty$.

Proposition B.3 (Backward concentration–remainder alternative). *Let U be a bounded mild ancient solution on $\mathbb{R}^3 \times (-\infty, 0)$. After passage to subsequences, at least one of the following conclusions can be selected; the first applicable conclusion in the displayed order may be used.*

- (i) *There are $s_k \downarrow -\infty$ satisfying (B.5).*
- (ii) *After removal of one spatially homogeneous mild mode, there are $z_k = (x_k, t_k)$, $t_k \rightarrow -\infty$, and $\eta > 0$ such that $\mathfrak{a}_U(z_k) \geq \eta$. The translates about z_k have a locally smooth subsequence converging to a bounded mild entire solution W with $\mathfrak{a}_W(0, 0) \geq \eta$.*
- (iii) *There are $t_k \rightarrow -\infty$, $R_k \rightarrow \infty$, and measurable sets $E_k \subset \{|x| > R_k\}$ such that*

$$(B.7) \quad 1 \leq \int_{E_k} |U(x, t_k)|^3 dx \leq 2,$$

whereas every unit parabolic recentering meeting $E_k \times (t_k - 1, t_k + 1)$ has quotient activity tending to zero. This is the diffuse remainder.

- (iv) *The selected exterior fields converge locally, after subtraction of time-dependent constants, to a spatially homogeneous velocity. The momentum equation then pairs it only with an affine pressure, so the limit is a Galilean/parasitic quotient state and has zero target activity.*

In particular, a bounded ancient profile which is terminal under nontrivial spacetime descendants, has no diffuse remainder, and is not a homogeneous quotient state satisfies alternative (i) and is zero by Theorem B.2.

Proof. Put $A(t) = \int_{\mathbb{R}^3} |U(x, t)|^3 dx \in [0, \infty]$. If $\liminf_{t \rightarrow -\infty} A(t) < \infty$, choose a backward sequence on which A is bounded; this is (i). Otherwise $A(t) \rightarrow \infty$ as $t \rightarrow -\infty$. Since U is bounded, its L^3 -mass on every fixed ball is bounded uniformly in time. We may therefore choose $t_k \rightarrow -\infty$, radii $R_k \rightarrow \infty$, and, by absolute continuity of the integral, exterior sets E_k satisfying (B.7).

Cover $E_k \times (t_k - 1, t_k + 1)$ by a fixed bounded-overlap family of unit parabolic cylinders. If the supremum of the quotient activity on those cylinders has a positive limsup, choose one such cylinder for each k . Bounded mild interior estimates give uniform velocity derivative bounds on every compact subcylinder after recentering. The local Calderón–Zygmund/harmonic decomposition gives pressure compactness modulo time functions. Because $t_k \rightarrow -\infty$, the translated time domains exhaust $\mathbb{R}$. A diagonal extraction therefore gives an actual bounded mild entire solution. Strong local L^3 convergence preserves the positive quotient activity and proves (ii).

If the supremum tends to zero, retain the normalized exterior restrictions in (B.7). When their quotient oscillation remains visible only after aggregation over an increasing number of unit cylinders, they give exactly (iii). Otherwise the local Poincaré quotients vanish on every cylinder in each connected overlap component. The associated local constants agree up to an error tending to zero; diagonal passage gives a spatially homogeneous limit. Substitution in the momentum equation shows that its pressure is affine in space modulo a function of time. This is (iv). The four outcomes exhaust the covering dichotomy.

Under the terminal assumptions in the last sentence, alternatives (ii)–(iv) are unavailable, so (i) holds and Theorem B.2 applies. $\square$

The proposition is deliberately phrased in terms of actual translated PDE limits. The diffuse alternative is measure-valued at this stage and is not declared an ordinary solution. The next subsections select its first scale of nonzero quotient oscillation and retain the observer and current through an infinite iteration.

The first part of that realization is a stopping-time argument. For a suitable solution W , a center $z = (x_0, t_0)$, and a constant vector c , let

$$Q_r^c(z) := \{(x, t) : t_0 - r^2 < t < t_0, |x - x_0 - c(t - t_0)| < r\}$$

and write

$$(B.8) \quad C_W^\circ(z, r) := \inf_{c \in \mathbb{R}^3} r^{-2} \int_{Q_r^c(z)} |W - c|^3 dx dt.$$

This is the velocity criterion applied after an exact constant Galilean transformation; it is invariant under translations, Galilean changes, and parabolic scaling.

Proposition B.4 (First bad scale gives an ordinary descendant). *Fix $0 < \varepsilon_0 < \varepsilon_{\text{reg}}$, where ε_{reg} is the velocity regularity threshold used in [Theorem 2.1](#). Let U be a bounded mild ancient solution and suppose that a diffuse remainder as in [\(B.7\)](#) contains points at which $U \neq 0$, after the homogeneous quotient has been removed. Then one of the following occurs.*

- (a) *A bounded-radius unit recentering carries a fixed positive quotient activity, and alternative (ii) of [Proposition B.3](#) applies.*
- (b) *There are parabolic centers z_j , radii $\rho_j \rightarrow \infty$, and constant Galilean parameters c_j such that the exact rescalings*

$$(B.9) \quad W_j(y, s) := \rho_j(U(x_j + \rho_j y + c_j \rho_j^2 s, t_j + \rho_j^2 s) - c_j)$$

with the correspondingly transformed pressure are suitable on expanding backward cylinders, have a locally smooth subsequence modulo homogeneous velocity and affine-pressure modes, and converge to an ordinary bounded mild ancient solution W_∞ . On the nonhomogeneous branch,

$$(B.10) \quad \int_{Q_1} |W_\infty - \bar{W}_{\infty, Q_1}|^3 \geq c(\varepsilon_0) > 0.$$

Every compact subset of the limit is obtained by an exact parabolic rescaling of the original solution; no measure-valued state is used.

- (c) *On the selected first-bad cylinders the local velocity bounds hold but the mean-subtracted pressures are not precompact in $L_{\text{loc}}^{3/2}$. In the local decomposition $P = P_{\text{loc}} + H$, the Calderón–Zygmund part is compact, so this failure is carried by a non-affine harmonic pressure tail sourced outside the selected cylinder.*

If the rescaled forward time depth tends to infinity, the limit is entire. If it stays bounded, the standard terminal self-similar change of variables places the limit in [\(5.4\)](#) with a constant coefficient $a_ < 0$. Thus both time-depth alternatives have nonpositive scale drift.*

Proof. For a smooth point z , one has $C_U^\circ(z, r) \rightarrow 0$ as $r \downarrow 0$. If $C_U^\circ(z, r) < \varepsilon_0$ on arbitrarily large backward cylinders, apply quantitative velocity ε -regularity after an almost minimizing Galilean transformation. Its derivative estimate gives

$$|\nabla U(z)| \leq \frac{C(\varepsilon_0)}{r^2}.$$

Thus a solution for which this happens at every point is spatially homogeneous. On the nonhomogeneous quotient branch there is consequently a point and a scale at which $C_U^\circ = \varepsilon_0$. Select the first such scale. If these first scales stay bounded, a bounded-radius cylinder has a positive oscillatory subsequence and gives (a).

Assume that the first scales tend to infinity. Apply the standard parabolic point-selection procedure. If a chosen cylinder contains a smaller cylinder at which the threshold is first reached and whose distance to the boundary, in its own units, is larger, replace the chosen

cylinder by that one. Each replacement decreases the radius by a fixed factor while increasing the available normalized boundary distance. An infinite replacement chain in a fixed compact physical cylinder would have radii tending to zero and centers converging to a smooth point, contradicting $C_U(z, r) \rightarrow 0$. Thus, for the j -th selected cylinder, the procedure terminates with a radius ρ_j such that every subcylinder of radius at most ρ_j lying within normalized parabolic distance j has Galilean-reduced velocity quantity at most ε_0 . Absence of alternative (a) forces $\rho_j \rightarrow \infty$.

Rescale by (B.9). Cover a fixed compact cylinder in the new variables by finitely many subunit cylinders. On each of them the velocity criterion is below ε_{reg} , so interior regularity gives bounds for every velocity derivative on a slightly smaller cover. The local pressure decomposition

$$P = P_{\text{loc}} + H, \quad -\Delta P_{\text{loc}} = \partial_i \partial_j (\chi W_i W_j), \quad \Delta H = 0,$$

makes the local part compact after subtracting spatial means. If the harmonic parts are not compact modulo affine functions, alternative (c) holds. Otherwise the affine part is exactly the pressure paired with the removed homogeneous velocity, and diagonal compactness produces an ordinary suitable mild limit. The threshold equality at the selected cylinder passes by strong local L^3 convergence. If its value were carried only by the homogeneous mode, the limit belongs to the quotient alternative. Otherwise Poincaré's inequality and the first-scale normalization give (B.10) after decreasing the constant once.

Let d_j^+ be the forward time depth of the rescaled physical domain. If $d_j^+ \rightarrow \infty$, the diagonal limit is entire. If d_j^+ is bounded, translate the terminal time to zero and use $t = -e^{-\tau}$, $\Lambda = e^{-\tau/2}$. The resulting equation is (5.8), up to a constant rescaling of τ , and hence has $a_* < 0$. These are the only two subsequential values of the time-depth ratio. $\square$

Proposition B.5 (Finite energy makes the non-affine pressure tail parabolically accessible). *Let u be a finite-energy solution on $(0, T)$, and let*

$$(B.11) \quad \widetilde{W}_j(y, s) := \lambda_j u(x_j + \lambda_j y, t_j + \lambda_j^2 s), \quad d_j := \frac{t_j}{\lambda_j^2} \rightarrow \infty,$$

where $t_j \geq t_* > 0$ and $\lambda_j \downarrow 0$. A constant Galilean velocity may be subtracted when local oscillation is measured; the finite-energy representative $\widetilde{W}_j$ is retained for the whole-space pressure formula. Put $R_j = d_j^{1/4}$, and let

$$H_j^{>R_j} := \mathcal{R}_i \mathcal{R}_k (\mathbf{1}_{\{|y| > R_j\}} \widetilde{W}_{j,i} \widetilde{W}_{j,k}).$$

For every fixed $R < \infty$ and compact interval $I \Subset (-\infty, 0]$, the part of the pressure generated by $|y| > R_j$, after subtraction of its constant and linear Taylor terms on B_R , satisfies

$$(B.12) \quad \|\nabla^2 H_j^{>R_j}\|_{L^\infty(B_R \times I)} \leq C_{R, I, t_*, \|u_0\|_2} d_j^{-3/4} \rightarrow 0$$

at every time at which the energy inequality holds. Moreover $R_j^2 = d_j^{1/2} = o(d_j)$. Consequently a failure of local pressure compactness modulo affine functions cannot be produced at spatial distances beyond the parabolic time depth. It must be sourced in the intermediate region

$$(B.13) \quad 2R < |y| < R_j, \quad -R_j^2 < s \leq 0,$$

up to a term converging to zero locally and an affine pressure mode.

Proof. The global energy inequality and (B.11) give, uniformly at admissible rescaled times,

$$(B.14) \quad \|\widetilde{W}_j(s)\|_2^2 \leq \lambda_j^{-1} \|u_0\|_2^2 = \sqrt{\frac{d_j}{t_j}} \|u_0\|_2^2 \leq t_*^{-1/2} \|u_0\|_2^2 d_j^{1/2}.$$

The pressure Poisson equation is unchanged by subtraction of a spatially constant velocity: the constant–constant term has zero second derivative, and the two cross terms vanish because the velocity is divergence free. Thus the tail may be estimated using the finite-energy representative.

The kernel of $\mathcal{R}_i \mathcal{R}_k$ is homogeneous of degree -3 , and its second spatial derivatives are bounded by $C|y|^{-5}$ away from the origin. Taylor’s formula on B_R therefore gives

$$\begin{aligned} \sup_{x \in B_R} |\nabla^2 H_j^{>R_j}(x, s)| &\leq C_R \int_{|y| > R_j} |y|^{-5} |\widetilde{W}_j(y, s)|^2 dy \\ &\leq C_R R_j^{-5} \|\widetilde{W}_j(s)\|_2^2 \leq C_{R, t_*, \|u_0\|_2} d_j^{-3/4}, \end{aligned}$$

because $R_j^5 = d_j^{5/4}$. The bound is uniform on a fixed compact rescaled time interval by the energy inequality. Interior estimates for harmonic functions control all lower local norms after the affine Taylor jet has been removed. Finally, $R_j^2 = d_j^{1/2} < d_j$ for large j , so the backward parabolic cylinders associated with the intermediate region lie inside the rescaled time domain. This proves the assertion. $\square$

Remark B.6 (What the pressure estimate does not prove). The accessibility estimate rules out a genuinely super-parabolic pressure source. It does not by itself turn the intermediate stress in (B.13) into a fixed positive Galilean-reduced L^3 oscillation on one cylinder. The ordered argument does not require that conversion: it retains the intermediate source as a pressure-tail coordinate. If a later first-bad-scale selection produces velocity oscillation, that ordinary descendant is then recorded; otherwise the pressure coordinate remains part of the marked critical tail treated by Theorem 11.14.

Remark B.7 (What the stopping time proves). Proposition B.4 removes diffuse velocity mass as a terminal label: it becomes an actual velocity descendant or a homogeneous quotient state. It also isolates the only pressure obstruction as a non-affine harmonic tail with exterior ancestry; it does not silently call that tail compact. An infinite sequence of such actual descendants need not have a stationary law satisfying all invariances in Definition 6.7. The marked construction in Section B.3 retains the successor witnesses and their currents and converts the resulting recurrent path into a critical tail.

Definition B.8 (Actual target-retaining descendant). An actual target-retaining descendant of a bounded ancient represented solution is a locally smooth limit, with compatible local pressure gauges, of exact spacetime translations, constant Galilean changes, and constant parabolic rescalings of that solution, for which a fixed quotient-oscillation floor holds on Q_1 . A first-bad-scale limit is included only through Proposition B.4. Thus every descendant is an ordinary Navier–Stokes solution; a diffuse measure, a failed pressure chart, or a formal boundary value is not a descendant.

Corollary B.9 (Finite ordinary descendant chains close). *Let U_0 be the record-point profile in Theorem B.1, after removal of the homogeneous quotient. Iterate Proposition B.3 and, on its diffuse branch, the ordinary compact alternative of Proposition B.4. If the resulting target-retaining chain terminates after finitely many actual descendants, then its last profile is impossible or its first non-affine intermediate pressure source is a typed pressure-tail output of (F3). Thus every finite chain is closed within the ordered source alternative; outside a typed first failure, only an infinite chain of actual target-retaining descendants can remain.*

Proof. At a terminal profile there is no further nonhomogeneous descendant and no diffuse remainder. If pressure compactness modulo affine functions fails, Proposition B.5 retains the first non-affine source with its exterior ancestry, which is the asserted (F3) output. Otherwise the homogeneous alternative has zero quotient target and the last node satisfies the backward

sequence- L^3 alternative of [Proposition B.3](#). It vanishes by [Theorem B.2](#), contradicting its retained quotient-oscillation floor. The same conclusion holds after a terminal self-similar change because its physical pullback is a bounded mild ancient solution and the critical L^3 -norm is invariant. $\square$

Lemma B.10 (Event law for an infinite realized chain). *Let K_η be a compact metric space of ordinary normalized represented solutions on which the quotient target $\mathcal{O}$ is continuous and $\mathcal{O} \geq \eta > 0$. Let $\mathcal{R} \subset K_\eta \times K_\eta$ be an actual-descendant relation. If*

$$(B.15) \quad w_0 \mathcal{R} w_1 \mathcal{R} w_2 \mathcal{R} \cdots ,$$

then the event-count measures

$$(B.16) \quad \nu_N := \frac{1}{N} \sum_{j=0}^{N-1} \delta_{w_j}$$

have weak- limit points, every one of which is supported on $\{\mathcal{O} \geq \eta\}$. The closure of the shift orbit of the given one-sided sequence carries a shift-invariant probability measure whose zeroth marginal has the same property. No closedness of the profile-only relation is asserted.*

Proof. Compactness of K_η gives weak-* compactness of $\mathcal{P}(K_\eta)$. Every measure in (B.16) is supported on the closed set $\{\mathcal{O} \geq \eta\}$, and the Portmanteau theorem preserves this support condition under weak-* limits. The closure of the shift orbit of $(w_j)_{j \geq 0}$ is compact in $K_\eta^\mathbb{N}$. Weak-* limits of the shift empirical measures are shift invariant because, for every continuous F ,

$$\frac{1}{N} \sum_{j=0}^{N-1} (F(S^{j+1}w) - F(S^j w)) = \frac{F(S^N w) - F(w)}{N} \longrightarrow 0.$$

Their zeroth marginals are limits of (B.16). The successor edge can be lost when its recentering witness escapes, so the conclusion deliberately concerns the event law and not a closed path relation. $\square$

Remark B.11 (The profile-only path space). If the unmarked relation happens to be closed, then the path space

$$(B.17) \quad \mathcal{P}_\eta := \{(v_j)_{j \geq 0} \in K_\eta^\mathbb{N} : v_j \mathcal{R} v_{j+1} \text{ for all } j\}$$

is nonempty and compact, is invariant under the left shift S , and carries an S -invariant Borel probability measure. Every coordinate marginal of that measure is target-retaining. The hypothesis is not available for the profile-only descendant relation: local convergence on fixed cylinders does not control observers whose centers escape. The marked construction below restores closedness by retaining those observers.

Proposition B.12 (A rooted target is not preserved by an invariant hull). *Let Q be a fixed bounded cylinder and define*

$$\mathcal{O}_Q(V) := \inf_{c \in \mathbb{R}^3} \int_Q |V - c|^3.$$

Suppose that spatial translates $T_{x_j} V$ converge strongly in $L^3(Q)$ to a spatially constant field as $|x_j| \rightarrow \infty$. Then

$$(B.18) \quad \mathcal{O}_Q(T_{x_j} V) \longrightarrow 0.$$

Consequently the closed covariant hull of a locally concentrated profile need not remain in $\{\mathcal{O}_Q \geq \eta\}$, even when the profile itself has $\mathcal{O}_Q(V) \geq \eta$. In particular, invariance of a probability law under the successor shift does not imply the spatial invariance required in [Definition 6.7](#) while retaining the root target.

Proof. The map $V \mapsto \mathcal{O}_Q(V)^{1/3}$ is 1-Lipschitz in $L^3(Q)$: apply the triangle inequality before taking the infimum over constants, and then interchange the two fields. Its value on a spatially constant field is zero, which proves (B.18). The remaining assertions follow because a covariantly invariant hull contains all finite spatial translates and their local limit points, whereas the successor shift acts on a sequence of separately recentered profiles. $\square$

B.3. Marked descendant structure for regenerated current. The successor shift and the covariant translation action are different dynamical systems. The preceding proposition prevents replacing one by the other. We instead retain the geometric and local-energy data which realize each successor. The resulting relation is closed because an escaping witness is recorded as an exterior, pressure, diffuse, or scale-tail output rather than discarded.

Fix once and for all a countable family $\mathcal{T}$ of nonnegative smooth parabolic cutoffs which is dense in the test-function Fréchet topology on each member of a compact-cylinder exhaustion. The stopping radii and times are chosen from the full-measure set for which the endpoint energy traces and the coarea limits of every test in $\mathcal{T}$ exist. Equivalently, the boundary currents below may be defined as limits of smooth collars. The local energy distribution, including its nonnegative Radon defect, then defines for every physical parabolic cell C the coordinates

$$(B.19) \quad \mathbf{q}_C(\psi) := (K_C^-(\psi), K_C^+(\psi), D_C(\psi) + M_C(\psi), J_C(\psi)), \quad \psi \in \mathcal{T},$$

where the endpoint energy traces and the complete signed current are oriented with respect to C . The suitable-energy defect is included in the third coordinate, so the balance is an equality.

Definition B.13 (Marked successor record). Let $C_j = Q_{r_j}(z_j)$ and $C_{j+1} = Q_{r_{j+1}}(z_{j+1}) \in Q_{r_j/2}(z_j)$ be two actual cells cut from the same suitable solution. A marked successor record is

$$(B.20) \quad \mathbf{m}_j = (\mathbf{w}_j, \mathbf{w}_{j+1}, C_j, \lambda_j, \xi_j, \theta_j, \gamma_j, \mathbf{q}_j, \mathbf{s}_j),$$

where

$$(B.21) \quad \lambda_j = \frac{r_{j+1}}{r_j}, \quad \xi_j = \frac{x_{j+1} - x_j}{r_j}, \quad \theta_j = \frac{t_{j+1} - t_j}{r_j^2}.$$

Here γ_j is the additive pressure gauge on the overlap, $\mathbf{q}_j$ is the countable vector $(\mathbf{q}_{C_j}(\psi))_{\psi \in \mathcal{T}}$, and $\mathbf{s}_j$ is the inverse-limit support coordinate: for every finite block ending at j , every first finite family of cylinders and tests, and every $m \geq 1$, it records a terminal-sequence index at least m , a composed observer, and a realization error at most m^{-1} in the strong velocity and pressure coordinates. Compatibility is restriction of the $(m+1)$ -st record to the first m coordinates. Concatenation requires equality of the outgoing window, incoming window, pressure gauge, and oriented current trace on the common overlap.

The topology of the marked records is strong local L^3 for velocity, strong local $L^{3/2}$ modulo the recorded gauge for pressure, weak local L^2 for gradients, weak-* convergence for the suitable defect, Euclidean convergence for (λ, ξ, θ) , and coordinatewise convergence of $\mathbf{q}$ and $\mathbf{s}$. The countable coordinates make this topology metrizable on bounded sets.

Definition B.14 (Uniformly record-bounded terminal extraction). A terminal extraction (U_n, P_n) on domains $\mathcal{D}_n$ is *uniformly record-bounded* if each pair is an exact parabolic rescaling of the same physical solution, the domains exhaust every compact cylinder used by the extraction, and

$$(B.22) \quad \sup_n \|U_n\|_{L^\infty(\mathcal{D}_n)} \leq M_*$$

for one $M_* < \infty$. The record-point sequence in Theorem B.1 has this property with $M_* = 1$ by (B.3).

Definition B.15 (Marked regenerated-current cascade). A marked regenerated-current cascade at level $\eta > 0$ is an infinite concatenated sequence $(\mathfrak{m}_j)_{j \geq 0}$ with the following properties. Write V_j for the velocity component of $\mathfrak{w}_j$.

(R1) The cells are strictly nested and

$$(B.23) \quad 0 < \lambda_j \leq \frac{1}{4}, \quad \mathcal{O}_{Q_1}(V_j) \geq \eta.$$

Every finite block is simultaneously realized in one member of the original terminal sequence by the recorded composed observers.

(R2) With $w_j = r_j^{-1}$, let $\mathscr{J}_j$ be the complete oriented local-energy current through the stopping surface between C_j and C_{j+1} , including its endpoint energy trace, and let $\mathscr{P}_j \geq 0$ be viscous production plus suitable defect in that shell. The scale-critical shell dissipation is

$$(B.24) \quad \mathscr{D}_j^{\text{crit}} := w_j \mathscr{P}_j.$$

It is invariant under Navier–Stokes scaling because physical local-energy production has homogeneity one in length. There are $N_k \uparrow \infty$ and $\delta_D > 0$ such that

$$(B.25) \quad \frac{1}{N_k} \sum_{j=0}^{N_k-1} \mathscr{D}_j^{\text{crit}} \geq \delta_D \quad (k \geq 1).$$

(R3) Every nonzero sequence-realized support profile of the remaining weighted current regenerates a child satisfying (B.23).

(R4) The marked records have compact closure; in particular there are $\lambda_* > 0$ and $B < \infty$ such that

$$(B.26) \quad \lambda_* \leq \lambda_j \leq \frac{1}{4}, \quad |\xi_j| + |\theta_j| \leq B.$$

(R5) The original terminal extraction is uniformly record-bounded in the sense of Definition B.14. Every finite initial block of length N has a simultaneous realization whose ungauged root velocity $U^{(N)}$ satisfies

$$(B.27) \quad \|U^{(N)}\|_{L^\infty(\mathcal{Q}_N)} \leq M_*.$$

Here $\mathcal{Q}_N$ contains the images, in root coordinates, of all unit cylinders in the block. The normalized velocity at every descendant is the exact parabolic rebase of $U^{(N)}$, up to an additive constant Galilean velocity. The realization accuracy on the terminal descendant may be chosen so that its velocity $V_N^{(N)}$ satisfies

$$(B.28) \quad \|V_N^{(N)} - V_N\|_{L^3(Q_1)} \leq \varepsilon_N, \quad \varepsilon_N \downarrow 0.$$

Proposition B.16 (Extraction of the marked alternative). *Consider an infinite nested sequence of target-retaining current events furnished by one finite-stage concentration–remainder construction from a uniformly record-bounded terminal extraction. Suppose that each event is accompanied by its actual successor witness and the coordinates in (B.20). Apply the following tests in order:*

- (E1) *if the Cesàro mean of the scale-critical shell dissipations has a subsequence tending to zero, select shells whose normalized dissipation tends to zero;*
- (E2) *if a nonzero sequence-realized support profile of the remaining current has no target-retaining child, select that profile at its first failed compactness, pressure, exterior, diffuse, affine, or critical-tail test;*
- (E3) *if the radii fail to tend to zero, select the compact non-collapsing scale limit; otherwise retain a subsequence with $r_{j+1}/r_j \leq 1/4$;*

- (E4) if, on this subsequence, the ratios have no positive lower bound, the normalized observer increments are unbounded, or the velocity, pressure gauge, suitable-defect, current, or realization coordinates have no compact closure, select the corresponding scale-tail, exterior, pressure/noncompact, defect, or realization-loss profile.

If none of these selected outputs occurs, the remaining events contain a marked regenerated-current cascade in the sense of [Definition B.15](#). Thus failure of the compact marked cascade is an explicit first-failure output; it is not an additional hypothesis hidden in the conclusion.

Proof. Put $A_N = N^{-1} \sum_{j < N} \mathcal{D}_j^{\text{crit}}$. If $\liminf_N A_N = 0$, choose $N_k \rightarrow \infty$ with $A_{N_k} \rightarrow 0$. Since these dissipations are nonnegative, for every k some $j_k < N_k$ satisfies $\mathcal{D}_{j_k}^{\text{crit}} \leq A_{N_k}$; this is test (E1). Otherwise choose $0 < \delta_D < \liminf_N A_N$; then (B.25) holds on every sufficiently long initial block.

Starting from the first retained event, follow a target-retaining child of each nonzero current-support profile. If no such child exists, the first failed test is exactly (E2). If a child always exists, recursive selection gives an infinite realized chain. If its radii do not tend to zero, a subsequence has a positive limiting radius; after normalization this is the compact non-collapsing scale output in (E3). If the radii tend to zero, choose successively the first later radius not exceeding one quarter of the preceding selected radius. Relabeling this subsequence gives the upper bound in (B.23); finite simultaneous realization is preserved under passage to a subsequence and composition of the intervening observers.

On the separated chain, failure of any lower scale bound, observer bound, or coordinate compactness has a first coordinate in the countable cylinder and test exhaustion on which compactness fails. Selecting that coordinate gives the output in (E4). Diagonal compactness gives the compact closure and the bounds in (B.26). The retained activity floor and the regeneration property are closed because velocity convergence is strong in the fixed local L^3 topology and because the complete successor witness is one of the marks. The record bound and the inverse-limit realization coordinate give (R5) by [Lemma B.19](#) below. Thus properties (R1)–(R5) hold, so the remaining chain is a marked regenerated-current cascade. $\square$

Lemma B.17 (Exact scale-critical shell balance). *For every finite initial block of a marked regenerated-current cascade,*

$$(B.29) \quad \sum_{j=0}^{N-1} \mathcal{D}_j^{\text{crit}} = w_0 \mathcal{J}_0 - w_{N-1} \mathcal{J}_N + \sum_{j=1}^{N-1} (w_j - w_{j-1}) \mathcal{J}_j.$$

Common parent–child currents cancel with opposite orientations. Viscous production and suitable defect remain nonnegative on the disjoint shell interiors.

Proof. Apply the suitable local energy identity, with its nonnegative defect included, to the disjoint regions between consecutive stopping surfaces. The normal trace on a common surface occurs once with each orientation and cancels. Writing the resulting shell identity as $\mathcal{P}_j = \mathcal{J}_j - \mathcal{J}_{j+1}$, multiplying by w_j , and summing gives

$$\sum_{j < N} w_j (\mathcal{J}_j - \mathcal{J}_{j+1}) = w_0 \mathcal{J}_0 - w_{N-1} \mathcal{J}_N + \sum_{j=1}^{N-1} (w_j - w_{j-1}) \mathcal{J}_j.$$

This is (B.29). The sign assertion follows because the dissipation and defect measures are restricted to disjoint interiors; they are not obtained by subtracting a parent localization from its children. $\square$

Lemma B.18 (Finite simultaneous realization). *Every finite set of marked events and every finite set of compact normalized cylinders on which their activity and current coordinates are*

tested are realized, with compatible pressure gauges, by one member of the original terminal extraction.

Proof. This is precisely the finite-block clause in (R1). More explicitly, the inverse-limit coordinate $\mathfrak{s}_j$ records, for each block length and each finite cylinder-and-test family, one terminal-sequence index and the composed observer realizing all of those coordinates. Restriction compatibility of $\mathfrak{s}_j$ preserves the same index when coordinates are discarded, and the overlap coordinate γ_j makes the pressure representatives agree up to their recorded additive gauge. Hence all members of the prescribed finite family occur in one realization. A diagonal choice over the countable exhaustion gives the final assertion. $\square$

Lemma B.19 (The record bound is inherited by finite marked blocks). *Let the marked events be generated from a uniformly record-bounded terminal extraction. If every finite block has the inverse-limit realization coordinate of Definition B.13, then clause (R5) of Definition B.15 holds with the same constant M_* as in (B.22).*

Proof. Fix N . The inverse-limit coordinate supplies terminal-sequence indices realizing the first N edges on the finite collection of cylinders needed for that block, with error tending to zero in every recorded strong local velocity coordinate. Choose one such index far enough along the sequence that the terminal velocity error on Q_1 is at most N^{-1} , and call the corresponding extraction member $U^{(N)}$. This is the ungauged root field: Galilean constants are retained in the composed observer and are not added to $U^{(N)}$. Since every image cylinder in the finite block lies in the domain of this realization, $\mathcal{Q}_N \subset \mathcal{D}_{k(N)}$, and hence

$$\|U^{(N)}\|_{L^\infty(\mathcal{Q}_N)} \leq \|U_{k(N)}\|_{L^\infty(\mathcal{D}_{k(N)})} \leq M_*.$$

Exact composition of translations, constant Galilean changes, and parabolic dilations gives the required descendant rebases. Taking $\varepsilon_N = N^{-1}$ proves (B.28) and therefore (R5). $\square$

Lemma B.20 (Quotient activity under a bounded parabolic rebase). *Let Q be a bounded spacetime cylinder, let U be defined on the image of an affine parabolic embedding $\Psi : Q \rightarrow \mathbb{R}^3 \times \mathbb{R}$, and suppose $\|U\|_{L^\infty(\Psi(Q))} \leq M$. For $0 < \lambda \leq 1$ and $h \in \mathbb{R}^3$, put*

$$(B.30) \quad V(y, s) := \lambda U(\Psi(y, s)) + h.$$

Then

$$(B.31) \quad \mathcal{O}_Q(V)^{1/3} \leq |Q|^{1/3} M \lambda.$$

Moreover, for any $V_1, V_2 \in L^3(Q)$,

$$(B.32) \quad \left| \mathcal{O}_Q(V_1)^{1/3} - \mathcal{O}_Q(V_2)^{1/3} \right| \leq \|V_1 - V_2\|_{L^3(Q)}.$$

Proof. Choose the constant $c = h$ in the infimum defining $\mathcal{O}_Q(V)$. Then

$$\mathcal{O}_Q(V) \leq \int_Q |V - h|^3 = \lambda^3 \int_Q |U \circ \Psi|^3 \leq |Q| M^3 \lambda^3,$$

which proves (B.31). For every constant c , Minkowski's inequality gives

$$\|V_1 - c\|_{L^3(Q)} \leq \|V_1 - V_2\|_{L^3(Q)} + \|V_2 - c\|_{L^3(Q)}.$$

Take the infimum over c , and then interchange V_1, V_2 , to obtain (B.32). $\square$

Lemma B.21 (Composition of marked descendant rebases). *For a finite concatenated block of marked successors, let*

$$(B.33) \quad \Lambda_N := \prod_{j=0}^{N-1} \lambda_j.$$

In every simultaneous realization of that block, its terminal normalized velocity has the form

$$(B.34) \quad V_N^{(N)}(y, s) = \Lambda_N U^{(N)}(\Psi_N(y, s)) + h_N \quad ((y, s) \in Q_1),$$

where $U^{(N)}$ is the ungauged root representative, h_N is a constant Galilean velocity, and $\Psi_N(Q_1) \subset Q_N$. In particular, if the block satisfies (R5), then

$$(B.35) \quad \mathcal{O}_{Q_1}(V_N^{(N)})^{1/3} \leq |Q_1|^{1/3} M_* \Lambda_N.$$

Proof. For one edge, the normalized child velocity is obtained from the normalized parent by an exact translation, parabolic dilation of ratio λ_j , and a constant Galilean change. It therefore has the form

$$V_{j+1}(y, s) = \lambda_j V_j(\Psi_j(y, s)) + g_j,$$

where Ψ_j is the affine parabolic embedding determined by (ξ_j, θ_j) and the Galilean spatial drift. Composition multiplies the amplitude factors and again gives an affine parabolic embedding; all spatially constant velocity terms combine into one constant vector. Induction proves (B.34), with amplitude factor Λ_N . The cells are nested, so the image of the terminal unit cylinder lies in Q_N . Apply Lemma B.20 with $M = M_*$ to obtain (B.35). $\square$

Theorem B.22 (Record-bounded marked cascades are impossible). *There is no marked regenerated-current cascade generated from a uniformly record-bounded terminal extraction with a positive activity floor.*

Proof. Suppose that a cascade exists at level $\eta > 0$. For each N , use (R5) to realize the first N edges with terminal velocity $V_N^{(N)}$ satisfying (B.28). By (R1), $\mathcal{O}_{Q_1}(V_N) \geq \eta$. On the other hand, Lemmas B.20 and B.21 give

$$\begin{aligned} \eta^{1/3} &\leq \mathcal{O}_{Q_1}(V_N)^{1/3} \\ &\leq \mathcal{O}_{Q_1}(V_N^{(N)})^{1/3} + \varepsilon_N \\ &\leq |Q_1|^{1/3} M_* \Lambda_N + \varepsilon_N. \end{aligned}$$

Since $\lambda_j \leq 1/4$, $\Lambda_N \leq 4^{-N} \rightarrow 0$. Letting $N \rightarrow \infty$ gives $\eta^{1/3} \leq 0$, a contradiction. $\square$

Lemma B.23 (Compact closed marked path space). *Let $(\mathbf{m}_j)_{j \geq 0}$ be a marked regenerated-current cascade. The closure of its shift orbit is a compact shift-invariant space X_{mark} . Concatenation and the activity floor are closed there. Every weak-* limit of the shift empirical measures is shift invariant and has zeroth marginal supported on $\{\mathcal{O}_{Q_1} \geq \eta\}$.*

Proof. Property (R4) gives relative compactness of the complete marked records, not merely of their profile coordinates. Take the closure in the countable product of the compact coordinate closures. The inverse-limit support coordinate is closed by Lemma B.18, so every point of this closure is still sequence-realized.

Concatenation is equality of outgoing and incoming overlap coordinates and is therefore closed. Strong local L^3 convergence makes the activity floor closed. Hence the shift-orbit closure is a compact shift-invariant space of concatenated marked paths. The endpoint calculation in the proof of Lemma B.10 gives shift invariance of every empirical weak-* limit, and the Portmanteau theorem preserves the activity floor. $\square$

Definition B.24 (Sequence-realized logarithmic critical tail). A sequence-realized logarithmic critical tail is a compatible projective family of annular limits on

$$\{(y, s) : e^{-L} < |y| < 1, -1 < s < 0\}, \quad L = 1, 2, \dots,$$

obtained from one diagonal subsequence of the original terminal extraction. The family includes velocity, pressure modulo its compatible additive gauge, the suitable defect, and all current

coordinates in $\mathcal{T}$. It is nonzero when one fixed annular cross-section has quotient oscillation bounded below by a positive constant. Loss of tightness as $L \rightarrow \infty$, survival of a pressure or defect coordinate, and strong coherent convergence are, respectively, its diffuse, defect, and coherent alternatives.

Theorem B.25 (Marked recurrence produces a realized critical tail). *Every marked regenerated-current cascade produces a nonzero sequence-realized logarithmic critical-tail profile. More precisely, there are a recurrent path $p \in X_{\text{mark}}$ and integers $n_k \rightarrow \infty$ for which the same non-affine marked profile returns through observers whose total scale tends to zero. The associated annular blocks are simultaneously realized by the original terminal sequence.*

Proof. Let ϖ be a shift-invariant weak-* limit on X_{mark} . Poincaré recurrence gives a recurrent path in the support of ϖ , still satisfying the activity floor, and integers $n_k \rightarrow \infty$ such that

$$(B.36) \quad S^{n_k} p \longrightarrow p \quad \text{in } X_{\text{mark}}.$$

This convergence includes velocity, pressure gauge, defect, current, observer, and realization coordinates.

Composing the first n observer maps gives

$$(B.37) \quad \Lambda_n = \prod_{j=0}^{n-1} \lambda_j,$$

$$(B.38) \quad X_n - X_0 = r_0 \sum_{j=0}^{n-1} \left(\prod_{i=0}^{j-1} \lambda_i \right) \xi_j,$$

$$(B.39) \quad T_n - T_0 = r_0^2 \sum_{j=0}^{n-1} \left(\prod_{i=0}^{j-1} \lambda_i^2 \right) \theta_j.$$

The bounds (B.26) make the last two series convergent, whereas

$$(B.40) \quad 0 < \Lambda_{n_k} \leq 4^{-n_k} \longrightarrow 0.$$

After recentering once at the limits in (B.38)–(B.39), (B.36) is a return of the same marked profile at logarithmic displacements tending to infinity.

Retain the complete blocks between scales 1 and Λ_{n_k} . Their component shells have disjoint logarithmic interiors. By Lemma B.18, every finite family of annular velocity, pressure, defect, and current tests is realized by one member of the original terminal extraction. A diagonal extraction therefore gives a sequence-realized logarithmic annular state. It is nonzero because the recurrent cross-section retains $\mathcal{O}_{Q_1} \geq \eta$.

Loss of logarithmic tightness defines the diffuse critical-tail component. Under logarithmic tightness, failure of strong annular compactness or survival of a pressure or suitable-defect coordinate defines the defect component. In the remaining case the annular convergence is strong, the pressure gauges are compatible, and the limiting state is a coherent sequence-realized critical tail. These alternatives are exhaustive by weak-* compactness of the logarithmic mass, local compactness on each fixed annulus, and the dichotomy between zero and nonzero defect coordinates. In every case the output is the nonzero critical-tail profile asserted in the theorem. $\square$

B.4. Closure of the amplitude-renormalizing residual. The preceding recurrence theorem is useful only after the current which moves the root has been put in the same scale-critical units as the retained activity. This subsection performs that accounting. It is important that the argument uses the successor shift on actual marked events. It never replaces that shift by the full spatial translation hull, and hence never loses a sparse active root by ordinary spatial averaging.

Lemma B.26 (A retained window has a fixed critical viscous price). *Let $\mathfrak{w} = (V, P, a, b; I)$ be an $(\eta_0, \mathbf{M})$ -normalized target-retaining window. There is $d_* = d_*(\eta_0, M_{R_*}, R_*, \zeta) > 0$ such that*

$$(B.41) \quad \int_I \int_{B_{R_*}} |\nabla V|^2 dy d\tau \geq d_*.$$

The number d_ is dimensionless under Navier–Stokes scaling.*

Proof. Put $f = \zeta(V - \bar{V}_\zeta)$. The support of ζ is connected, and the weighted mean has been removed. The local Poincaré–Sobolev inequality, followed by interpolation between L^2 and L^6 , gives for almost every $\tau \in I$

$$\|f(\tau)\|_3^3 \leq C_\zeta \|V(\tau)\|_{L^2(B_{R_*})}^{3/2} \|\nabla V(\tau)\|_{L^2(B_{R_*})}^{3/2}.$$

The harmless terms containing $\nabla \zeta$ are absorbed by applying the weighted Poincaré inequality to $V - \bar{V}_\zeta$ before multiplication by ζ . Integrating in time and using Hölder’s inequality and (10.5) yields

$$\eta_0 \leq C_I^\circ(V; I) \leq C_\zeta M_{R_*}^{3/2} |I|^{1/4} \left(\int_I \int_{B_{R_*}} |\nabla V|^2 \right)^{3/4}.$$

Since $|I| = 1$, rearrangement proves (B.41), for example with $d_* = (\eta_0 / (C_\zeta M_{R_*}^{3/2}))^{4/3}$. Every quantity in the displayed inequality is written in normalized variables, so the lower bound is scale critical. $\square$

Definition B.27 (Sparse marked current path). A sparse marked current path is an infinite concatenated path of the records in Definition B.13 which satisfies clauses (R1)–(R4) of Definition B.15, the uniform normalized window bounds (10.5), and the following ordered conditions.

- (S1) Every ordinary occupation average over the normalized spatial translation subgroup assigns zero mass to the set $\{\mathcal{O}_{Q_1} \geq \eta\}$.
- (S2) Event-count averages along the successor path are not covariantly balanced. Equivalently, a fixed element of the countable test family detects a nonzero translation, scale, pressure, or moving-boundary current along a subsequence.
- (S3) The first loss of velocity compactness, pressure-gauge compactness, observer tightness, defect tightness, or simultaneous realization, and every nonzero residual pressure or suitable-energy defect coordinate, is retained as a marked output and is not identified with an ordinary velocity profile.

Thus sparsity describes loss of the root under volume averaging, whereas the marked path continues to see every active event.

Lemma B.28 (Positive density and vanishing event current are terminal). *An infinite target-retaining successor path either is impossible by covariant rigidity or has a subsequence which is a sparse marked current path.*

Proof. Let $\nu_N = N^{-1} \sum_{j < N} \delta_{\mathfrak{w}_j}$ be the event-count measures. Compactness of the complete marked records gives a weak-* limit after passage to a subsequence, and the endpoint calculation makes its path law invariant under the successor shift. Strong local L^3 convergence preserves the activity floor.

If active roots have positive upper density in a Følner sequence of the normalized spatial translation group, average the event-count law and its spatial translates. The endpoint calculation gives stationarity under the successor shift, while the Følner calculation gives spatial covariance. A weak-* limit assigns positive mass to the closed active set. The pressure cancellation in Lemma 6.3, followed by Theorem 6.8, makes its velocity spatially homogeneous almost surely, contradicting that positive mass.

If the active roots have zero covariant density but a subsequence of ν_N is covariantly balanced on the countable dense test algebra, its weak-* limit is again stationary under the successor shift and invariant under normalized spatial translations: first pass the balance to the limit on the dense algebra and then use continuity of the action. The same rigidity contradiction applies. In the remaining case ordinary averages lose the target and some test detects a persistent covariance defect. The infinitesimal pressure component of that defect is exactly the rooted correction in [Lemma 6.4](#); all other components are the translation, dilation, cutoff, and exterior coordinates already present in $\mathbf{q}_j$. Retaining the first noncompact coordinate gives (S3). Hence the remaining path satisfies (S1)–(S3). $\square$

Lemma B.29 (Nesting–antichain alternative for realized active events). *Partially order simultaneously realized active parabolic cylinders by strict inclusion after shrinking each cylinder by a fixed factor. For an arbitrarily long finite family, exactly one of the following occurs along a subsequence:*

- (i) *the lengths of inclusion chains tend to infinity;*
- (ii) *the family contains antichains of arbitrarily large cardinality.*

Every fixed finite chain or antichain is realized in one member of the original terminal sequence, with compatible pressure gauges and observer maps.

Proof. If the heights of the finite partially ordered sets are unbounded, choose a chain whose length tends to infinity and diagonalize. Otherwise their heights are bounded by some integer L . Assign to each cylinder the maximum length of an inclusion chain ending there. This partitions a family of m cylinders into at most L antichains, one of which has cardinality at least m/L . Hence the second alternative occurs.

For simultaneous realization, proceed inductively through the chosen finite family. Compose the exact translation, Galilean, and parabolic observer maps along the ancestry of each event. Only finitely many compact cylinders are prescribed, so local convergence permits one common terminal index on which all approximations hold. Normalize the pressures on overlaps and extend the gauges over their finite union. This gives one realization of the whole finite family and proves the final assertion. $\square$

Lemma B.30 (Noncompact observer marks give a first geometric output). *Consider a simultaneously realized family satisfying the normalized bounds (10.5). Local velocity, normalized-pressure, suitable-defect, and endpoint-energy coordinates are precompact on the fixed countable cylinder exhaustion. Consequently, loss of compactness of the complete marked family has a first coordinate of one of the following types:*

- (i) *normalized centers escape, producing an exterior, diffuse, pressure/noncompact, or separated recentering datum;*
- (ii) *logarithmic scale increments escape every bounded interval, producing a sequence-realized logarithmic critical-tail datum;*
- (iii) *center and logarithmic-scale increments are tight, and the shortest bounded-variation interpolation in the finite-dimensional observer group makes the translation and dilation currents tight signed marks.*

Proof. The local energy inequality and (10.5) give compact endpoint-energy coordinates and weak-* compact suitable-defect measures. Strong local L^3 compactness gives the local nonlinear terms, while the Calderón–Zygmund/harmonic decomposition gives the normalized-pressure coordinates. Thus only center and logarithmic-scale parameters can fail tightness before one of these PDE coordinates has already supplied the first failed coordinate.

Unbounded center displacement is recorded at its first escaping cylinder and gives alternative (i). Unbounded logarithmic displacement is tested on the annular cylinder exhaustion; diagonal

extraction with its realization coordinate gives alternative (ii). In the complementary case join successive observer parameters by shortest bounded-variation paths. Their distributional derivatives are finite signed measures with total variation controlled by the endpoint displacements. Pairing them with the fixed smooth cutoff gradients therefore gives tight translation and dilation currents. Diagonal compactness over the test family proves alternative (iii). $\square$

Theorem B.31 (Reduction of transverse active families). *Let $\mathcal{A}_N$ be simultaneously realized active antichains with $|\mathcal{A}_N| \rightarrow \infty$. After an ordered subsequence selection, one of the following occurs:*

- (i) *a first center, pressure, exterior, defect, or logarithmic-scale coordinate produces a sequence-realized datum of the corresponding type;*
- (ii) *for every $\delta > 0$, all but a uniformly bounded number of events belong to finitely many δ -clusters in center-scale space; each nested sequence of occupied clusters has one local profile limit, and its canonical observer interpolation produces a compact coalescent marked path.*

Thus the transverse alternative yields a first classified output or returns to the same marked-current analysis as a longitudinal active path.

Proof. Apply [Lemma B.30](#). Its first two alternatives give clause (i), so assume that centers and logarithmic scales lie in a fixed compact parameter set. After one normalization, all cylinders lie in a fixed parent cylinder and their radii lie in $[\lambda, 1]$ for some $\lambda > 0$.

Fix $\delta > 0$ and choose a maximal subfamily whose core cylinders are pairwise separated by at least δ in the normalized parabolic metric. The radii lower bound gives a uniform overlap constant. If the quotient activity of a selected sequence tends to zero, its compact limit is spatially homogeneous and hence target-null. Otherwise every selected member has a common positive activity floor. The local Poincaré estimate in [Lemma B.26](#) assigns each such core a fixed positive normalized viscous price. Since the cores coexist in one realization and have bounded overlap, their prices are bounded by the local energy on one fixed enlargement of the parent cylinder. Therefore the cardinality of the separated subfamily is bounded independently of N .

Maximality places every remaining event in one of finitely many δ -clusters. Let $\delta_k \downarrow 0$ and diagonalize. Local compactness and pressure-gauge compatibility show that each nested sequence of occupied clusters has one local profile limit; two inequivalent limits would remain separated at some fixed δ_k . The bounded-variation observer interpolation from [Lemma B.30](#) retains the complete oriented currents and produces the compact coalescent marked path in clause (ii). The two clauses exhaust the antichain alternative. $\square$

Lemma B.32 (Exact signed balance along the moving root). *Let $I_j = [\tau_j, \tau_{j+1})$ be consecutive represented intervals of one absolutely continuous chart, and let $\phi \in C_c^\infty$ be nonnegative and equal to one on the retained core. Set*

$$K_j^- = \lim_{\tau \downarrow \tau_j} \int \frac{|V(y, \tau)|^2}{2} \phi(y) dy, \quad K_j^+ = \lim_{\tau \uparrow \tau_{j+1}} \int \frac{|V(y, \tau)|^2}{2} \phi(y) dy, \quad D_j = \nu \int_{I_j} \int |\nabla V|^2 \phi,$$

and let $M_j \geq 0$ be the suitable-energy defect on the represented cell. Then

$$(B.42) \quad K_j^+ - K_j^- + D_j + M_j = H_j,$$

where the signed supply is

$$(B.43) \quad \begin{aligned} H_j = & \int_{I_j} \int \left(\frac{|V|^2}{2} + P \right) V \cdot \nabla \phi + \nu \int_{I_j} \int \frac{|V|^2}{2} \Delta \phi \\ & - \int_{I_j} a(\tau) \int \frac{|V|^2}{2} (\phi + y \cdot \nabla \phi) dy d\tau - \int_{I_j} \int \frac{|V|^2}{2} b(\tau) \cdot \nabla \phi. \end{aligned}$$

All pressure, translation, dilation, and cutoff terms are signed. If the endpoint trace of one cell is the initial trace of the next, then

$$(B.44) \quad \sum_{j=0}^{N-1} (D_j + M_j) = K_0^- - K_{N-1}^+ + \sum_{j=0}^{N-1} H_j.$$

Proof. Test (5.4) against $V\phi$. Incompressibility turns convection and pressure into the first flux in (B.43); integration by parts gives the viscous production and cutoff term. Finally,

$$\int V \cdot (V + y \cdot \nabla V) \phi = - \int \frac{|V|^2}{2} (\phi + y \cdot \nabla \phi), \quad \int V \cdot (b \cdot \nabla V) \phi = - \int \frac{|V|^2}{2} b \cdot \nabla \phi.$$

The suitable local-energy inequality becomes equality after adjoining its nonnegative Radon defect. This proves (B.42). Summing consecutive cells cancels each common energy trace with the opposite orientation and gives (B.44). No partition of test functions is used, so dissipation is not canceled in this gluing. $\square$

Lemma B.33 (Scale-critical balance on nested regenerated cells). *Let $\psi_0 \geq \psi_1 \geq \dots \geq \psi_N \geq 0$ be compatible physical localizations for a nested regenerated family with radii $r_0 > r_1 > \dots > r_N$, and put $w_j = r_j^{-1}$. Denote by $\mathcal{J}_j$ the complete oriented local-energy carrier through the stopping surface of ψ_j , and set*

$$\mathcal{P}_j := w_j \left[\nu \iint |\nabla u|^2 (\psi_j - \psi_{j+1}) + \mu_{\text{loc}} (\psi_j - \psi_{j+1}) \right] \geq 0.$$

Then $\mathcal{P}_j$ is invariant under Navier–Stokes scaling and

$$(B.45) \quad \sum_{j=0}^{N-1} \mathcal{P}_j = w_0 \mathcal{J}_0 - w_{N-1} \mathcal{J}_N + \sum_{j=1}^{N-1} (w_j - w_{j-1}) \mathcal{J}_j.$$

Consequently, a positive mean normalized price has precisely the same linear growth as the weighted scale current on the right. If the mean price tends to zero, a selected normalized shell has zero gradient and zero suitable defect, unless a first compactness, pressure, observer, or realization coordinate has already failed.

Proof. The physical local-energy equality on the shell $\psi_j - \psi_{j+1}$ gives

$$\nu \iint |\nabla u|^2 (\psi_j - \psi_{j+1}) + \mu_{\text{loc}} (\psi_j - \psi_{j+1}) = \mathcal{J}_j - \mathcal{J}_{j+1},$$

with the common interface carrying opposite orientations. Multiply by w_j , sum in j , and apply discrete summation by parts to obtain (B.45).

Under the parabolic change of variables based at radius r_j , physical viscous dissipation and the suitable defect both equal r_j times their normalized values. Multiplication by $w_j = r_j^{-1}$ therefore makes $\mathcal{P}_j$ scale invariant. If $N^{-1} \sum_{j < N} \mathcal{P}_j \rightarrow 0$, choose a shell whose price does not exceed the mean. Normalize that shell. Compactness and lower semicontinuity give zero gradient and zero defect in its limit. Failure of any limit interface is, by the fixed first-coordinate order, the stated marked output. If the mean stays positive, the identity itself gives the same linear growth for the weighted scale current. $\square$

Remark B.34 (Physical energy and critical expenditure). For a cell of radius r_j , a fixed normalized price d_* costs only $r_j d_*$ in physical energy. When $r_{j+1} \leq r_j/4$, the series $\sum_j r_j d_*$ converges. Equation (B.45) retains the missing factor r_j^{-1} : persistent normalized expenditure appears as persistent scale current, which is compactified together with the observer and converted into a realized logarithmic tail.

Lemma B.35 (Sublinear critical current cannot sustain the sparse path). *Let (C_j) be a simultaneously realized finite block of a sparse marked current path, selected so that $C_{j+1} \subseteq C_j/2$ and $r_{j+1} \leq r_j/4$. If the complete signed moving-root supply in (B.43) is $o(N)$ on blocks of length N , then the path is impossible or has a first marked output of (S3).*

Proof. Interpolate the selected cells by the absolutely continuous chart carried by the marked observer coordinates, and choose consecutive half-open represented intervals. The endpoint traces and pressure gauges agree because concatenation is part of the marked record. Choose $\phi = 1$ on the fixed target core. By Lemma B.26, $D_j + M_j \geq D_j \geq \nu d_*$ on every interval. The normalized local energy bounds give $0 \leq K_j^\pm \leq K_*$, uniformly in j .

Apply (B.44). If the complete signed supply is sublinear, then

$$N \nu d_* \leq \sum_{j < N} (D_j + M_j) \leq 2K_* + o(N),$$

which is impossible as $N \rightarrow \infty$. If the moving chart, endpoint trace, pressure gauge, or interpolation needed for the identity loses compactness, its first failure is exactly the marked output in (S3). Thus there is no unrecorded nested-core alternative: the price is paid on disjoint represented time cells, and all transport across their moving spatial boundaries is retained in H_j . $\square$

Definition B.36 (Concentration-free realized critical tail). A sequence-realized logarithmic critical tail in the sense of Definition B.24 is concentration free when every fixed positive-threshold active observer has first been retained as an actual successor. The residual annular state therefore contains only diffuse velocity mass, a pressure or suitable-energy defect, or a coherent strong annular limit. This is an ordered definition: an active observer is selected before the residual tail is formed.

Lemma B.37 (Realization of logarithmic shell support). *Let (W_n, P_n, a_n, b_n) be represented normalized states on time windows modeled on one compact interval J_* , and suppose that for some $R_n \rightarrow \infty$ and $\varepsilon_* > 0$,*

$$(B.46) \quad M_n := \int_{J_*} \int_{|y| > R_n} |W_n|^3 dy ds \geq \varepsilon_*.$$

The probability measures

$$d\nu_n = M_n^{-1} |W_n|^3 \mathbf{1}_{J_* \times \{|y| > R_n\}} dy ds$$

have, after dyadic shell decomposition and passage to a subsequence, a weak- limit on a compact metrizable logarithmic shell-state space. Every point in the support of that limit is realized by actual translated unit-shell cylinders from the same sequence (W_n, P_n) . The construction uses spatial translation and bounded time recentering only; the shell radius is a logarithmic coordinate and is not used as a Navier–Stokes dilation.*

For every realized support point, either a first pressure, observer-current, local-energy, defect, or domain coordinate fails, or the point is represented on a fixed model cylinder by a suitable local solution with the corresponding compact-cylinder bounds.

Proof. Decompose $\{|y| > R_n\}$ into the shells

$$A_{n,k} = \{2^k R_n < |y| \leq 2^{k+1} R_n\}, \quad k \geq 0,$$

and cover each occupied spacetime shell by a family of unit parabolic cylinders with uniformly bounded overlap. Translate the center of each selected cylinder to the origin and recenter its time interval inside a fixed model cylinder Q_{sh} . Translation, bounded time recentering, and subtraction of the pressure mean preserve the equation, incompressibility, the local energy inequality, and local pressure reconstruction. In particular, no factor depending on $2^k R_n$ enters the local equation.

Record the logarithmic shell index, translated center, local velocity and normalized pressure on a countable compact-cylinder exhaustion, suitable defect measures, observer currents, and the physical realization coordinate. Compactify each scalar or finite-measure coordinate by adjoining its first failure value. The represented part is sequentially compact by the local energy bounds, the pressure decomposition, and diagonal extraction. The countable product is therefore compact and metrizable. Pushing ν_n forward through the shell cover and taking a weak-* subsequence gives a probability measure Π_{sh} .

Fix $z \in \text{supp } \Pi_{\text{sh}}$ and a nested neighborhood basis $U_m \downarrow \{z\}$. Since $\Pi_{\text{sh}}(U_m) > 0$, weak-* convergence and the portmanteau inequality give $\nu_n(U_m) > 0$ along a further subsequence. Choose an actually occupied shell cylinder whose state lies in U_m , with the indices increasing in m . These genuine translated states converge to z , proving sequence realization. If z lies on a compactified failure face, the first such coordinate is the asserted output. Otherwise all local coordinates remain represented and their distributional limits give the suitable local solution claimed in the statement. $\square$

Lemma B.38 (Strong compactness on a realized shell). *Let (W_n, P_n, a_n, b_n) be a realized shell sequence on Q_{sh} , and assume that the pressure gauge, represented equation, local energy inequality, and compact-cylinder bounds have passed on every $Q' \in Q_{\text{sh}}$. Then exactly one of the following occurs:*

- (i) *the local modulation currents, an upper local critical bound, a pressure or defect coordinate, or a domain coordinate has a first failure;*
- (ii) *after a subsequence, $W_n \rightarrow W$ strongly in $L^3_{\text{loc}}(Q_{\text{sh}})$, and the local Calderón–Zygmund pressure parts converge strongly in $L^{3/2}_{\text{loc}}$.*

Thus a non-Dirac Young or viscous-defect limit is recorded at a first concrete coordinate and cannot remain an unrepresented tail.

Proof. Assume that alternative (i) does not occur and fix $Q' \in Q'' \in Q_{\text{sh}}$. The local energy inequality gives

$$W_n \text{ bounded in } L_s^\infty L_y^2(Q'') \cap L_s^2 H_y^1(Q'').$$

The equation (5.4), the local pressure bound, and the passed modulation-current coordinate give

$$\partial_s W_n \text{ bounded in } L_s^{4/3} W_y^{-1,3/2}(Q'').$$

Indeed, viscosity belongs to $L_s^2 H_y^{-1}$; convection is controlled in $L_s^{5/3} W_y^{-1,5/3}$ by the energy interpolation $W_n \in L^{10/3}(Q'')$; the normalized pressure is bounded in $L^{3/2}$ and therefore its gradient in $W^{-1,3/2}$; and the two observer terms are exactly the retained modulation-current coordinate. Aubin–Lions gives strong convergence in $L^2(Q')$. The same energy interpolation supplies a uniform $L^{10/3}(Q')$ bound, so interpolation upgrades the convergence to strong $L^3(Q')$. A diagonal choice over $Q' \in Q_{\text{sh}}$ proves the velocity assertion.

On nested balls write $P_n = P_n^{\text{loc}} + H_n + c_n(s)$, where P_n^{loc} is the Calderón–Zygmund pressure generated by $W_n \otimes W_n$, H_n is harmonic, and c_n is the pressure gauge. Strong L^3 convergence

gives strong $L^{3/2}$ convergence of P_n^{loc} . Harmonic interior estimates compactify H_n after the fixed mean normalization; failure of that compactness is precisely the first pressure coordinate in alternative (i). Finally, lower semicontinuity in the local energy inequality leaves a nonnegative viscous defect. If it is nonzero, the first positive test in the fixed countable test family records the defect in alternative (i); if every such test vanishes, the defect is zero. This proves the dichotomy. $\square$

Lemma B.39 (Bounded-origin rigidity of coherent logarithmic tails). *Let W be a coherent sequence-realized logarithmic critical tail obtained from uniformly bounded record-point representatives. Suppose that all pressure, defect, observer, and domain-realization failures have already been retained as first-coordinate outputs. Then W has a locally bounded representative on full macroscopic cylinders, including cylinders meeting the origin. Moreover its amplitude-normalized logarithmic field*

$$Z(\rho, \omega, s) := e^\rho W(e^\rho \omega, s)$$

has the following property: every sequence-realized element Z_ of its logarithmic/time-translation hull either produces a first-coordinate output or satisfies*

$$(B.47) \quad Z_*(\rho, \cdot, \cdot) \longrightarrow 0 \quad \text{locally as } \rho \rightarrow -\infty.$$

Consequently no nonzero homogeneous, log-periodic, or aperiodic minimal coherent tail can retain quotient activity.

Proof. Let W_n be the affine-normalized macroscopic representatives producing the coherent tail. The record bound gives $|W_n| \leq 2M_*$ after choosing the minimizing homogeneous velocity gauges in the closed ball of radius M_* . The domain-realization coordinate ensures that every fixed full macroscopic cylinder lies in the domain of W_n for all sufficiently large n ; otherwise that coordinate is already the asserted first output. Banach–Alaoglu and a diagonal extraction therefore give a weak-* locally bounded limit W^{full} on full cylinders. Strong convergence on punctured annuli identifies its restriction there with W .

Consider a sequence-realized amplitude-normalized translate $Z(\rho + a_k, \omega, s + b_k)$. The realization mark either remains compact, in which case the same full-cylinder argument gives a locally bounded representative W_* , or its first failed pressure, observer, defect, or domain coordinate is retained. In the compact case, with $r = e^\rho$,

$$|Z_*(\rho, \omega, s)| = r |W_*(r\omega, s)| \leq C_K r$$

on every compact angular–time set K . This proves (B.47).

If Z_* is independent of ρ , the decay forces $Z_* = 0$. If it has a nonzero period L , then $Z_*(\rho) = Z_*(\rho - kL)$; choosing the sign of k so that $\rho - kL \rightarrow -\infty$ again gives $Z_* = 0$. Finally, let $\mathcal{M}$ be a minimal aperiodic subsystem of the sequence-realized logarithmic hull. Every member obeys the decay alternative unless a first-coordinate output has already occurred. Hence zero belongs to the hull of every member. Since $\{0\}$ is a closed invariant subsystem, minimality gives $\mathcal{M} = \{0\}$. Each case contradicts retained nonzero quotient activity. $\square$

Lemma B.40 (Persistent current gives a successor or a critical tail). *Suppose a sparse marked current path has positive mean critical price and is not covered by Lemma B.35. Then, after a subsequence, exactly one of the following occurs:*

- (C1) *positive limiting current is supported on a bounded logarithmic window which contains a target-retaining active observer;*
- (C2) *after all such observers have been removed, a nonzero concentration-free sequence-realized critical tail remains;*
- (C3) *a first pressure, exterior, defect, observer, or realization coordinate loses compactness, or a nonzero residual pressure/defect coordinate survives, and is the marked output in (S3).*

Proof. For a block of length N , put mass $N^{-1}H_j^+$ on the logarithmic interval between r_j and r_{j+1} . By (B.44),

$$\sum_{j < N} H_j^+ \geq \sum_{j < N} H_j = \sum_{j < N} (D_j + M_j) + K_{N-1}^+ - K_0^- \geq N\nu d_* - 2K_*.$$

Thus these positive measures have total mass bounded below. Decompose H_j into its already recorded pressure, nonlinear transport, cutoff, translation, and dilation coordinates. Since the positive part of their sum is bounded by the sum of their absolute values, one coordinate carries a fixed fraction of the mass.

If this coordinate is not tight on the countable cylinder-and-test exhaustion, select its first failure; this is (C3). Under tightness, Lemma B.18 realizes every finite collection of its annular tests in one member of the original terminal sequence. Diagonal compactness therefore gives a sequence-realized support state, not merely an abstract current measure. If positive support meets the closed active set, the first such window gives (C1). Otherwise remove the active windows. The remaining positive mass prevents the projective annular state from vanishing and gives (C2). The three outputs are disjoint by their order and exhaustive by tightness. $\square$

Lemma B.41 (Concentration-free critical tails have no terminal component). *A concentration-free sequence-realized critical tail arising from a finite-energy suitable solution is target-null or produces a marked output already listed in (S3). In particular, it cannot be the terminal support of the persistent current in Lemma B.40.*

Proof. Apply the following alternatives in order on the logarithmic annular projective family.

If velocity mass is not tight on bounded logarithmic windows, choose the first scale at which the Galilean-reduced quantity (B.8) reaches a fixed number below the ε -regularity threshold. The point-selection proof of Proposition B.4 gives an actual target-retaining successor, a homogeneous quotient, or a non-affine harmonic pressure tail. The first alternative is absent by concentration-free selection, the second is target-null, and the third is the pressure output in (S3).

Assume next that the velocity is logarithmically tight. Apply Lemma B.37 to the normalized annular mass. Thus every point of the limiting shell support is represented by an actual translated cylinder from the original sequence; no abstract support measure is substituted for a Navier–Stokes state. On each represented shell, Lemma B.38 gives a first pressure, modulation, defect, or domain coordinate in (S3), or strong local L^3 compactness. In the latter case the Young measure is Dirac. If a positive compact quotient activity survives, its first value on the fixed dyadic activity ladder is an actual target-retaining successor, contrary to the concentration-free selection. Consequently the remaining strongly compact shell state has zero compact activity.

Pass now to the pressure and suitable-energy coordinates. On a fixed annulus, use the actual pressure in the local energy identity and subtract only its time-dependent gauge. Strong velocity convergence passes the transport term. Lower semicontinuity leaves a nonnegative viscous defect. A positive convex test detecting that defect supplies a first defect coordinate in (S3). If all positive convex tests vanish, the matrix-valued viscous defect vanishes; polarization removes its signed coordinates. The local pressure decomposition then writes the remaining pressure as a compact Calderón–Zygmund part plus a harmonic part. The former converges strongly in $L^{3/2}$; a noncompact non-affine harmonic part is precisely the pressure output in (S3). Hence outside (S3) every pressure and suitable-energy defect coordinate vanishes.

It remains to consider a coherent strong annular state. Apply Lemma B.39. A failure of full-cylinder or amplitude-normalized hull realization is the first observer, pressure, defect, or domain coordinate in (S3). When all realization coordinates pass, that lemma excludes the homogeneous, log-periodic, and aperiodic minimal coherent alternatives by bounded-origin decay.

Thus the coherent component is target-null as well. These alternatives exhaust weak annular compactness, and the assertion follows. $\square$

Theorem B.42 (Sparse moving-root path closure). *Every sparse marked current path arising from the collapsed same-origin residual reaches a typed local output in (F3) or is impossible.*

Proof. Sublinear current is excluded by Lemma B.35. The weighted identity Lemma B.33 shows that the surviving quantity is the scale-critical current: vanishing mean price gives a target-null zero-gradient shell, while positive mean price gives persistent weighted current. On the remaining path apply Lemma B.40. Alternative (C3) is the retained typed output required by (J4) and (S3), and gives the first conclusion of the theorem. Assume henceforth that it does not occur. Alternative (C2) is impossible by Lemma B.41.

Thus every sufficiently long consecutive block has its positive limiting current supported on regenerated active observers. Retain each observer, its complete oriented current, and its realization coordinate. Compactness gives the closed marked path space of Lemma B.23. Poincaré recurrence for the successor shift and exact composition of the observer maps then give a nonzero sequence-realized critical tail by Theorem B.25.

Apply the ordered annular reduction in the proof of Lemma B.41. A target-null component is impossible because the recurrent cross-section retains the activity floor; a pressure, defect, exterior, or realization failure is (F3). A nonzero concentration-free component is excluded by that lemma. The only remaining output is a target-retaining first-bad observer. Append this observer, with its complete marks, to the successor path and repeat.

If this procedure terminates, one of the preceding alternatives applies. If it does not, choose at every stage the first mesoscopic scale at which the fixed activity threshold is reached. All smaller scales are then below that threshold. The translation-modulus argument in the second paragraph of the proof of Lemma B.41 shows that the associated minimal-scale blow-downs are strongly L^3_{loc} -compact; the pressure and defect alternatives have already been placed in (F3). Their diagonal limit is therefore a nonzero coherent logarithmic tail. The bounded-origin/minimal-hull argument in the final paragraph of that proof forces this tail to be zero, a contradiction. Hence endless regeneration is also impossible. This eliminates the persistent-current case and proves the theorem. $\square$

Theorem B.43 (Closure of the collapsed same-origin residual). *Every collapsed same-origin residual in Definition 11.10 either reaches a typed local output in (F3) or is impossible. Consequently no collapsed residual remains after the typed (F3) outputs have been removed in the ordered current alternative.*

Proof. Apply Lemma B.28 to its infinite active successor path. Positive-density and covariantly balanced paths contradict covariant rigidity. The only remaining path is sparse, and is impossible by Theorem B.42. Every failure used in that proof is one of the typed successors in (J4) and therefore reaches (F3). If no such output occurs, the sparse-path contradiction applies. Thus no pressure, exterior, scale, defect, or realization loss is discarded, and the two alternatives in the statement are exhaustive. $\square$

Proof of Theorem 11.13. Apply Theorem 11.7 to the compatible marked records cut from the canonical orbit. The retained quotient floor excludes (F1). Alternative (F2) is impossible because a finite nonnegative Radon measure cannot dominate a fixed positive charge on infinitely many pairwise disjoint events. In (F3), choose the first failed coordinate in the fixed countable exhaustion. Its detecting test, physical realization coordinate, and pressure gauge survive the diagonal selection; Proposition 11.9 and Lemma B.40 therefore give the sequence-realized datum in clause (i) of the theorem.

It remains to consider the ordinary inverse-limit output (F4). Compact nonexpanding recurrence is target-null by [Proposition 11.8](#), and sublinear signed current is impossible by [Lemma B.35](#). The ordered reduction [Corollary 11.12](#) consequently leaves the collapsed same-origin path, with every observer and oriented current retained as a mark. A first loss of a marked coordinate again gives clause (i). If no coordinate is lost, the closed path space in [Lemma B.23](#) carries a recurrent successor path, and [Theorem B.25](#) produces the nonzero sequence-realized logarithmic tail in clause (ii). These choices are ordered, disjoint, and exhaustive, and every limit retains the common inverse-limit realization by the original solution. $\square$

B.5. Terminal concentration and exclusion of the record profile. The record-point extraction reduces a finite-time breakdown of a classical solution to a bounded mild ancient profile. We now close that profile without assuming the existence of a separately constructed continuous orbit. All limits below are taken with the pressure gauge, suitable-energy defect, and realization coordinate in [Definition B.13](#).

Definition B.44 (Terminally indecomposable profile). A bounded mild ancient profile U , represented modulo spatially homogeneous velocity and time-dependent pressure, is terminally indecomposable if it retains positive quotient activity on one compact cylinder and has none of the following outputs:

- (A1) an actual target-retaining descendant;
- (A2) a second asymptotically separated target-retaining profile;
- (A3) a diffuse exterior remainder;
- (A4) a noncompact velocity or non-affine pressure-gauge remainder;
- (A5) a nonzero sequence-realized logarithmic critical tail.

The alternatives are tested at every positive rational activity threshold.

Lemma B.45 (Indecomposability produces a backward L^3 sequence). *Let U be a terminally indecomposable bounded mild ancient profile obtained from the record-point extraction. Then there are $s_k \downarrow -\infty$ such that*

$$(B.48) \quad \sup_k \|U(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. Assume the contrary. Then $\int_{\mathbb{R}^3} |U(x, s)|^3 dx \rightarrow \infty$ as $s \rightarrow -\infty$. Choose $s_k \downarrow -\infty$ and $R_k \uparrow \infty$ so that

$$1 \leq \int_{|x| > R_k} |U(x, s_k)|^3 dx \leq 2.$$

Cover the exterior slice, together with a unit backward time interval, by a fixed bounded-overlap family of unit parabolic cylinders. If one cylinder has quotient activity bounded below, recentering there and using the local pressure decomposition gives either an actual target-retaining descendant or a first non-affine pressure/compactness output. These are (A1) or (A4).

Suppose instead that the supremum of the unit-cylinder activity tends to zero. The normalized exterior restrictions are then diffuse unless their mass first becomes visible at a larger multiplicative scale. The former is (A3). In the latter case choose the first dyadic scale at which a fixed quotient threshold below ε_{reg} is reached. The point-selection and pressure argument of [Proposition B.4](#) produces an actual descendant, a spatially homogeneous quotient, a pressure/noncompact output, or a realized critical-tail window. The homogeneous quotient carries no activity; the other three outputs are (A1), (A4), and (A5). Every alternative contradicts terminal indecomposability. Hence (B.48) holds. $\square$

Lemma B.46 (No terminally indecomposable active profile). *No terminally indecomposable bounded mild ancient profile obtained from the record-point extraction retains positive quotient activity.*

Proof. By Lemma B.45, the profile satisfies the hypothesis of the endpoint ancient Liouville theorem Theorem B.2. Hence $U \equiv 0$, contradicting its positive quotient activity. $\square$

Lemma B.47 (Finite separated families reduce to an infinite chain or an indecomposable profile). *Let $U_1, \dots, U_J$ be a finite family of pairwise asymptotically separated target-retaining profiles simultaneously realized from one record-point sequence. Then some U_j is terminally indecomposable, or the family generates an infinite chain of actual target-retaining descendants.*

Proof. Assume that no U_j is terminally indecomposable. Choose for each profile its first output in the order (A1)–(A5). Outputs (A3)–(A5) are retained by their own coordinates and enter the diffuse, pressure, or critical-tail alternatives below. On the remaining profiles, every vertex has an actual target-retaining successor.

Whenever such a successor is asymptotically separated from all profiles already selected, adjoin it to the family. If this happens indefinitely, the successors form the asserted infinite chain. Otherwise one obtains a maximal finite family. Draw an arrow from each vertex to the member equivalent to its selected successor. Every vertex has an outgoing arrow, so the finite directed graph contains a cycle. Periodically repeating that cycle gives an infinite actual descendant chain. Simultaneous realization and pressure-gauge compatibility are preserved under composition by Lemma B.18. $\square$

Theorem B.48 (Terminal concentration alternative). *Let U be a nonzero bounded mild ancient profile produced by Theorem B.1, with its complete record-point ancestry. After removal of its spatially homogeneous quotient, one of the following occurs:*

- (T1) *a terminally indecomposable active profile;*
- (T2) *a finite separated family of active profiles;*
- (T3) *an infinite chain of actual target-retaining descendants;*
- (T4) *a diffuse exterior remainder;*
- (T5) *a noncompact velocity or non-affine pressure remainder;*
- (T6) *a nonzero sequence-realized logarithmic critical tail.*

Every output is realized by the original record-point sequence; a measure or pressure remainder is not declared to be an ordinary velocity profile.

Proof. Apply Proposition B.3 to U . Its backward L^3 alternative makes $U = 0$ by Theorem B.2. Its homogeneous alternative is target-null. A positive compact oscillation gives an actual target-retaining profile. On the diffuse output, apply Proposition B.4; this gives another actual descendant, a homogeneous quotient, a pressure/noncompact remainder, or the first critical-tail window.

Iterate this procedure, always choosing the first output in the displayed order and retaining the realization marks. If it stops at one active profile with no further output, that profile is terminally indecomposable and (T1) holds. If only finitely many inequivalent active profiles are produced, they form (T2). Infinitely many successive active profiles give (T3). Loss of exterior tightness, local velocity or pressure compactness, or logarithmic scale tightness gives respectively (T4)–(T6). The countable cylinder-and-test exhaustion in Definition 11.6 makes the first-failure selection exhaustive, and its inverse-limit coordinate gives simultaneous realization. Thus no additional output is possible. $\square$

Theorem B.49 (Exclusion of record-point ancient profiles). *No nonzero bounded mild ancient profile with the complete ancestry and uniform record bound furnished by Theorem B.1 exists.*

Proof. Assume that such a profile exists and apply [Theorem B.48](#). Alternative (T1) is excluded by [Lemma B.46](#). Alternative (T2) reduces by [Lemma B.47](#) to (T1) or (T3).

For (T3), positive-density and covariantly balanced chains are excluded by [Lemma B.28](#). The remaining sparse marked path reaches a typed first-failure output or is impossible by [Theorem B.42](#). A typed first failure is one of (T4)–(T6), so it is retained rather than discarded.

For (T4), normalize a positive exterior restriction. Positive unit-scale quotient activity gives an actual descendant and returns to (T2) or (T3). Vanishing unit activity gives the first bad scale in [Proposition B.4](#), hence a homogeneous quotient, (T5), or (T6). Thus diffuse mass is not terminal.

In (T5), the local pressure decomposition separates a compact Calderón–Zygmund part from a harmonic remainder. The far harmonic Taylor jet is affine by [Proposition B.5](#) and is target-null. A non-affine intermediate remainder has a first bad parabolic scale and therefore gives an actual descendant or a critical-tail window by [Proposition B.4](#). Hence (T5) is not terminal.

Finally, remove all active observers from (T6). A remaining nonzero tail is excluded by [Lemma B.41](#); if an active observer remains, it is an actual descendant and returns to (T2)–(T3). Endless return is the marked regenerated case in the proof of [Theorem B.42](#) and produces the same excluded coherent tail. Every alternative is impossible. This contradicts the assumed nonzero profile. $\square$

Proof of Theorem 11.14. First remove every bounded logarithmic window carrying the fixed quotient activity threshold. If none remains, the tail is concentration free and [Lemma B.41](#) shows that it is target-null unless a first pressure, exterior, scale, trace, defect, or realization coordinate is visible. Retain that coordinate with its realizing observer. The stopping-time construction [Proposition B.4](#) and the pressure accessibility estimate [Proposition B.5](#) give, in the fixed order, an ordinary active successor, a homogeneous or affine target-null state, or the next typed marked coordinate. The first alternative enters the active case below; the second is already target-null; and indefinite repetition of the third is the sparse marked path excluded by [Theorem B.42](#). Thus no nonordinary coordinate is silently promoted to a velocity profile.

Suppose instead that active logarithmic windows remain. Retain their centers, scales, Galilean parameters, pressure gauges, defects, and oriented currents. The marked space is closed by [Lemma B.23](#). Apply [Lemma B.28](#) to the resulting infinite active successor path. Positive-density or covariantly balanced recurrence is target-null by the stationary covariant identity. In the sparse case the root and its complete boundary current remain marks, so no spatial average is used. The support alternative [Lemma B.40](#) gives a first failed coordinate, a concentration-free tail, or another active successor. The first is reduced at its first visible parabolic scale exactly as in the preceding paragraph; the second is reduced by [Lemma B.41](#) to the target-null case or the same ordered first-coordinate reduction; and endless active regeneration is excluded by [Theorem B.42](#). Thus the rooted and unrooted recurrence cases are both target-null.

For completeness, the weak annular limits used above exhaust the usual five forms. By [Lemma B.37](#), every support point is an actual translated-cylinder limit. Loss of logarithmic tightness is the diffuse case. On a tight shell, [Lemma B.38](#) either records the first Young, pressure, modulation, or viscous-defect coordinate, or makes the Young measure Dirac by strong L^3 convergence. A coherent homogeneous limit has zero quotient activity, while coherent log-periodic and aperiodic minimal limits are zero by [Lemma B.39](#). Thus no additional tail remains. $\square$

Theorem B.50 (Exclusion of bounded regenerated current). *No uniformly record-bounded marked regenerated-current cascade exists. Before bounded realization, every failed center, pressure, scale, exterior, defect, or trace coordinate is retained by Theorem 11.13.*

Proof. The bounded cascade is impossible by [Theorem B.22](#). The first-failure assertion is exactly the ordered coordinate selection in the proof of [Theorem 11.13](#). $\square$

Remark B.51 (Scope of the marked closure). The finite-depth estimate closes the record-bounded regenerated-current case before a logarithmic invariant law is needed. The quotient observable removes the accumulated constant Galilean velocity, while exact composition retains the product scale Λ_N . The required root bound is inherited from (B.22) by [Lemma B.19](#). Pressure created by motion of the root remains in $\mathbf{q}_j$ and in the exact shell balance (B.29); it is not used in the amplitude estimate.

B.6. A single forward orbit.

Proposition B.52 (Coherent interpolation of selected observers). *Let (x_j, t_j, r_j) be the chronologically ordered, target-retaining cells selected from the record-point sequence and its first-bad-scale descendants. After passing to a subsequence, they are represented by one absolutely continuous chart (X, Λ, t) on $[0, \infty)$ for which*

$$t' = \Lambda^2, \quad \Lambda(\tau_j) = r_j \downarrow 0, \quad X(\tau_j) = x_j,$$

the selected unit intervals have disjoint interiors, and (10.9) holds. The assertion is unchanged when $|x_j| \rightarrow \infty$.

Proof. First pass to a subsequence with $r_{j+1} \leq r_j/4$ and with physical time images strictly ordered. On a unit neighborhood of each selected time keep $X = x_j$, $\Lambda = r_j$, and use $t' = \Lambda^2$. In the intervening intervals interpolate $\log \Lambda$ monotonically and X linearly after choosing the interval length large enough that $a = \Lambda'/\Lambda$ and $b = X'/\Lambda$ are integrable. Define $t(\tau)$ by integrating $t' = \Lambda^2$, adjusting only the lengths of the inactive intervals; their total physical duration can be made equal to the prescribed gaps because increasing a represented interval continuously increases its physical duration from zero to infinity.

The change of variables (5.1)–(5.2) is applied to the same physical solution at every τ . Composition of affine parabolic changes therefore gives (10.9) algebraically. Constant Galilean modes are removed only in the observable (10.2), so interpolation does not alter the target on a selected window. No bound on X is used, which covers escaping centers. $\square$

B.7. Compactness and persistence.

Proposition B.53 (Compactness and retention on the canonical orbit). *The windows in [Proposition B.52](#) satisfy (10.5). Their closure is sequentially compact in (10.7), normalized pressures converge strongly in $L_{\text{loc}}^{3/2}$, and the target floor (10.6) passes to every limit.*

Proof. On every fixed normalized cylinder, the record bound and the first-bad-scale selection give a uniform velocity bound after an inner-cylinder reduction. The local energy inequality then gives uniform $L_\tau^\infty L_y^2 \cap L_\tau^2 H_y^1$ bounds. The equation gives $\partial_\tau V$ bounded in $L_\tau^{4/3} H_y^{-1}$ after the L^1 bounds for a, b from [Proposition B.52](#). Aubin–Lions compactness and interpolation yield strong local L^3 convergence.

Write $P = P_{\text{loc}} + H$ on nested balls. Calderón–Zygmund estimates and strong L^3 velocity convergence give strong $L^{3/2}$ convergence of P_{loc} . After subtracting $(P)_{B_1}$, harmonic interior estimates compactify H . If a non-affine harmonic component fails these bounds, [Proposition B.5](#) locates its first parabolic scale and it is selected as the next window, rather than remaining in the closure under consideration. Diagonal extraction over the compact-cylinder exhaustion proves sequential compactness of the closure. Finally, C_Γ° is continuous under strong L^3 convergence by [Proposition A.2](#), so the positive floor persists. $\square$

Proof of Theorem 10.2. Apply Theorem B.1 at successive record times. At a terminal singular point, apply the velocity regularity criterion after each exact constant Galilean transformation. If the Galilean-reduced quantity C^{circ} in (B.8) were below its regularity threshold on all sufficiently small tilted cylinders, the corresponding Galilean representative would be locally bounded and the original point would be regular. Hence C^{circ} crosses a fixed level below that threshold on a sequence of shrinking cylinders. Choose its first crossing on each dyadic exhaustion. The point-selection argument of Proposition B.4 supplies target-retaining normalized windows with a common quotient floor and the stated local bounds; any first pressure or realization failure is retained in the same ordered marked record rather than discarded.

Use Proposition B.52 to join these selected charts by absolutely continuous center and scale parameters. Every chart is an exact change of variables of the same suitable continuation, so the composition law holds and the physical time images approach (T). Finally, Proposition B.53 gives the asserted product compactness, pressure convergence, and persistence of the quotient floor. Passing once to the diagonal subsequence in that proposition produces the single orbit and the pairwise disjoint windows required in clauses (i)–(iv). $\square$

APPENDIX C. GAUGE, PRESSURE, AND ADJOINT ESTIMATES

C.1. Scale–center normalization.

Proposition C.1 (Local scale–center slice). *On every retained window for which the Jacobian (7.9) is invertible, the equations $\mathfrak{F} = 0$ determine a unique local scale and center. Along a represented solution these parameters are absolutely continuous and $a, b \in L^1_{loc}$. A degenerating sequence either loses the quotient target or has a first-bad-scale descendant with a nondegenerate slice.*

Proof. The first assertion is the implicit-function theorem applied to (7.8). Differentiate the four moment equations along the weakly regularized flow. The coefficient matrix is (7.9); its inverse is bounded on a compact nondegenerate window. The right-hand side consists of the localized equation pairings and the derivatives of both fixed-cutoff means in (7.7), all in L^1 . Hence the scale and center are absolutely continuous and their logarithmic velocities a, b are locally integrable. If the determinant tends to zero while the quotient activity persists, select the first smaller parabolic scale at which the fixed activity threshold is met. Proposition B.4 gives an ordinary nonhomogeneous limit or a retained pressure/scale output. Repeating the moment selection on that first scale yields a nondegenerate slice; a homogeneous limit is target-null. $\square$

C.2. Local pressure decomposition.

Proposition C.2 (Pressure normalization and compactness). *On nested inner balls one has*

$$P = P_{loc} + H, \quad -\Delta P_{loc} = \partial_i \partial_j (\zeta V_i V_j), \quad \Delta H = 0,$$

with compatible spatial means. On compact normalized families the local part is compact in $L^{3/2}$, and the harmonic part is compact modulo its affine Taylor jet. A non-affine failure occurs at an accessible first parabolic scale.

Proof. Apply the whole-space pressure formula after multiplying the velocity by a cutoff equal to one on the inner ball. Calderón–Zygmund estimates give $\|P_{loc}\|_{3/2} \lesssim \|V\|_3^2$, and strong local L^3 convergence gives compactness of this part. The remainder is harmonic, so interior derivative estimates control it after subtraction of its constant and linear Taylor terms. The constant is the pressure gauge and the linear term pairs with the removed homogeneous velocity. If a non-affine harmonic remainder is not compact, Proposition B.5 places its source inside the available parabolic depth, and Proposition B.4 selects its first visible scale. Affine parabolic chart changes preserve this decomposition and compatible means. $\square$

C.3. Adjoint localization on marked blocks.

Proposition C.3 (Covariant adjoint localization). *On every finite marked block, the normalized backward adjoint problem (8.10) has a positive solution of unit mass. Under the chart changes of the canonical observer it transforms covariantly, and the localized kinetic identity is Theorem 12.4. Every cutoff, pressure, translation, and scale term appearing there is exactly a coordinate of the signed current H_j in Theorem 11.2.*

Proof. Existence, positivity, normalization, and weak stability are Lemma 8.2. Covariance follows by changing variables in the backward equation with $t' = \Lambda^2$; the mass normalization removes the Jacobian. Testing the suitable local-energy inequality by the transported cutoff and subtracting the weighted mean equation gives Theorem 12.4. Comparing its displayed terms with (11.2) identifies them one by one with the complete moving-root current. Pressure-gauge invariance follows from $\int V \cdot \nabla \Phi = 0$. $\square$

APPENDIX D. CRITICAL FLUX–DISSIPATION BALANCE

This appendix supplies the localization algebra behind Theorem 11.2. The local energy equation separates positive dissipation and defect from signed transport. Compatible smooth localizations give an exact refinement identity, while disjoint spacetime cells give cancellation of oppositely oriented internal traces. The outer scale-space boundary remains in the complete signed current H_j .

D.1. Compatible nested parabolic cells. Let $\{\psi_\alpha\}_{\alpha \in \mathcal{T}_0}$ be a finite family of nonnegative smooth local-energy tests on one represented time interval. It is compatible when every nonterminal test is the sum of its children. The family may be obtained from a smooth partition of a finite nested collection of parabolic cells; only the partition identity is used below.

D.2. Summed local energy identity.

Proposition D.1 (Refinement identity for local-energy tests). *For a compatible finite family, the sum of the signed currents of the terminal tests equals the signed current of the root test. The same equality holds separately for kinetic energy, dissipation, and suitable-energy defect. Consequently replacing a parent by its children changes none of the terms in the balance and introduces no additional pressure or convection contribution. The identity is invariant under $P \mapsto P + c(\tau)$.*

Proof. Iterate (11.4) from the leaves to the root. Each term in (11.2) is linear in the test and its derivatives, so the asserted equality holds term by term. Pressure-gauge invariance follows from $\int V \cdot \nabla \psi = 0$ for every compactly supported test. $\square$

This proposition identifies the aggregate root flux but does not discard its outer boundary term. In the single proof spine that term is precisely the signed financing current. Sublinear growth is excluded by the critical price; persistent growth is compactified by Theorem 11.13. Thus no separate flux-domination alternative is required.

APPENDIX E. TARGET-VISIBLE COERCIVITY AND ZERO-DISSIPATION LIMITS

The dissipation need not control every material direction uniformly. The singular target detects only the Galilean-invariant oscillatory component of the velocity. Compactness and the local Dirichlet form are coercive on that quotient: a sequence on which the form vanishes converges to a spatially constant velocity and therefore disappears from the target. This gives the required coercivity-or-reduction alternative without estimating the rotation of the material eigenframe.

E.1. Zero-dissipation limits are target-null.

Theorem E.1 (Target-nullity of zero-dissipation limits). *Let $\mathfrak{w}_j = (V_j, P_j, a_j, b_j; [0, 1])$ be normalized windows in the compact closure specified by [Theorem 10.2](#). If*

$$(E.1) \quad \int_0^1 \int_{\mathbb{R}^3} \zeta^2 |\nabla V_j|^2 dy d\tau \longrightarrow 0,$$

then, after passage to a subsequence,

$$(E.2) \quad V_j \longrightarrow c(\tau) \quad \text{strongly in } L^3(\text{supp } \zeta \times [0, 1])$$

for a spatially constant vector field c . Consequently

$$(E.3) \quad \mathcal{C}_\Gamma^\circ(c; [0, 1]) = 0.$$

In particular, no sequence satisfying the fixed target floor [\(10.6\)](#) can satisfy [\(E.1\)](#).

Proof. The compactness in [\(10.7\)](#) gives, after extraction,

$$V_j \rightarrow V_\infty \quad \text{strongly in } L^3(\text{supp } \zeta \times [0, 1]), \quad \nabla V_j \rightharpoonup \nabla V_\infty \quad \text{weakly in } L_{\text{loc}}^2.$$

Weak lower semicontinuity and [\(E.1\)](#) imply

$$\int_0^1 \int \zeta^2 |\nabla V_\infty|^2 dy d\tau = 0.$$

The set $\{\zeta > 0\}$ is connected, so for almost every τ there is a vector $c(\tau)$ with $V_\infty(y, \tau) = c(\tau)$ on that set. This proves [\(E.2\)](#). The weighted mean of a constant is the same constant, hence the Galilean-invariant oscillatory integrand vanishes and [\(E.3\)](#) follows. Strong L^3 -convergence passes the fixed-cutoff oscillatory functional to the limit by [Proposition A.2](#). The positive target floor would therefore pass to a limit with zero target value, which is impossible. $\square$

E.2. Coercivity on the target quotient.

Theorem E.2 (Target-visible coercivity-or-reduction). *Let $\mathcal{K}$ be the compact closure of the selected windows of one canonical orbit, and let*

$$\mathcal{K}_{\eta_0} := \{\mathfrak{w} \in \mathcal{K} : \mathcal{C}_\Gamma^\circ(V; [0, 1]) \geq \eta_0\}.$$

Exactly one of the following alternatives holds:

- (i) $\mathcal{K}_{\eta_0} = \emptyset$, so the compact orbit closure has lost the singular target in the Galilean quotient;
- (ii) $\mathcal{K}_{\eta_0} \neq \emptyset$, and there exists $\delta_\nu > 0$ such that

$$(E.4) \quad \nu \int_0^1 \int \zeta^2 |\nabla V|^2 dy d\tau \geq \delta_\nu \quad (\mathfrak{w} \in \mathcal{K}_{\eta_0}).$$

For the canonical target-retaining orbit, alternative (i) is excluded by [\(10.6\)](#); hence [\(E.4\)](#) holds on every selected window.

Proof. The target functional is continuous under the strong local L^3 convergence, so $\mathcal{K}_{\eta_0}$ is compact. Assume first that it is nonempty. If the infimum of the left side of [\(E.4\)](#) were zero, a minimizing sequence would satisfy [\(E.1\)](#). By [Theorem E.1](#), its limit would have zero Galilean-invariant oscillation, contradicting membership in $\mathcal{K}_{\eta_0}$. The infimum is therefore positive. If this set is empty, alternative (i) holds; otherwise the positive infimum gives alternative (ii). $\square$

The preceding theorem is the material reduction needed by the regularity argument. The null space consists of spatially constant velocities on the retained connected core, which are precisely the local Galilean directions removed by the target observable. Material stretching, eigenvalue crossing, and rotation of compressed directions may be arbitrarily large inside this null class; none remains visible to the singular target.

E.3. Divergence on a persistent target sequence.

Corollary E.3 (Divergence of normalized target-visible dissipation). *For the infinite pairwise disjoint target-retaining windows of a canonical orbit,*

$$(E.5) \quad \sum_{j=1}^{\infty} \nu \int_{I_j} \int \zeta^2 |\nabla V|^2 dy d\tau = \infty.$$

If a nonnegative measure μ satisfies

$$(E.6) \quad \mu(I_j) \geq c_0 \nu \int_{I_j} \int \zeta^2 |\nabla V|^2 dy d\tau, \quad c_0 > 0,$$

then μ cannot have finite mass on the terminal tail.

Proof. Every summand in (E.5) is at least δ_ν by Theorem E.2; there are infinitely many windows. If (E.6) holds, countable additivity on the disjoint half-open windows gives

$$\mu([\tau_0, \infty)) \geq \sum_j \mu(I_j) \geq c_0 \sum_j \nu \int_{I_j} \int \zeta^2 |\nabla V|^2 dy d\tau = \infty.$$

□

For the corrected moving-cutoff production, (E.6) holds with $c_0 = 1/2$ because $\Phi = 1$ on $\text{supp } \zeta$. Thus this realization needs no estimate of the material condition number and no classification of rotating eigendirections. A different physical measure may carry scale-dependent coefficients; its lower-bound series must then be proved divergent separately. The critical localized-energy balance avoids that loss because its inward feeding and dissipation are compared before returning to physical Stokes units.

APPENDIX F. LOCALIZED EQUALITY AND THE ROLE OF THE GAUSSIAN

For the moving-cutoff realization, the proof of Theorem 12.6 uses only the coercive fixed-cutoff Dirichlet floor and connectedness of $\{\zeta > 0\}$. These properties are closed under the convergence proved for the canonical orbit. The self-contained signed-current realization has a larger apparent equality class; its reduction to target-null states is the content of Appendix E.

The Gaussian statement has a separate and more limited role. For a smooth positive whole-space probability density, vanishing of $\mathfrak{D}_G$ gives $\nabla^2 f = (2\nu\theta)^{-1}I$, and normalization and integrability then give the heat kernel (8.18) up to translation. A localized Dirichlet adjoint density is not that whole-space Gaussian. The manuscript therefore uses Gaussian rigidity only as the equality model for the formal adjoint entropy, not as a premise in the local oscillation contradiction.

REFERENCES

- [1] D. Albritton and T. Barker, *On local Type I singularities of the Navier–Stokes equations and Liouville theorems*, J. Math. Fluid Mech. **21** (2019), Article 43.
- [2] V. I. Arnold, *Sur la géométrie différentielle des groupes de Lie de dimension infinie et ses applications à l’hydrodynamique des fluides parfaits*, Ann. Inst. Fourier (Grenoble) **16** (1966), no. 1, 319–361.
- [3] L. Caffarelli, R. Kohn, and L. Nirenberg, *Partial regularity of suitable weak solutions of the Navier–Stokes equations*, Comm. Pure Appl. Math. **35** (1982), no. 6, 771–831.
- [4] D. G. Ebin and J. Marsden, *Groups of diffeomorphisms and the motion of an incompressible fluid*, Ann. of Math. (2) **92** (1970), no. 1, 102–163.
- [5] S. Gustafson, K. Kang, and T.-P. Tsai, *Interior regularity criteria for suitable weak solutions of the Navier–Stokes equations*, Comm. Math. Phys. **273** (2007), no. 1, 161–176.
- [6] J. Leray, *Sur le mouvement d’un liquide visqueux emplissant l’espace*, Acta Math. **63** (1934), 193–248.

- [7] F.-H. Lin, *A new proof of the Caffarelli–Kohn–Nirenberg theorem*, Comm. Pure Appl. Math. **51** (1998), no. 3, 241–257.
- [8] L. Onsager, *Reciprocal relations in irreversible processes. I*, Phys. Rev. **37** (1931), 405–426.
- [9] G. Perelman, *The entropy formula for the Ricci flow and its geometric applications*, arXiv:math/0211159.
Email address: `guillem@fragile.tech`